Exit, Voice, and Disclosure:

Liquidity and Takeover Premia under Stake-Triggered Ownership Rules

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Abstract

Ownership-disclosure rules trigger on the size of an investor's stake, so the disclosure threshold quietly determines how much of an activist's value creation the market observes directly versus infers from trading. I model an informed blockholder choosing among exit, passive holding, quiet engagement, and public engagement that triggers mandatory disclosure; a market maker prices order flow; a bidder conditions entry on resulting signals. Because the liquidity that lets the activist accumulate cheaply also lets her exit cheaply, minority shareholders' takeover gains are hump-shaped in liquidity—single-peakedness certified over a baseline region—peaking where engagement is worthwhile yet partly concealed. By shifting the activism premium from an inferred onto an observed basis, stricter disclosure attenuates takeover premia's sensitivity to liquidity—sharply in partial equilibrium, conditionally in general equilibrium. The engagement-premium wedge is itself microfounded through a free-rider tender game. The 5% U.S. and 3% U.K. thresholds act as distinct governance policies.

1 Introduction

When an activist investor discovers underperformance at a portfolio company, she faces a consequential choice: sell her stake and move on, engage quietly behind the scenes, or accumulate shares aggressively enough to cross a regulatory disclosure threshold. Each path carries different risks and rewards, and each sends different signals to the market. How she resolves this dilemma shapes not only her own returns but whether a takeover bid emerges and what minority shareholders ultimately receive. This tension is the classic exit-versus-voice problem of Hirschman (1970), transplanted into financial markets. On one side, liquidity facilitates governance by letting activists accumulate stakes cheaply and reducing free-riding (Maug, 1998). On the other, liquidity makes selling painless, tempting blockholders to exit rather than bear the costs of engagement (Coffee, 1991; Bhide, 1993). The tension deepens once we recognize that prices are not passive scorekeepers: when takeover decisions depend on stock prices, trading has real effects through feedback channels (Bond et al., 2012; Edmans, Goldstein, et al., 2012). Consequently, the blockholder's choice between exit and voice reverberates beyond her portfolio into the real economy. Existing models isolate these forces rather than combining them. Relative to models of order-flow feedback (Edmans, Goldstein, et al., 2015), disclosure timing (Ordóñez-Calafí and Bernhardt, 2022), and activism under corporate control (Corum and Levit, 2019; Cetemen et al., 2026), this paper lets stake-triggered disclosure partition order-flow inference into disclosed and nondisclosed branches and feed that inference into endogenous bidder entry, yielding a single equilibrium object in which endogenous governance choice, order-flow inference, and takeover feedback are jointly determined.

To fill this gap, I develop a model in which a blockholder observes a private signal about firm value and chooses among four actions: Exit (sell her stake), Hold (retain without engaging), Quiet Voice (engage below the disclosure threshold), or Public Voice (buy additional shares, engage, and trigger mandatory disclosure). A competitive market maker sets prices from discrete order flow and the disclosure flag, while a potential bidder conditions entry on these market signals. Engagement raises standalone value and strengthens resistance to acquisition, increasing the premium a bidder must offer. The key modeling choice is that disclosure is stake-triggered: activism is directly observable only when the blockholder crosses a regulatory threshold, creating two distinct information regimes within a single equilibrium.

The model yields a central result and a supporting comparative static. The comparative static comes first because it exposes the inference channel on which the central result operates: liquidity has a nonmonotone effect on expected minority takeover gains. At low liquidity, noise trading is scarce, making the blockholder's orders highly informative. The market maker largely infers her action, which deters quiet engagement and suppresses bids. As liquidity rises, noise camouflages the activist's trades, encouraging voice and improving bid incidence. However, at high liquidity, exit becomes so cheap that engagement incentives collapse entirely. The net effect is hump-shaped: minority shareholders benefit most at intermediate liquidity, where voice is both incentive-compatible and partially concealed. The central result is that stake-triggered disclosure attenuates this inference channel. Lowering the disclosure threshold shifts probability mass from Quiet Voice to Public Voice, compressing the activism premium into the observable regime and making it less sensitive to liquidity. The model predicts that the sensitivity of takeover premia to liquidity is lower in strict-disclosure jurisdictions. This attenuation holds sharply in partial equilibrium; once the blockholder's engagement incentives adjust in general equilibrium, it splits into a transparency channel and a deterrence channel whose net sign is governed by primitives (Proposition 10). Throughout, I state the mechanism where it holds sharply and mark precisely where the general-equilibrium sign turns on primitives.

These theoretical mechanisms speak directly to active policy debates. The United States Schedule 13D threshold of 5% and the United Kingdom FCA threshold of 3% create meaningfully different information environments. The framework provides a rigorous way to evaluate how such cross-country differences affect takeover outcomes and minority shareholder welfare, revealing that liquidity regulation operates implicitly as governance policy.

The model's predictions are motivated by well-documented empirical patterns. Brav et al. (2008) show that hedge fund activism often targets firms that subsequently receive acquisition offers. Greenwood and Schor (2009) find that a substantial fraction of activist returns is attributable to takeover outcomes. The free-rider problem that Grossman and Hart (1980) identified in tender offers makes takeover premia the natural welfare object for dispersed minority shareholders, underscoring the importance of understanding how liquidity shapes these outcomes. The same free-rider logic does double duty here: solving the tender game at the bargaining disagreement node is what pins the engagement premium wedge from primitives (Appendix D), and the resulting appropriability coefficient predicts *where* activism raises premia and where — as Celentano and Levine (2025) estimate on average — measured premia fall.

The paper proceeds as follows. Section 2 reviews the literature. Section 3 presents the model, and Section 4 characterizes the equilibrium. Section 5 develops comparative statics with numerical analysis. Section 6 presents testable implications. Sections 7 and 8 analyze welfare and disclosure extensions. Section 9 concludes.

2 Related Literature

This paper connects three strands of the finance literature (blockholder governance, market microstructure, and corporate takeovers) and shows that their interaction produces equilibrium effects that no single strand generates alone.

Three recent papers are especially close, and the contribution is sharpest when set against them. Ordóñez-Calafí and Bernhardt (2022) study the optimal *design* of the beneficial-ownership disclosure threshold, weighing the adverse-selection cost it imposes on uninformed investors against the activism it disciplines; here the threshold is instead taken as an institutional primitive whose equilibrium role is to partition order-flow inference and, through that channel, shape takeover premia. Corum and Levit (2019) model activists as catalysts of corporate control via contracting with bidders, but abstract from the secondary-market inference and disclosure regimes that are central here. Finally, Cetemen et al. (2026) provide a dynamic account in which multiple activists time their trades sequentially and coordinate through market signals, producing leader-follower dynamics; this paper is its static counterpart, with a single blockholder whose threshold-crossing disclosure splits order-flow inference into disclosed and nondisclosed branches that in turn feed bidder entry.

2.1 Exit vs. Voice in Corporate Governance

The exit-voice trade-off originates with Hirschman (1970). Early applications treated the two as substitutes, arguing that liquidity weakens governance by allowing blockholders to flee rather than fight (Coffee, 1991; Bhide, 1993). Maug (1998) reversed this logic, demonstrating that liquidity can facilitate voice by easing stake accumulation. Edmans (2009) recast exit itself as a governance mechanism, showing that trading on bad private information punishes underperforming managers through equity-linked compensation. Edmans and Manso (2011) extend this to multiple blockholders, while Edmans, V. W. Fang, et al. (2013) provide empirical evidence that liquidity shifts blockholders toward exit and away from voice. Levit et al. (2024) further show that trading reshapes the shareholder base and interacts with governance through voting, complementing the exit-voice trade-off studied here.[1]

2.2 Market Microstructure and Price Feedback

Microstructure models explain how prices aggregate private information through trading (Kyle, 1985; Glosten and Milgrom, 1985). I build on the tractable discrete order-flow framework of Edmans, Goldstein, et al. (2015), which permits closed-form Bayesian updating and clear price fixed points. Financial prices also shape real economic decisions through feedback channels. Edmans, Goldstein, et al. (2012) demonstrate that stock prices affect takeover outcomes: low prices signal weak fundamentals and invite bids, while high prices deter them. Bond et al. (2012) survey this broader feedback literature; Goldstein (2023) provides a more recent synthesis of how market information feeds back into real corporate decisions.[2] None of these frameworks, however, allow an informed blockholder to choose between exit and voice before prices form.

2.3 Shareholder Activism and Takeovers

Empirical evidence firmly links activism to corporate control. Brav et al. (2008) document foundational stylized facts regarding activist target selection and subsequent acquisition offers. Greenwood and Schor (2009) show that a substantial fraction of activist returns traces to these takeover outcomes. Corum and Levit (2019) model this control channel directly, showing that activists hold a comparative advantage over bidders in pressuring entrenched incumbents to sell; relative to their static contracting focus, the present paper embeds the activist's governance choice in a market-microstructure setting where order-flow inference and disclosure shape bidder entry. The canonical Grossman and Hart (1980) model establishes the free-rider problem that makes the takeover premium the relevant welfare metric for dispersed minority shareholders.

A rapidly maturing structural literature quantifies activism along complementary margins, and the present paper's positioning is sharpest against it. Gantchev (2013) estimates the sequential costs of campaigns; Albuquerque et al. (2022) structurally estimate the binary 13D-versus-13G choice jointly with announcement returns, decomposing activist returns into treatment, stock-picking, and selection — direct evidence that prices *capitalize* activism at disclosure, the denominator channel

[1] A common feature of these governance models is continuous trading. The present paper departs from that tradition by adopting discrete order flow, which keeps the inference problem tractable when integrating both disclosure regimes and price feedback.

[2] Dow et al. (2017) provide foundations for endogenous information production in settings where prices affect real investment.

of Proposition 14; Johnson and Swem (2021) estimate a dynamic reputation model of activism whose single state variable is the activist's reputation, the accumulation stage collapsed into a lognormal campaign-cost draw standing in for price impact and trading costs; and Celentano and Levine (2025) estimate an equilibrium model of activism, takeovers, and managerial discipline — finding that activism *lowers* bid premia by 13.7% relative to a no-activist counterfactual through board bargaining, a finding Proposition 14 rationalizes through the price channel without reversing the engagement wedge, with cross-sectional content (q, γ, ψ) that a presence-only channel does not carry. None of these models the trading-and-disclosure environment itself — liquidity, order-flow inference, and the stake-triggered disclosure rule are this paper's state variables, and the premium those papers treat as an estimated constant is here an equilibrium object of tender mechanics (Appendix D) whose sign and size vary in observable primitives. Collin-Dufresne and Fos (2015) provide the reduced-form complement, showing measured price impact falls exactly when informed blockholders trade against liquidity.[3]

2.4 Relation to This Paper

The closest antecedent is Edmans, Goldstein, et al. (2015), who develop a discrete order-flow model in which prices feed back into takeover decisions. That framework shares this paper's tractable Bayesian structure and price fixed points, but differs in three critical respects. First, the blockholder in Edmans, Goldstein, et al. (2015) has an exogenous information advantage and does not choose between alternative governance strategies; in this paper, the blockholder endogenously selects among Exit, Hold, Quiet Voice, and Public Voice, making the governance mode itself a strategic variable. Second, there is no disclosure regime: because the blockholder in Edmans, Goldstein, et al. (2015) does not acquire a stake, there is no threshold-crossing event that partitions the information environment. Here, stake-triggered disclosure splits equilibrium inference into disclosed and nondisclosed branches with sharply different informational content. Third, the absence of endogenous voice means Edmans, Goldstein, et al. (2015) cannot generate the nonmonotone relationship between liquidity and takeover premia or the disclosure-attenuation mechanism that constitutes this paper's central contribution. Among theories of disclosure-threshold *design*, the closest competitor is Ordóñez-Calafí and Bernhardt (2022), who study how the level of the beneficial-ownership disclosure threshold trades off the adverse-selection cost borne by uninformed investors against the disciplining benefit of activism. They characterize the optimal threshold in a setting without takeover feedback; this paper takes the threshold as the institutional primitive and shows how, in equilibrium, it splits order-flow inference into disclosed and nondisclosed branches and thereby governs the sensitivity of takeover premia to liquidity.

[3] Back et al. (2018) embed an activist in a continuous-time Kyle model to study liquidity and efficiency, but they do not incorporate the feedback loop between order-flow inference and bidder entry conditional on disclosure. Cetemen et al. (2026) study a dynamic complement in which sophisticated activists time their trades sequentially and use market signals to coordinate, generating leader-follower dynamics among blockholders; this paper instead focuses on a single blockholder's static choice among exit, quiet voice, and threshold-triggering public voice, and how disclosure partitions order-flow inference that feeds bidder entry.

3 Model

The model has four stages that capture the blockholder's information advantage, the market's inference problem, and the bidder's response.

3.1 Timeline

Time is discrete, $t \in \{0, 1, 1.5, 2\}$.

$t = 0$: Nature draws a standalone fundamental v. The blockholder observes a private signal s about v.

$t = 1$: The blockholder chooses an action (q, a) from the feasible set $\{(-1, 0), (0, 0), (0, 1), (+1, 1)\}$, where $q \in \{-1, 0, +1\}$ is the trade and $a \in \{0, 1\}$ is engagement. A noise trader submits $z \in \{-1, 0, +1\}$. The market maker observes total order flow $X = q + z$ and disclosure indicator D, then sets a competitive price $P(X, D)$.

$t = 1.5$: A potential bidder observes $(P(X, D), D)$ and a private synergy shock ξ, then decides whether to initiate a takeover attempt.

$t = 2$: Payoffs are realized: either the takeover is consummated at the offered price, or the firm remains standalone with (possibly) improved fundamentals due to engagement.

The feasible set $\{(-1, 0), (0, 0), (0, 1), (+1, 1)\}$ is not an arbitrary restriction. The two omitted combinations—*engaged exit* $(-1, 1)$ and *silent buy* $(+1, 0)$—are never chosen by an optimizing blockholder, so the four-action menu is a result rather than an assumption.

Lemma 1 (Dominance of the excluded actions)**.** *Under the maintained assumptions, with the disclosure rule* $D = \mathbf{1}\{q = +1\}$ *and a strictly positive engagement cost* $C(s) > 0$ *(Assumption* (A2)*):*

1. *Engaged exit* $(q, a) = (-1, 1)$ *is strictly dominated by plain Exit* $(-1, 0)$ *at every signal* s*.*
2. *Silent buy* $(q, a) = (+1, 0)$ *is weakly dominated, and is dominated strictly whenever engagement is worthwhile at the disclosed posterior; equivalently, any action with* $q = +1$ *optimally sets* $a = 1$*.*

Hence every best response lies in $\{E, H, Q, P\} = \{(-1, 0), (0, 0), (0, 1), (+1, 1)\}$*.*

Proof. (i) Engaged exit. Selling the entire stake sets the end-of-period holding to $h = 1 + q = 0$, so the blockholder retains no claim on the firm's terminal value Y and captures none of the engagement-driven improvements $\delta\tilde{\Delta}$ (standalone) or $\delta(\tilde{m} - m_0)$ (takeover premium). The two actions $(-1, 1)$ and $(-1, 0)$ generate the same order flow $q = -1$, the same disclosure outcome $D = 0$, and the same execution proceeds $-qP(X, 0)$, so they share an identical payoff *except* that $(-1, 1)$ additionally incurs the engagement cost $a\,C(s) = C(s) > 0$. Therefore $U(-1, 1; s) = U(-1, 0; s) - C(s) < U(-1, 0; s)$ for all s, and engaged exit is strictly dominated.

(ii) Silent buy. Because disclosure is stake-triggered, $q = +1$ forces $D = 1$ regardless of a; a buy cannot be concealed. Conditional on $q = +1$ and the disclosed information set, the engagement decision affects the payoff only through the term $\delta h\,[(1 - p)\,a\tilde{\Delta} + p\,a(\tilde{m} - m_0)] - a\,C(s)$ with $h = 2$.

Since engaging weakly raises both the standalone value (by $\tilde{\Delta} \geq 0$) and the required takeover premium (by $\tilde{m} - m_0 > 0$ under Assumption (A3)), the marginal value of engagement conditional on having already bought and disclosed is $\delta\, 2\, [(1-p)\tilde{\Delta} + p(\tilde{m} - m_0)] - C(s)$, which is positive on the high-signal region where any buy occurs (the same single-crossing logic that delivers the cutoff k_D in Proposition 1). Thus a blockholder who buys optimally also engages: $(+1, 0)$ is weakly dominated by $(+1, 1)$, and strictly so wherever the disclosed engagement benefit exceeds $C(s)$. The exclusion of $(+1, 0)$ is therefore pinned jointly by the disclosure technology (a buy always discloses) and by the optimality of engaging once disclosed; we do not claim it is dominated for arbitrary primitives, only under the maintained assumptions. □

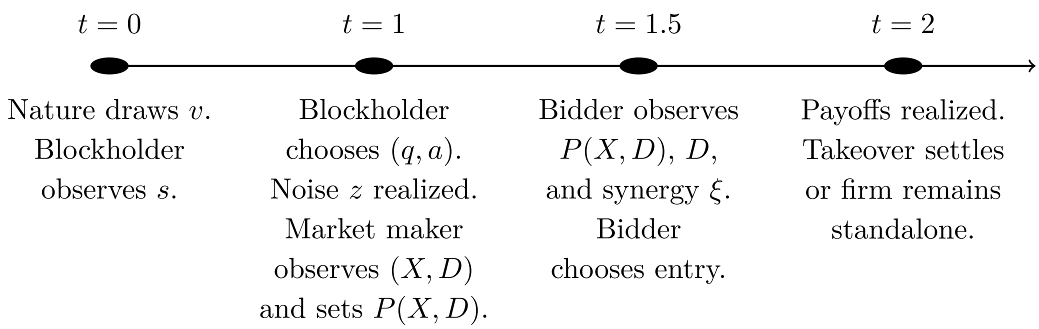

Figure 1: Timeline of the model. The four stages capture the blockholder's information advantage, the market's inference problem, and the bidder's response.

Table 1 in Appendix A provides a quick reference for notation. The remainder of this section introduces each model component in turn, beginning with the blockholder's information environment and ending with the equilibrium concept.

3.2 Fundamentals and Information

The standalone fundamental is $v \sim \mathcal{N}(\mu, \sigma_v^2)$. The blockholder observes a private signal

$$s = v + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

independent of v. Let $\sigma_s^2 \equiv \sigma_v^2 + \sigma_\varepsilon^2$. The blockholder's posterior mean is linear:

$$\hat{v}(s) \equiv \mathbb{E}[v \mid s] = \mu + \beta(s - \mu), \quad \beta \equiv \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\varepsilon^2} \in (0, 1).$$

The bidder draws an independent synergy shock $\xi \sim \mathcal{N}(0, \sigma_\xi^2)$. This signal structure gives the blockholder a transient information advantage that she can exploit through trading, engagement, or both.

3.3 Trading and Liquidity

The blockholder has an initial stake of one share (normalization). Her order $q \in \{-1, 0, +1\}$ implies an end-of-$t = 1$ holding of $h = 1 + q$ shares.

Noise trading is discrete:

$$z \in \{-1, 0, 1\}, \quad \mathbb{P}(z = 0) = 1 - \kappa, \quad \mathbb{P}(z = \pm 1) = \kappa/2,$$

where $\kappa \in (0, 1)$ indexes noise-trading intensity (market liquidity). Higher κ means more frequent nonzero noise imbalance and therefore weaker inference from order flow.

Total order flow observed by the market maker is

$$X = q + z \in \{-2, -1, 0, 1, 2\}.$$

The blockholder's trading decision is intertwined with her engagement choice, to which I now turn.

3.4 Disclosure Rule

Let $D \in \{0, 1\}$ be a public indicator for whether a stake acquisition crosses a disclosure threshold. I model disclosure as *stake-triggered*: in the discrete framework,

$$D = 1 \iff (q = +1).$$

That is, **if the blockholder acquires an additional share, a threshold-crossing disclosure occurs**. This captures the institutional logic that crossing an ownership threshold triggers a regulatory filing (e.g., Schedule 13D), independent of whether the filing explicitly reflects intent. Below-threshold engagement remains hidden.

Timing of disclosure. The disclosure indicator D is determined simultaneously with the blockholder's action (q, a) and is observed by the market maker at the same time as order flow X. Specifically, the sequence within $t = 1$ is:

(i) The blockholder chooses (q, a), which determines $D = \mathbf{1}\{q = +1\}$.

(ii) Noise trader submits z; total order flow $X = q + z$ is realized.

(iii) The market maker observes (X, D) jointly and sets price $P(X, D)$.

This timing ensures that disclosure does not reveal additional information beyond what is captured by (X, D) jointly.

3.5 Engagement Technology (Voice)

If the blockholder chooses engagement $a = 1$, she pays a private, signal-dependent cost $C(s) > 0$ at $t = 1$. This cost is observed by the blockholder when she observes s but not by the market maker or bidder. I assume engagement costs are decreasing in the signal: a blockholder with more favorable private information finds it easier to build a compelling case for change. I parameterize the cost as

$$C(s) = C_0 \exp\left(-\chi \frac{s - \mu}{\sigma_s}\right),$$

where $C_0 > 0$ is the cost at the prior mean and $\chi \geq 0$ controls cost sensitivity to signal quality (Assumption (A2)). The assumption that engagement costs are decreasing in the signal reflects the economic mechanics of persuasion. A blockholder with more favorable private information about fundamentals finds it easier to build a compelling case for change. With a strong underlying asset, the activist can point to concrete operational improvements, favorable peer comparisons, or verifiable evidence of undervaluation, substantially lowering the friction of persuading the board or rallying fellow institutional shareholders. The exponential form is chosen for analytical convenience to ensure non-negativity, but the qualitative results require only strict positivity and a monotone decrease in s. This monotonicity assumption is economically substantive. If costs were instead increasing in s (for example, because high-value firms feature entrenched managers who are harder to dislodge), the cutoff ordering would potentially invert, pushing the blockholder toward exit on good news and voice on bad news. The decreasing cost structure captures the standard paradigm where a strong fundamental thesis is required for a viable activist campaign.

Appendix E makes the role of Assumption (A2) exact. At fixed prices the gross (pre-cost) Quiet-over-Hold value gap is *signal-free* — Hold and Quiet generate the same order flow and disclosure flag, so their payoffs differ only through the engagement lottery, whose value depends on the cell objects, not on s (eq. (43)); the gross slope is $\Lambda(s) \equiv 0$. The Hold/Quiet margin therefore single-crosses precisely in the *persuasion regime* $C'(s) < \Lambda(s) \equiv 0$ — that is, exactly under (A2) (Proposition 15). A flat cost ($\chi = 0$) makes the Hold/Quiet gap constant in s, and the equilibrium sits on the two-cutoff collapse face (footnote below); the four-action architecture, existence, and comparative statics survive, but the interior Hold/Quiet boundary degenerates. The ordering can invert under an *entrenchment regime* ($C'(s) \geq 0$ on a nondegenerate set, e.g. better-informed blockholders facing harsher managerial entrenchment precisely when their case is strong), which carries the opposite, falsifiable empirical signature (Definition 2, Remark 14). The role of $\chi > 0$ is thus the interior Hold/Quiet margin and the tilt of engagement toward higher-signal types. The interior peak of $\Delta^{\min}(\kappa)$ survives a flat cost, with endpoint symmetry $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$ derived from the noise law (Appendix E).

This monotone, strictly positive specification ensures that neither engagement nor passivity is always optimal when the blockholder holds her initial stake, yielding an interior cutoff structure (Assumption (A1)).[4]

Engagement succeeds with probability $\rho \in (0, 1]$ and affects outcomes in two ways when successful:

Outside option improvement. If engagement succeeds and no takeover is consummated, the standalone payoff is increased by $\Delta > 0$ at $t = 2$. If engagement fails, standalone payoff remains v.

Bargaining/resistance premium. If engagement succeeds and a takeover is consummated, the required premium increases from m_0 to m_1, where $m_1 > m_0 \geq 0$. The wedge captures bargaining power, defensive tactics, or any friction that raises the price needed to complete the acquisition

[4] Sufficient conditions for interior cutoffs include $C(\mu) < \delta(\tilde{\Delta} + (\tilde{m} - m_0))$ (engagement is worthwhile at the prior mean) and $\lim_{s \to -\infty} C(s) > \delta(\tilde{\Delta} + (\tilde{m} - m_0))$ (engagement is dominated for sufficiently low signals), together with the monotonicity of $C(s)$. I also assume that stake building (Public Voice) is not always dominated when engagement occurs, so that disclosure is triggered for sufficiently high signals. When $\chi > 0$, cost heterogeneity supports an interior Hold region; when $\chi = 0$, costs are constant and the equilibrium collapses to the two-cutoff case.

when an engaged blockholder is present. Failed engagement yields the baseline premium m_0. I interpret m_0 and m_1 as *per-share takeover premia* above the market price (so the consummated offer satisfies $b = P + m$); they are distinct from $\hat{v}(s)$, the posterior mean of fundamentals.

Risk-neutral simplification. Since success is unobserved before $t = 2$ and all parties are risk-neutral, I define expected engagement effects:

$$\tilde{\Delta} \equiv \rho\Delta, \qquad \tilde{m} \equiv m_0 + \rho(m_1 - m_0).$$

Then engagement ($a = 1$) yields expected standalone improvement $\tilde{\Delta}$ and expected premium wedge $\tilde{m}$. I assume the premium wedge satisfies $\tilde{m} > m_0$, which requires $\rho(m_1 - m_0) > 0$ (Assumption (A3)). This ensures that engagement has real consequences for takeover outcomes; an engaged blockholder can extract higher premia through an improved bargaining position or defensive tactics.

Tender-game micro-foundation of the premium wedge. Assumption (A3) is the conclusion of an explicit two-layer micro-foundation rather than a free-standing primitive. In Appendix C I show that if the bid-conditional premia (m_0, m_1) arise from generalized Nash bargaining between the target's control bloc (bargaining weight $\theta \in (0, 1)$) and the bidder over the takeover surplus $\bar{S}+\xi$, and if engagement raises the bloc's disagreement payoff by an appropriable amount $\lambda\,\rho\,\Delta_{\text{eng}} > 0$ (Condition 1), then the wedge is the closed form $m_1 - m_0 = (1-\theta)\lambda\rho\Delta_{\text{eng}}$, invariant to the synergy draw ξ (Theorem 1). In Appendix D I then close the step earlier drafts left open: solving the complete free-rider tender game at the disagreement node (in the institution of Grossman and Hart (1980), with the outside-option reading of Binmore et al. (1986)) *derives* the appropriability coefficient,

$$\lambda \;=\; 1 - q\,(1-\gamma)\,\psi \;\in\; [0, 1],$$

where q is the equilibrium fringe-raid probability (the symbol is local to this tender-game discussion and to Appendix D; it is unrelated to the blockholder's trade direction q of Section 3), $\gamma \in [0, 1]$ the portability of the activist's improvement under an alternative acquirer, and ψ a pivotality factor (Proposition 13). Assumption (A3) then holds *except on a characterized boundary*: it fails only when fringe raids are certain, fully supersede the activist's improvement, and cannot be blocked (Theorem 2). The same appendix shows that the *measured* bid premium — the offer over a market price that already capitalizes activism — declines in activism evidence whenever λ is small even though $m_1 \geq m_0$ throughout (Proposition 14), reconciling the model with the negative premium effect estimated by Celentano and Levine (2025). The reduced form used in the body is the constant-θ, λ-at-baseline special case.

3.6 Bidder Entry and Bidding Terms

The bidder follows a price-contingent entry rule as in standard takeover models (Grossman and Hart, 1980; Fishman, 1988; Bulow et al., 1999; Edmans, Goldstein, et al., 2012). The bidder observes the public pair $(P(X, D), D)$ and a private synergy shock ξ. On the nondisclosed branch $D = 0$, I assume the equilibrium price schedule separates order-flow realizations ($P^*(X, 0) \neq P^*(X', 0)$ for all distinct $X, X' \in \{-2, -1, 0, 1\}$) so that the bidder can infer X from $(P, 0)$ (Assumption (A7)).[5] I therefore use $p(X,0)$ and $\bar{m}(X,0)$ as shorthand for $p(P(X,0),0)$ and $\bar{m}(P(X,0),0)$. On the disclosed branch $D=1$, disclosure pins down Public Voice and hence $a=1$, so X reveals only the noise realization and carries no information about fundamentals. Under Assumption A5 (described below), this implies that equilibrium beliefs and the pricing fixed point are identical across $X \in \{0,1,2\}$ on the disclosed branch, so $P^*(X,1)$ and $p(X,1)$ are constant across X (see Appendix B.3). I therefore continue to write $p(X,1)$ as shorthand for $p(P^*(X,1),1)$.

I assume that synergies and costs satisfy $\bar{S}-K > m_0-\mu$ and $\bar{S}-K < m_1+(\mu+3\sigma_v)$ (Assumption (A4)). These bounds ensure that takeover bids occur with interior probability: the bidder is neither always deterred nor always willing to bid regardless of prices.

Conditional on (P,D), the bidder's net deal surplus is

$$\Pi_B(X,D) = \bar{S} - P(X,D) + \xi - \bar{m}(X,D) - K,$$

where $\bar{S}$ is baseline synergy, $K>0$ is a fixed bidding cost, and $\bar{m}(X,D)$ is the expected premium wedge conditional on (X,D). The market price enters surplus dollar-for-dollar because the bidder must pay the target's market value (plus the premium wedge) to complete the acquisition.

The premium wedge ultimately depends on whether the blockholder has engaged. I define the premium required to complete the acquisition as a function of the *realized* engagement action a:

$$m^R(a) \equiv m_0 + a\cdot(\tilde{m}-m_0), \qquad a\in\{0,1\}.$$

I interpret $m^R(a)$ as the (ex ante) premium required at closing, determined by the blockholder's engagement-driven resistance. When engagement is not publicly disclosed, the market maker and bidder do not observe a and instead form the posterior $\pi(X,D)=\mathbb{P}(a=1\mid X,D)$. The corresponding *expected* premium wedge conditional on (X,D) is therefore

$$\bar{m}(X,D) \equiv \mathbb{E}[m^R(a)\mid X,D] = m_0 + (\tilde{m}-m_0)\cdot\pi(X,D), \quad \pi(X,D)\equiv\mathbb{P}(a=1\mid X,D).$$

I define $\bar{m}(P,D) \equiv \bar{m}(X(P,D),D)$ and the takeover completion probability conditional on observed (P,D) as

$$p(P,D) \equiv 1-\Phi\left(\frac{\bar{m}(P,D)+K-\bar{S}+P}{\sigma_\xi}\right), \tag{1}$$

and use the shorthand $p(X,D)\equiv p(P(X,D),D)$ in equilibrium. The bidder initiates a takeover attempt if and only if $\Pi_B(X,D)\ge 0$, so $p(P,D)$ is the probability of a consummated transaction. Here $\Phi(\cdot)$ is the standard normal cumulative distribution function.

If the takeover is consummated, the *realized* per-share offer is

$$b(X,D,a) = P(X,D) + m^R(a).$$

[5] Injectivity on the $D=0$ branch is a standard regularity condition in discrete-order-flow models: different X imply different posteriors about fundamentals and hence different competitive prices. Assumption (A7) avoids a notational distinction between conditioning on (P,D) and conditioning on (X,D).

3.7 Terminal Payoff and Price Formation

Let $\delta = \exp(-r\tau)$ (approximately $1/(1+r)$ for short horizons) be the discount factor between the pricing/entry date and payoff/settlement at $t = 2$, where τ is the time between these dates (Cochrane, 2005; Duffie, 2010).[6]

If no takeover is consummated, per-share terminal payoff is $y^{\text{stand}} = v + a\tilde{\Delta}$. If a takeover is consummated, per-share payoff is $y^{\text{M\&A}} = b(X, D, a) = P(X, D) + m^R(a)$.

Therefore, the per-share terminal payoff is

$$Y = \mathbf{1}\{\text{bid}\} \cdot (P(X, D) + m^R(a)) + (1 - \mathbf{1}\{\text{bid}\}) \cdot (v + a\tilde{\Delta}),$$

where $\mathbf{1}\{\text{bid}\}$ denotes a *consummated* takeover rather than a mere offer. Bargaining, board resistance, and regulatory frictions are captured in reduced form by the premium wedge $m^R(\cdot)$ and the fixed cost K. A simple robustness extension would multiply $p(P, D)$ by a completion probability $\alpha(P, D) \in (0, 1]$ in the pricing fixed point, without altering the main mechanism.

Conditional on (X, D), $\mathbb{E}[m^R(a) \mid X, D] = \bar{m}(X, D)$, so the expected takeover payoff equals $P(X, D) + \bar{m}(X, D)$; this keeps the pricing fixed-point equations below unchanged.

The market maker is competitive and sets

$$P(X, D) = \delta\, \mathbb{E}[Y \mid X, D]. \tag{2}$$

Equation (2) is the key equilibrium discipline: the price reflects the possibility of takeover and the premium wedge, and takeover decisions depend on that same price. I assume that for each information set (X, D) and each feasible posterior $\pi(X, D)$, the competitive pricing fixed point admits a unique solution $P^*(X, D)$, and that takeover entry is not too sensitive to prices: $\delta \cdot \sup_P |\partial p/\partial P| < 1$ (Assumption (A5)).[7]

3.8 Blockholder Payoff and Equilibrium Concept

Conditional on signal s, the blockholder chooses (q, a) to maximize expected present value. If she submits q and ends with $h = 1 + q$ shares, her $t = 1$ value is

$$U(q, a \mid s) = \mathbb{E}\Big[- q \cdot P(X, D) + \delta \cdot h \cdot Y - a \cdot C(s) \,\Big|\, s, q, a \Big],$$

where $-qP(X, D)$ is the net cash flow from trading at $t = 1$ (selling $q = -1$ yields $+P$; buying $q = +1$ yields $-P$).

Definition 1 (Perfect Bayesian Equilibrium)**.** *A* Perfect Bayesian Equilibrium *consists of:*

(i) a blockholder strategy mapping s into (q, a),

(ii) beliefs $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$ consistent with Bayes' rule,

[6] If one measures payoffs in time-1 present-value units, δ can be set to one and later reintroduced by scaling. I keep δ explicit throughout.

[7] Assumption (A5) bounds the strength of the price-to-entry feedback loop and ensures that comparative statics are well-defined. Under the Gaussian bidding rule, a sufficient condition *for the sensitivity clause* is $\delta/\sigma_\xi < 1/\phi(0)$, where $\phi(\cdot)$ is the standard normal density; the uniqueness clause is the separate inner-price contraction stated as Assumption (A5a) below, whose baseline margins are discussed in Remark 2.

(iii) a bidder entry strategy satisfying (1)*, and*

(iv) a competitive price schedule $P(X, D)$ *satisfying* (2)*.*

Because noise trading z has full support on $\{-1, 0, +1\}$, every (X, D) pair that is feasible under an on-path action occurs with positive probability. In particular, when $D = 0$, $X \in \{-2, -1, 0, 1\}$; when $D = 1$, only Public Voice is feasible so $X \in \{0, 1, 2\}$. Consequently, Bayes' rule pins down $\pi(X, D)$ for all reached information sets; off-path beliefs are irrelevant.[8]

4 Equilibrium Characterization

4.1 Threshold Structure with Three Cutoffs

The blockholder's optimal strategy has a natural ordering: she exits on bad news, holds passively on mildly negative news, engages quietly on moderate news, and goes public on good news. The next result makes this precise.

Proposition 1 (Monotone Cutoff Structure)**.** *Under the Standing Assumptions, there exists a Perfect Bayesian Equilibrium in which the blockholder follows cutoff rules with (weakly) ordered thresholds* $k_1 \le k_0 \le k_D$*:*

$$(q, a) = \begin{cases} (-1, 0) & s < k_1 \quad \textit{(Exit)}, \\ (0, 0) & k_1 \le s < k_0 \quad \textit{(Hold)}, \\ (0, 1) & k_0 \le s < k_D \quad \textit{(Quiet Voice)}, \\ (+1, 1) & s \ge k_D \quad \textit{(Public Voice)}. \end{cases}$$

The corresponding disclosure outcomes are: $D = 1$ *if and only if* $q = +1$ *(Public Voice), and* $D = 0$ *otherwise.*

Proof. See Appendix B.1.

The ordering reflects how the blockholder's incentives shift with her private signal. For very low signals, she holds unfavorable information about fundamentals and prefers to sell, monetizing her negative information before it reaches prices.

For somewhat better signals, fundamentals are not bad enough to warrant selling at a discount, but engagement is not yet worthwhile given its cost. The blockholder holds passively.

For intermediate signals, engagement improves expected value enough to justify the campaign cost $C(s)$, but stake building remains unattractive: buying additional shares costs more than the benefit of higher ownership. The blockholder engages quietly.

For the highest signals, buying additional shares allows the blockholder to internalize more of the engagement gains, and the discrete stake increase triggers public disclosure. This is the Public Voice region.

[8] Throughout, I maintain Assumptions (A1)–(A7) (the Standing Assumptions). These ensure interior action regions, nontrivial takeover outcomes, and well-behaved equilibrium existence. I verify each assumption numerically in the baseline calibration. Assumption (A6) (contraction) ensures uniqueness of the monotone-cutoff equilibrium; existence is established unconditionally via Brouwer's theorem (Proposition 4), while uniqueness is verified numerically following Edmans, Goldstein, et al. (2015).

When engagement costs are constant (i.e., $\chi = 0$), the Hold region may collapse, yielding a simpler two-cutoff structure with $k_0 = k_1$. The analysis accommodates this case by allowing weak inequalities among cutoffs.

4.2 Action Probabilities

Given cutoffs (k_1, k_0, k_D), define standardized cutoffs $\alpha_i \equiv (k_i - \mu)/\sigma_s$ for $i \in \{1, 0, D\}$. The unconditional action probabilities are:

$$\begin{aligned}
\omega_E &\equiv \mathbb{P}(s < k_1) = \Phi(\alpha_1), \\
\omega_H &\equiv \mathbb{P}(k_1 \leq s < k_0) = \Phi(\alpha_0) - \Phi(\alpha_1), \\
\omega_Q &\equiv \mathbb{P}(k_0 \leq s < k_D) = \Phi(\alpha_D) - \Phi(\alpha_0), \\
\omega_P &\equiv \mathbb{P}(s \geq k_D) = 1 - \Phi(\alpha_D),
\end{aligned}$$

corresponding to Exit, Hold, Quiet Voice, and Public Voice.

4.3 Conditional Means of Fundamentals

Let $\lambda_L(\alpha) \equiv \phi(\alpha)/\Phi(\alpha)$ and $\lambda_U(\alpha) \equiv \phi(\alpha)/(1 - \Phi(\alpha))$ denote inverse Mills ratios. The conditional means of v in each signal region are:

$$\begin{aligned}
\mu_E &\equiv \mathbb{E}[v \mid s < k_1] = \mu - \beta\sigma_s\lambda_L(\alpha_1), \\
\mu_H &\equiv \mathbb{E}[v \mid k_1 \leq s < k_0] = \mu + \beta\sigma_s\frac{\phi(\alpha_1) - \phi(\alpha_0)}{\Phi(\alpha_0) - \Phi(\alpha_1)}, \\
\mu_Q &\equiv \mathbb{E}[v \mid k_0 \leq s < k_D] = \mu + \beta\sigma_s\frac{\phi(\alpha_0) - \phi(\alpha_D)}{\Phi(\alpha_D) - \Phi(\alpha_0)}, \\
\mu_P &\equiv \mathbb{E}[v \mid s \geq k_D] = \mu + \beta\sigma_s\lambda_U(\alpha_D).
\end{aligned}$$

4.4 Bayesian Posteriors

Disclosure creates a sharp asymmetry in what the market can learn: when activism is disclosed, inference is trivial; when it is not, the market must extract engagement probabilities from noisy order flow. The next result characterizes the posterior beliefs that sustain equilibrium pricing.

Proposition 2 (Posterior Engagement Probabilities)**.** *For any information set (X, D) that is reached with positive probability in equilibrium, the posterior probability of engagement $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$ satisfies:*

(a) ***Disclosed states:*** *If $D = 1$, then $\pi(X, 1) = 1$ for all compatible $X \in \{0, 1, 2\}$.*

(b) ***Nondisclosed states:*** *If $D = 0$, let $p_0 \equiv 1 - \kappa$ and $p_1 \equiv \kappa/2$. Then:*

$$\begin{aligned}
\pi(1,0) &= \frac{\omega_Q}{\omega_H + \omega_Q}, \\
\pi(-1,0) &= \frac{\omega_Q\, p_1}{(\omega_H + \omega_Q)\, p_1 + \omega_E\, p_0}, \\
\pi(0,0) &= \frac{\omega_Q\, p_0}{(\omega_H + \omega_Q)\, p_0 + \omega_E\, p_1}, \\
\pi(-2,0) &= 0.
\end{aligned}$$

Proof. See Appendix B.2.

When $D = 1$, the inference problem collapses entirely: disclosure reveals that the blockholder chose Public Voice, which implies engagement with certainty. The market knows activism has occurred, though the ultimate success of the campaign remains uncertain until $t = 2$.

When $D = 0$, the market faces a genuine inference problem. It observes only aggregate order flow X and must form beliefs about whether the blockholder is engaged (Quiet Voice) or not (Exit or Hold). The key insight is that Hold and Quiet Voice both involve $q = 0$, so they generate identical order-flow distributions; the market cannot distinguish between them from order flow alone. Instead, the posterior split between these actions is pinned down by their prior probabilities (ω_H, ω_Q), which depend on the equilibrium cutoffs.

This is where liquidity enters the inference channel: higher κ changes the noise distribution and thereby affects how order flow X is interpreted. The posteriors $\pi(X, 0)$ inherit this liquidity dependence, transmitting it to prices and takeover premia.

4.5 Conditional Expectation of Fundamentals Given (X, D)

For pricing, we need $\mathbb{E}[v \mid X, D]$. Since v is independent of noise z, only the action regions compatible with (X, D) matter.

For $D = 1$: Only Public Voice is possible:

$$\mathbb{E}[v \mid X, D = 1] = \mu_P \quad \text{for } X \in \{0, 1, 2\}.$$

For $D = 0$: Mix over Exit, Hold, and Quiet Voice with Bayesian weights. Let $p_0 \equiv 1 - \kappa$ and $p_1 \equiv \kappa/2$. Then:

$$\begin{aligned}
\mathbb{E}[v \mid -2, 0] &= \mu_E, \\
\mathbb{E}[v \mid 1, 0] &= \frac{\omega_H \mu_H + \omega_Q \mu_Q}{\omega_H + \omega_Q}, \\
\mathbb{E}[v \mid -1, 0] &= \frac{(\omega_H \mu_H + \omega_Q \mu_Q)\, p_1 + \omega_E \mu_E\, p_0}{(\omega_H + \omega_Q)\, p_1 + \omega_E\, p_0}, \\
\mathbb{E}[v \mid 0, 0] &= \frac{(\omega_H \mu_H + \omega_Q \mu_Q)\, p_0 + \omega_E \mu_E\, p_1}{(\omega_H + \omega_Q)\, p_0 + \omega_E\, p_1}.
\end{aligned}$$

4.6 Equilibrium Prices

The competitive pricing condition (2) defines $P(X, D)$ implicitly, since the bid probability $p(X, D)$ depends on $P(X, D)$ through (1). For each information set (X, D) and posterior $\pi(X, D)$, the equilibrium price $P^*(X, D)$ satisfies

$$P = \delta\Big((1 - p(P, D)) \cdot \hat{V}(X, D) + p(P, D) \cdot (P + \bar{m}(X, D))\Big), \tag{3}$$

where $\hat{V}(X, D) \equiv \mathbb{E}[v \mid X, D] + \tilde{\Delta} \cdot \pi(X, D)$ is the expected standalone value (including engagement effects), and $p(P, D) = 1 - \Phi\big((\bar{m}(P, D) + K - \bar{S} + P)/\sigma_\xi\big)$ is the bid probability. Under Assumption (A5), this fixed-point equation admits a unique solution for each (X, D).

The fixed-point structure captures a key economic feedback: the price depends on takeover probability, which in turn depends on price. Higher prices deter bidders (making takeover less likely), which reduces the takeover component of value, which feeds back into the price. Assumption (A5) ensures that this feedback loop does not generate multiplicity.

Rearranging (3), any fixed point satisfies

$$P^*(X, D) = \frac{\delta\big((1 - p^*)\hat{V}(X, D) + p^* \, \bar{m}(X, D)\big)}{1 - \delta p^*}, \qquad p^* \equiv p(P^*(X, D), D).$$

When takeover is unlikely ($p^* \approx 0$), this reduces to $P^*(X, D) \approx \delta \, \hat{V}(X, D)$.

4.7 Price Decomposition and Activism Premium

Building on the pricing fixed point (3), I decompose the equilibrium price into interpretable components. The decomposition isolates the channels through which blockholder engagement feeds into asset prices.

Proposition 3 (Price Decomposition)**.** *The unique equilibrium price $P^*(X, D)$ satisfies*

$$P^*(X, D) = \delta\Big((1 - p^*) \cdot \mathbb{E}[v \mid X, D] + (1 - p^*) \cdot \tilde{\Delta} \cdot \pi(X, D) + p^* \cdot (P^*(X, D) + \bar{m}(X, D))\Big),$$

where $p^* \equiv p(P^*(X, D), D)$, $\pi(X, D) = \mathbb{P}(a = 1 \mid X, D)$, and $\bar{m}(X, D) = m_0 + (\tilde{m} - m_0)\pi(X, D)$.

The terms involving $\pi(X, D)$ constitute an "activism premium":

- ***Standalone channel:*** $(1 - p^*)\tilde{\Delta}\pi(X, D)$ *reflects expected value improvement from engagement when no takeover occurs.*
- ***Takeover channel:*** $p^* \cdot (\tilde{m} - m_0)\pi(X, D)$ *reflects the expected* incremental *takeover premium attributable to engagement (over and above the baseline m_0).*

Proof. See Appendix B.5.

Activism affects prices through two distinct channels. The *standalone channel* operates when no takeover occurs: an engaged blockholder improves firm value by $\tilde{\Delta}$ in expectation, and this improvement is capitalized into the share price.

The *takeover channel* operates when a bid arrives: the bidder must pay a higher premium to acquire a firm with an engaged blockholder, and this higher premium is also reflected in the pre-bid price.

The way these channels operate differs sharply between disclosed and nondisclosed states. In disclosed states ($D = 1$), engagement is known with certainty, so $\pi(X, 1) = 1$ and the activism premium is fully priced. Liquidity affects how often $D = 1$ occurs and can influence prices through feedback, but it does not affect the activism premium *within* disclosed states.

In nondisclosed states ($D = 0$), the posterior $\pi(X, 0)$ depends on κ through Bayesian inference, so the activism premium inherits liquidity sensitivity. This asymmetry between disclosed and nondisclosed states is central to understanding how disclosure attenuates the liquidity–premia relationship.

4.8 Bid Incidence and Premia

The bid probability $p(X, D)$ defined in (1) has two key monotonicity properties. First, higher prices deter bids: $\partial p(X, D)/\partial P(X, D) < 0$. This is intuitive: a higher market price raises the acquisition cost and makes the deal less attractive to the bidder.

Second, when the activism premium is positive ($\tilde{m} > m_0$), higher inferred engagement also reduces bid incidence: $\partial p(X, D)/\partial \pi(X, D) < 0$. The bidder anticipates paying a higher premium when facing an engaged blockholder, which makes entry less attractive.

These monotonicity properties have a direct implication for the role of disclosure. When $D = 1$, the bidder *knows* activism has occurred, so bid deterrence is maximal: the full activism premium $\tilde{m}$ enters the bidder's calculation. When $D = 0$, deterrence is probabilistic and depends on the inferred engagement probability $\pi(X, 0)$. Consequently, disclosed activism deters bids more strongly than inferred activism, all else equal. (Proof: see Appendix B.6.)

4.9 Equilibrium Cutoff Equations

The equilibrium cutoffs (k_1, k_0, k_D) are pinned down by indifference conditions at each boundary. Let U_E, $U_H(s)$, $U_Q(s)$, and $U_P(s)$ denote the expected payoffs from Exit, Hold, Quiet Voice, and Public Voice, computed by integrating over noise realizations and disclosure outcomes. (The exit payoff U_E does not depend on s because the sell order $q = -1$ fully liquidates the position, so the blockholder's terminal holding is zero.) The three cutoffs satisfy:

$$U_E = U_H(k_1), \tag{4}$$

$$U_H(k_0) = U_Q(k_0), \tag{5}$$

$$U_Q(k_D) = U_P(k_D). \tag{6}$$

At k_1, the blockholder is indifferent between exiting and holding passively. At k_0, she is indifferent between Hold and Quiet Voice. At k_D, she is indifferent between Quiet Voice and Public Voice. These indifference conditions, combined with single-crossing of payoffs in the signal s, pin down the equilibrium strategy. (Derivation: see Appendix B.9.)

4.10 Equilibrium Existence and Uniqueness

Under Assumptions (A1)–(A5), any monotone-cutoff Perfect Bayesian Equilibrium must satisfy Bayes-consistent beliefs and the pricing fixed point (3) together with the cutoff indifference conditions (4)–(6).

A pure strategy for the blockholder partitions the signal line into the four action regions by an ordered triple of cutoffs. Fix a finite bracket $[\underline{s}, \overline{s}]$ containing all indifference points (its existence is established in Lemma 2) and define the *cutoff polytope*

$$\Theta \;=\; \big\{\, \mathbf{k} = (k_1, k_0, k_D) \in [\underline{s}, \overline{s}]^3 \;:\; \underline{s} \le k_1 \le k_0 \le k_D \le \overline{s} \,\big\}. \tag{7}$$

Θ is the intersection of a cube with the half-spaces $\{k_1 \le k_0\}$ and $\{k_0 \le k_D\}$, hence a compact, convex, nonempty polytope. The boundary faces $\{k_1 = k_0\}$ and $\{k_0 = k_D\}$ are *collapse faces* on which Hold (respectively Quiet Voice) carries zero probability; the baseline equilibrium lives on the face $k_1 = k_0$ (Hold collapses).

The best-response map $T : \Theta \to \mathbb{R}^3$ sends a conjecture $\mathbf{k}$ to the indifference cutoffs it implies. Given $\mathbf{k}$, the market maker's pricing function $P(\cdot,\cdot)$ and the bidder's takeover probability $p(\cdot)$ are determined, and the blockholder's payoff to action (q, a) at signal s is, from (42),

$$U(q, a; s) \;=\; A(q, a) \;+\; B(q, a)\, \hat{v}(s) \;-\; a\, C(s), \tag{8}$$

where, writing $\beta = \sigma_v^2/\sigma_s^2 \in (0, 1]$ for the posterior-mean slope and absorbing the conjecture into the intercepts $A(\cdot)$,

$$B(E) \;=\; 0, \qquad B(H) \;=\; 1, \qquad B(Q) \;=\; 1, \qquad B(P) \;=\; 1 + \eta, \quad \eta > 0, \tag{9}$$

the extra slope η on Public Voice arising from the extra share held into the terminal date and from the takeover term, which is nondecreasing in $\hat{v}$. Because the per-share slope in (42) is $\delta\, h\, \mathbb{E}_z[1 - p(X, D)]$ (slopes here are normalized by the common Hold/Quiet factor), the second share contributes $2\, \mathbb{E}_z[1 - p(X, 1)]$ against $\mathbb{E}_z[1 - p(X, 0)]$, so $\eta > 0$ is *conditional*: it holds whenever the disclosed bid probability satisfies $\bar{p}_1 \equiv \mathbb{E}_z\, p(X, 1) \le \frac{1}{2}$ — Lemma 4's condition, which holds with a wide margin at the baseline (disclosure deters bids: $\bar{p}_1 \approx 0.013$). The component T_1 (T_2, T_3) is the Exit/Hold (Hold/Quiet, Quiet/Public) indifference signal.

Lemma 2 (Boundedness of best responses). *Under Assumptions* (A1), (A2), (A4)*, there is a finite bracket* $[\underline{s}, \overline{s}]$*, independent of the conjecture* $\mathbf{k} \in \Theta$*, such that every indifference signal of* T *lies in* $[\underline{s}, \overline{s}]$.

Proof. Each indifference condition equates two payoffs of the form (8). The difference between two such payoffs is

$$D_{ab}(s) = \big[A(a) - A(b)\big] + \big[B(a) - B(b)\big]\hat{v}(s) - \big[\mathbf{1}\{a \in \{Q, P\}\} - \mathbf{1}\{b \in \{Q, P\}\}\big]C(s). \tag{10}$$

The intercepts $A(\cdot)$ are bounded uniformly over $\mathbf{k} \in \Theta$: prices lie in $[\,0, \delta\, \overline{Y}\,]$ with $\overline{Y}$ the finite maximal terminal value, and the takeover probability $p \in [0, 1]$. The slope gap $B(a) - B(b)$ is one of $\{\pm 1, \pm\eta, \pm(1 + \eta)\}$ and is nonzero for every adjacent pair $(E, H), (H, Q), (Q, P)$ except (H, Q), treated separately. For a nonzero slope, $\hat{v}(s) = \mu + \beta(s - \mu) \to \pm\infty$ as $s \to \pm\infty$, while $C(s) = C_0 \exp(-\chi(s - \mu)/\sigma_s)$ is bounded on the half-line where $\hat{v}$ dominates: as $s \to +\infty$ the linear term forces $D_{ab} \to +\infty$ for the higher-slope action; in the (E, H) comparison the cost term is absent, and in the (Q, P) comparison the common engagement cost cancels, so the slope gap $\eta > 0$ on $\hat{v}$ governs the crossing, which is finite. The (H, Q) comparison has $B(H) = B(Q)$ and reduces to

$C(s) = A(Q) - A(H)$; since C is continuous, strictly positive, strictly decreasing (Assumption (A2), $\chi > 0$) with range $(0, \infty)$, this has a unique finite solution whenever $A(Q) - A(H) > 0$, and otherwise Quiet Voice is never optimal and the cutoff sits at the bracket endpoint. Taking $\underline{s}$ below and $\overline{s}$ above all (finitely many, uniformly bounded) crossings gives the claim. $\square$

Lemma 3 (Self-map and ordering preservation)**.** *Under Assumptions* (A1)*,* (A2)*,* (A4)*,* (A5a) *(so that the cell-level price is single-valued) and the slope ordering* $\eta > 0$ *of Lemma 4, the map* T *is continuous and maps* Θ *into itself,* $T(\Theta) \subseteq \Theta$*, including the collapse faces.*

Proof. Continuity. The action probabilities $\omega_\bullet$ are continuous in $\mathbf{k}$ (compositions of the smooth signal CDF); the posteriors, prices and takeover probability are continuous in $(\mathbf{k}, \omega)$ on each information cell; and each D_{ab} in (10) crosses zero transversally (strictly monotone in s at the crossing, by the slope/cost argument of Lemma 2), so the implicit-function theorem gives a solution continuous in $\mathbf{k}$. On a collapse face the relevant D_{ab} is degenerate; there we define T by the limiting boundary cutoff, the continuous extension. The operative 0/0 object $\bar{\pi} = \omega_Q/(\omega_H + \omega_Q)$ extends continuously through the collapse face because numerator and denominator are Lipschitz and the denominator is bounded below away from the double-collapse corner, where l'Hôpital's rule gives a direction-independent limit.

Into the bracket. By Lemma 2 every component of $T(\mathbf{k})$ lies in $[\underline{s}, \overline{s}]$.

Ordering. We must show $T_1(\mathbf{k}) \leq T_2(\mathbf{k}) \leq T_3(\mathbf{k})$. This is the content of the global monotone–best-response lemma (Lemma 5): the upper envelope $\max_{(q,a)} U(q, a; s)$ is a piecewise combination of the affine-in-$\hat{v}$ functions (8) with totally ordered slopes $B(E) < B(H) = B(Q) < B(P)$, hence the optimal action is nondecreasing in s and the indifference signals are automatically ordered. On a collapse face the ordering holds with equality, so T maps each face into itself or its interior. Thus $T(\Theta) \subseteq \Theta$. $\square$

Proposition 4 (Existence of Monotone Equilibrium)**.** *Under Assumptions* (A1)*,* (A2)*,* (A4)*,* (A5a) *and the slope ordering* $\eta > 0$ *of Lemma 4, an equilibrium in (weakly ordered) cutoff strategies exists.*

Proof. Θ in (7) is nonempty, compact and convex, and by Lemma 3 $T : \Theta \to \Theta$ is continuous. Brouwer's fixed-point theorem yields $\mathbf{k}^\star \in \Theta$ with $T(\mathbf{k}^\star) = \mathbf{k}^\star$, an equilibrium. The fixed point may lie on a collapse face, in which case the corresponding action carries zero probability; this is precisely the baseline ($k_1 = k_0$, Hold collapses). $\square$

Lemma 4 (Public Voice dominates Quiet Voice in slope)**.** *Public Voice is strictly more sensitive to the signal than Quiet Voice,* $B(P) > B(Q)$*, whenever the disclosed-flow takeover probability satisfies* $p(X, 1) \leq \frac{1}{2}$ *for the realized disclosed flow* X*; a sufficient primitive condition is*

$$\bar{S} - K - P(X, 1) \;\leq\; \tilde{m}. \tag{11}$$

Proof. The payoff slopes in $\hat{v}$ differ between Public and Quiet Voice through two channels. (1) *Holdings:* by (42) the slope of action (q, a) in $\hat{v}$ is $\delta\,(1 + q)\,\mathbb{E}_z[1 - p(X, D)]$; Public Voice holds two shares into the terminal date against Quiet Voice's one, contributing $\delta\big[2\,\mathbb{E}_z(1 - p(X, 1)) - \mathbb{E}_z(1 - p(X, 0))\big]$, which is strictly positive whenever $\bar{p}_1 \equiv \mathbb{E}_z\, p(X, 1) \leq \frac{1}{2}$ (then $2(1 - \bar{p}_1) \geq 1 \geq 1 - \bar{p}_Q$, strictly unless both bind). (2) *Takeover:* both carry the engagement term $(\tilde{m} - m_0)\,p$; under disclosure ($D = 1$ for Public Voice) the takeover probability is $p(X, 1) = 1 - \Phi(\mathcal{T})$ with

$\mathcal{T} = (\bar{m}(X,1) + K - \bar{S} + P(X,1))/\sigma_\xi$, and $\bar{m}(X,1)$ rises with the disclosed posterior, which rises with s. Hence $\partial p/\partial\hat{v} = \phi(\mathcal{T})(\partial\bar{m}/\partial\hat{v})/\sigma_\xi \geq 0$, so the takeover channel adds a nonnegative amount to the Public slope. Therefore

$$\eta = \underbrace{\delta\big[2(1-\bar{p}_1) - (1-\bar{p}_Q)\big]}_{>0 \text{ iff } \bar{p}_Q > 2\bar{p}_1 - 1} + \underbrace{(\tilde{m} - m_0)\,\frac{\phi(\mathcal{T})}{\sigma_\xi}\,\frac{\partial\bar{m}}{\partial\hat{v}}}_{\geq 0} \;>\; 0 \quad \text{when } \bar{p}_1 \leq \tfrac{1}{2},$$

giving $B(P) > B(Q)$ under the displayed condition. The condition $p(X,1) \leq \frac{1}{2} \iff \mathcal{T} \geq 0 \iff$ (11) (using $\bar{m}(X,1) = \tilde{m}$ on the disclosed branch) additionally places the takeover channel on the rising part of Φ, making the slope gap robust to perturbations of the disclosed price.[9] $\square$

Lemma 5 (Global monotone best response)**.** *Fix any measurable bounded pricing function $P(\cdot,\cdot)$ and any takeover probability $p(\cdot) \in [0,1]$, hence any intercepts $A(q,a)$ in* (8)*. Under Assumptions* (A1), (A2) *with the slope ordering* (9)*, $B(E) < B(H) = B(Q) < B(P)$, the action correspondence $a^\star(s) \in \arg\max_{(q,a)} U(q,a;s)$ is nondecreasing in s in the order $E \prec H \prec Q \prec P$, and the upper envelope $V(s) = \max_{(q,a)} U(q,a;s)$ is convex in $\hat{v}(s)$. Consequently the best response is a cutoff partition with ordered thresholds $k_1 \leq k_0 \leq k_D$.*

Proof. Each $U(q,a;\cdot)$ in (8) is affine in $\hat{v}$ once the engagement cost is folded in: for non-engagement actions (E,H) it is affine in $\hat{v}$ with slope $B \in \{0,1\}$; for the two engagement actions (Q,P) the comparison has the cost cancel, leaving an affine-in-$\hat{v}$ difference with slope $B(P) - B(Q) = \eta > 0$. The upper envelope of a finite family of affine functions of $\hat{v}$ is convex and piecewise affine, and its argmax is the action with the largest slope among those still attainable, which is nondecreasing in $\hat{v}$, hence (as $\hat{v}$ is strictly increasing in s, Assumption (A1)) nondecreasing in s. With totally ordered slopes $B(E) < B(H) = B(Q) < B(P)$ the envelope is traversed in the order $E, \{H \text{ or } Q\}, P$. The tie $B(H) = B(Q)$ is broken by the intercepts: Q adds the engagement net benefit $A(Q) - A(H)$ less $C(s)$, decreasing in s, so among the two equal-slope actions H is chosen for low s (high cost) and Q for high s (low cost)—again a nondecreasing switch. Thus the optimal action is nondecreasing in s and the thresholds are ordered. No conjecture about the *form* of the partition is used; only the belief-determined intercepts and the primitive slope ordering enter, on and off the equilibrium path, so the argument is not self-confirming. $\square$

Remark 1 (Why the tie $B(H) = B(Q)$ is benign)**.** *Hold and Quiet Voice have equal slopes and are separated purely by the decreasing cost schedule C (Assumption* (A2)*), so the $H \to Q$ switch is monotone; if the engagement net benefit is too small the Q region is empty and the equilibrium sits on the collapse face $k_0 = k_D$. This is the mechanism behind the baseline Hold collapse and is fully consistent with Lemma 5.*

Dropping (A7): conditioning the bidder on the price

Lemma 6 (Price-conditioned bidder; (A7) redundant)**.** *Suppose the bidder conditions only on the equilibrium price P (and not separately on D or the stake). Then the bidder's takeover decision*

[9] (11) is the transparent primitive bound and holds comfortably at the baseline, where the disclosed takeover probability is $p^\star = 0.0131$ ($\mathcal{T} = +2.22$): disclosure *deters* bids ($p(X,1) < p(X,0)$ at every reached cell; cf. the bid-incidence discussion in Section 5), so $2(1-\bar{p}_1) \approx 1.97$ and the holdings channel secures $\eta > 0$ with a wide margin.

and probability coincide with those under Assumption (A7)*. Hence* (A7) *can be dropped.*

Proof. The competitive market maker sets $P(X,D) = \delta\,\mathbb{E}[Y \mid X,D]$. Because the map $(X,D) \mapsto P$ is injective on the realized support at any nondegenerate calibration (distinct order-flow/disclosure cells command distinct prices, as $\mathbb{E}[Y \mid \cdot]$ separates cells whenever their posteriors differ), the price carries all the information in (X,D) that is payoff-relevant to the bidder. The bidder bids iff $\Pi_B = \bar{S} - P + \xi - \bar{m} - K \geq 0$ and, integrating over $\xi \sim \mathcal{N}(0,\sigma_\xi^2)$,

$$p(P) = 1 - \Phi\left(\frac{\bar{m}(P) + K - \bar{S} + P}{\sigma_\xi}\right), \tag{12}$$

with $\bar{m}(P) = \mathbb{E}[m^R \mid P]$. By the law of iterated expectations $\mathbb{E}[m^R \mid P] = \mathbb{E}[\mathbb{E}[m^R \mid X,D] \mid P] = \mathbb{E}[m^R \mid X,D]$ on the price-injective support, so $\bar{m}(P) = \bar{m}(X,D)$ and (12) equals the expression under (A7). On the measure-zero set where two cells share a price the bidder uses the price-conditional posterior, the correct Bayesian object, which weakly dominates any cell-specific rule. Thus the bidder problem and p are unchanged, and (A7) is redundant. □

Splitting (A5): inner-price and cross-cutoff contraction

Assumption 1 ((A5a): inner-price contraction)**.** *On each information cell the price map $\Psi(P) = \delta\,\mathbb{E}[Y \mid X,D;P]$ is a contraction in P:*

$$\left|\Psi'(P)\right| = \delta\Big[\underbrace{p^\star}_{(i)\ direct} + \underbrace{\left|P + \bar{m} - \hat{V}\right| \tfrac{\phi(0)}{\sigma_\xi}}_{(ii)\ feedback}\Big] < 1. \tag{13}$$

Assumption 2 ((A5b): cross-cutoff stability)**.** *The induced prices are Lipschitz in the cutoffs with a modulus small enough that the composed outer map (cutoffs → prices → cutoffs) does not amplify perturbations beyond the bound used in Assumption* (A6)*.*

Remark 2 (Baseline margins for (A5a), stated honestly)**.** *The sufficient rule* (13) *must hold cell by cell, and the binding cell at the baseline is the* Exit *cell* $(X,D) = (-2,0)$*, where the bid probability is largest:* $p^\star = 0.806$*, so the direct term is* $\delta\,p^\star \approx 0.766$*, and the worst-case feedback term (ii) adds* $\delta\,|P + \bar{m} - \hat{V}|\,\phi(0)/\sigma_\xi \approx 0.27$*, for a sum* $\approx 1.04 > 1$*. So the conservative sufficient rule does* not *certify* $|\Psi'| < 1$ *at every cell.* ***We therefore state Assumption* (A5a) *as a maintained assumption****, not a derived inequality. It is nonetheless mild: the rule bounds* $\phi(\mathcal{T})$ *by* $\phi(0)$ *and treats both channels as additive at their worst case; on every* other *reached cell the sum is well below one (on the disclosed branch, where* $p^\star = 0.013$*, it is* ≈ 0.18*); and the inner fixed point is found numerically without difficulty at every* κ *(residuals* $\sim 10^{-10}$ *at fully converged points; Remark 3). The reader should read* (13) *as the transparent primitive condition, binding only on the high-p Exit cell under worst-case bounds.*

Remark 3 (Uniqueness as a numerical regularity, not a proposition)**.** *We do not claim global uniqueness as a theorem. Assumption* (A6) *(contraction of T) would deliver it, but we cannot derive* $\|DT\|_\infty < 1$ *from primitives without further structure on the cross-cutoff Jacobian. At the baseline calibration the equilibrium located by the solver is numerically unique: a multistart search with* 30 *random ordered seeds at* $\kappa \in \{0.2, 0.5, 0.8\}$ *converges in all* 30/30 *cases to a single fixed point* *per κ (residuals $\sim 4 \times 10^{-10}$), and at the hump peak $\kappa^\dagger = 0.59$ the cold-start and both warm-started solves agree to $\Delta^{\min}$-spread 6.9×10^{-9}. We label this a* numerical regularity *at the calibration, not a uniqueness theorem; the comparative statics below are shown to hold across all fixed points the multistart locates, following the methodological precedent of Edmans, Goldstein, et al. (2015).*

4.11 Nonmonotonic Minority Takeover Gains

I now turn to the paper's central object of interest: expected minority takeover gains. These gains capture what a passive shareholder can expect to earn from the takeover channel, and the model predicts that they respond nonmonotonically to market liquidity. Define

$$\Delta^{\min}(\kappa) \equiv \mathbb{E}\big[m^R(a) \cdot \mathbf{1}\{\text{bid}\}\big], \tag{14}$$

as the expected takeover premium (per share) earned by a representative minority share. The expectation is taken over (v, s, z, ξ) induced by equilibrium strategies. This quantity captures the unconditional expected takeover premium from the takeover channel, averaged over all states of the world (including states in which no bid occurs).

Expected minority takeover gains decompose naturally into two components:

$$\Delta^{\min}(\kappa) = \underbrace{m_0 \cdot \mathbb{P}(\text{bid})}_{\text{baseline premium}} + \underbrace{(\tilde{m} - m_0) \cdot \mathbb{E}\big[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}\big]}_{\text{activism-driven premium}}. \tag{15}$$

The first term is the baseline premium weighted by overall bid probability. The second term, which I denote $\Delta^{\text{act}}(\kappa)$, captures the additional premium attributable to blockholder engagement. To avoid ambiguity, Δ (without superscript) denotes the engagement improvement parameter, while $\Delta^{\min}(\kappa)$ and $\Delta^{\text{act}}(\kappa)$ are equilibrium objects defined above.

This decomposition separates the effect of liquidity on *whether* a bid occurs from its effect on the *size* of the premium conditional on bidding. (Derivation: see Appendix B.10.)

Two opposing channels drive the liquidity dependence. Higher liquidity can raise bid incidence by reducing adverse-selection costs and shifting prices, but it also changes equilibrium engagement incentives and the inference-driven component of premia. A full closed-form characterization of $\Delta^{\min}(\kappa)$ is generally unavailable because κ affects equilibrium cutoffs and prices jointly. The following results establish nonmonotonicity from primitives, without relying on assumed boundary conditions.

Because the equilibrium cutoffs themselves depend on κ, the total derivative of $\Delta^{\min}$ splits into two economically distinct channels:

$$\frac{\mathrm{d}\,\Delta^{\min}}{\mathrm{d}\kappa} = \underbrace{\left.\frac{\partial \Delta^{\min}}{\partial \kappa}\right|_{k \text{ fixed}}}_{\text{(A) posterior / pricing channel}} + \underbrace{\sum_{i \in \{1,0,D\}} \frac{\partial \Delta^{\min}}{\partial k_i} \frac{\mathrm{d}k_i}{\mathrm{d}\kappa}}_{\text{(B) GE cutoff-shift channel}}. \tag{16}$$

The substantive claim—that $\Delta^{\min}(\kappa)$ is single-peaked (a "hump") in the interior of $[0, 1]$—requires controlling *both* channels. I sign channel (A) under an explicit primitive condition and pin down the endpoints exactly; channel (B) I cannot sign in closed form and supply it as a numerically verified regularity. The dividing line is sharp: Lemma 7 and Lemma 8 are *proved*; the control of channel (B), Hypothesis 1, is *not* proved but stated as a primitive regularity condition and corroborated numerically (Section 5). Proposition 5 is therefore a conditional statement.

Under the stake-triggered disclosure rule, $D = 1$ if and only if the blockholder buys ($q = +1$); a realized $D = 0$ pools the Hold and Quiet-Voice regions. In the liquidity limits the realized $D = 0$ posterior support collapses to the two-point set

$$\{0,\ \bar{\pi}\}, \qquad \bar{\pi} \ = \ \frac{\omega_Q}{\omega_H + \omega_Q}, \tag{17}$$

where $\bar{\pi}$ is the conditional probability of an active (Quiet-Voice) type given $D = 0$. (At the baseline, Hold collapses, so $\omega_H \approx 0$ and $\bar{\pi} = 1$ at interior κ, while $\bar{\pi} \approx 0.96$ at the extreme $\kappa \in \{0, 1\}$ where Hold reappears infinitesimally.) Writing π for the activist posterior weight, let

$$h(\pi) \ = \ \pi\, p(\pi), \qquad p(\pi) \ = \ 1 - \Phi\!\left(\frac{\bar{m}(\pi) + K - \bar{S} + P}{\sigma_\xi}\right), \tag{18}$$

be the bidder's entry probability at posterior π, with $\bar{m}(\pi) = m_0 + \pi\,(\tilde{m} - m_0)$. The contribution of the $D = 0$ event to $\Delta^{\min}$ is governed by the expectation of h over the realized law. The operative integrand is $h(\pi) = \pi\, p(\pi)$—the activist probability times the entry probability—because value is at stake only in the engaged state and the tower identity $\mathbb{E}[\pi\, \mathbf{1}\{D = 0\}] = \omega_Q$ controls precisely the π-weight. Because the realized law is the two-point measure (17) rather than a smooth density, the curvature that matters is the *secant* curvature of h across the chord $[0, \bar{\pi}]$, summarized by the second difference

$$\mathcal{C}(\bar{\pi}) \ := \ h(0) \ - \ 2\, h\!\left(\tfrac{\bar{\pi}}{2}\right) \ + \ h(\bar{\pi}), \tag{19}$$

with h secant-concave on $[0, \bar{\pi}]$ iff $\mathcal{C}(\bar{\pi}) < 0$.

Lemma 7 (Endpoint symmetry of minority gains)**.** *As $\kappa \to 0^+$ and as $\kappa \to 1^-$ the realized $D = 0$ posterior law converges to the* same *two-point measure on $\{0, \bar{\pi}\}$, and consequently*

$$\lim_{\kappa \to 0^+} \Delta^{\min}(\kappa) \ = \ \lim_{\kappa \to 1^-} \Delta^{\min}(\kappa). \tag{20}$$

Proof. The price function is $P(X, D) = \delta\, \mathbb{E}[Y \mid X, D]$ and order flow is $X = q + z$ with $\mathbb{P}(z = 0) = 1 - \kappa$, $\mathbb{P}(z = \pm 1) = \kappa/2$. Fix the cutoffs. The four non-disclosed order-flow cells carry posteriors $\{0, \pi_1, \pi_2, \bar{\pi}\}$ (Appendix B.13, Table 3); at $\kappa = 0$ ($p_1 = 0$) only the atoms 0 and $\bar{\pi}$ carry mass, and at $\kappa = 1$ ($p_0 = 0$) the four cells regroup—by the measure-preserving relabeling $X \mapsto -X$ that the symmetry $\mathbb{P}(z = +1) = \mathbb{P}(z = -1)$ induces—onto the *same* two atoms $0, \bar{\pi}$ with the same masses (ω_E, $\omega_H + \omega_Q$). The disclosure decision $D = \mathbf{1}\{q = +1\}$ is unaffected by z, so the disclosed branch is κ-invariant. Hence the integrand h in (18) is integrated against the same measure at both ends, giving (20). The tower identity $\mathbb{E}[\pi\, \mathbf{1}\{D = 0\}] = \omega_Q$ holds exactly at every κ, confirming the two-point structure. The two limits coincide because the equilibrium indifference system is identical at the two extremes (the limiting price/bid-indexed payoffs and the cost $C(\cdot)$ agree), so all equilibrium objects share the common limit. □

Remark 4 (Numerical corroboration and the refutation of voice-collapse)**.** *Solving directly at the extremes (multistart solver, tightened 10^{-5} gate) gives $\Delta^{\min}(10^{-6}) = 0.07020822$ and $\Delta^{\min}(1 - 10^{-6}) = 0.07020753$, a gap of 6.86×10^{-7}, with the two equilibrium cutoff triples coinciding to four* *decimals. On fully converged near-endpoints the gap is* 3.56×10^{-5}*. This* confirms *the corrected endpoint symmetry and* refutes *the earlier "voice-collapse" claim: voice does not collapse as* $\kappa \to 1$*, and the* $\kappa \to 1$ *posterior is the two-point law* $\{0, \bar{\pi}\}$*, not the prior. Because the exact endpoints are degenerate grid-edge solves (large residuals),* (20) *is read as a* limit *statement.*

Lemma 8 (Sign of the posterior/pricing channel under (C*))**.** *Hold the cutoffs fixed. The interior of channel (A) in* (16) *inherits the sign of the chord second difference* (19)*. In particular, if the primitive condition*

$$\textit{(C*)} \qquad \mathcal{C}(\bar{\pi}) \;=\; h(0) - 2\,h\!\left(\tfrac{\bar{\pi}}{2}\right) + h(\bar{\pi}) \;<\; 0 \quad \textit{along the equilibrium path} \tag{21}$$

holds, then channel (A) is strictly hump-shaped in κ *at fixed cutoffs: it rises from the* $\kappa \to 0$ *endpoint, attains an interior maximum, and falls back to the (equal)* $\kappa \to 1$ *endpoint. The pointwise sufficient form is full-interval concavity of* h *on* $[0, \bar{\pi}]$*, equivalently* $\max_{\pi \in [0,\bar{\pi}]} \pi\, T(\pi)\, T'(\pi) \le 2$*, where* $T(\pi) = (\bar{m}(\pi) + K - \bar{S} + P(\pi))/\sigma_\xi$ *(Appendix B.13, eq.* (47)*).*

Proof. At fixed cutoffs, the only κ-dependence in channel (A) enters through the weight the realized $D = 0$ law (17) places on its interior atoms relative to the surviving end atoms $\{0, \bar{\pi}\}$; as κ sweeps $[0, 1]$, that weight starts and ends on the chord ends (Lemma 7) and is interior in between. Because the conditional first moment is pinned by the tower identity ($\mathbb{E}[\pi\, \mathbf{1}\{D = 0\}] = \omega_Q$) for every κ, the affine part of h contributes a κ-free constant, and the entire interior motion of $\mathbb{E}_\kappa[h]$ equals the probability-weighted gap between h and its chord ℓ on $[0, \bar{\pi}]$ (an exact finite-support Jensen identity, no remainder). If h lies above its chord ($\mathcal{C}(\bar{\pi}) < 0$, secant concavity) the gap is nonnegative and strictly positive when interior atoms carry mass, so channel (A) strictly humps above the equal endpoints; strict inequality in (21) yields a strict interior maximum. The converse (secant convexity) yields an interior trough. □

I now isolate exactly what I cannot prove. Channel (B) in (16),

$$B(\kappa) \;:=\; \sum_{i \in \{1,0,D\}} \frac{\partial \Delta^{\min}}{\partial k_i} \frac{\mathrm{d}k_i}{\mathrm{d}\kappa}, \tag{22}$$

is the general-equilibrium feedback of κ on $\Delta^{\min}$ operating *through* the movement of the equilibrium cutoffs, and it resists a clean sign for two reasons. First, the *envelope theorem does not annihilate* $B(\kappa)$: $\Delta^{\min}$ is *minority* welfare, not the blockholder's objective, so the equilibrium first-order conditions that define $k(\kappa)$ zero the derivative of the *blockholder's* value with respect to k but say nothing about $\partial \Delta^{\min}/\partial k_i$, which is generically nonzero. Second, the factors $\mathrm{d}k_i/\mathrm{d}\kappa$ are themselves unsigned: the cutoff path is non-monotone (the equilibrium $k_1 = k_0$ runs $0.657 \to 0.822 \to 0.743$ at $\kappa = 0.2, 0.5, 0.8$), and the *conditional* mean of the $D = 0$ law drifts in full equilibrium ($\mathbb{E}[\pi \mid D = 0]$ traces $0.68 \to 0.58 \to 0.63$ at $\kappa = 0.2, 0.5, 0.8$, and back up toward the endpoints, because $\mathbb{P}(D = 0)$ drifts) even though the *unnormalized* moment $\mathbb{E}[\pi\, \mathbf{1}\{D = 0\}] = \omega_Q$ is invariant. The fixed-cutoff invariance used in Lemma 8 therefore does not transfer mechanically to equilibrium. I make the following explicit regularity hypothesis, in the style of Edmans, Goldstein, et al. (2015), who likewise close a feedback result under a primitive condition with numerical confirmation of the residual.

Hypothesis 1 (Cutoff-shift dominance; numerically verified, not proved)**.** *Along the equilibrium path the GE cutoff-shift channel $B(\kappa)$ in* (22) *does not overturn the single-peakedness established for channel (A) under* (C*)*; equivalently,* $\mathrm{d}\Delta^{\min}/\mathrm{d}\kappa$ *changes sign exactly once on* $(0,1)$*. Under Assumption* (A6) *the cutoff map $\kappa \mapsto k(\kappa)$ is C^1 (Appendix B.14), so $B(\kappa)$ is well defined and bounded by an explicit, computable equilibrium quantity; on any sub-interval where the residual integral falls below the channel-(A) amplitude, the equilibrium hump is a* theorem *rather than a regularity (Appendix B.14).*

Proposition 5 (Conditional liquidity hump in minority gains)**.** *Suppose the primitive secant-concavity condition* (C*) *of* (21) *holds along the equilibrium path and the cutoff-shift dominance Hypothesis 1 holds. Then the minority's expected disciplining gain $\Delta^{\min}(\kappa)$ is single-peaked on* $(0,1)$*: it is symmetric at the endpoints in the limit sense,* $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$ *(Lemma 7), strictly increasing then strictly decreasing, with a unique interior maximizer* $\kappa^\dagger \in (0,1)$*. Conversely, if* (C*) *fails—h secant-convex on the chord* $[0, \bar{\pi}]$*—the same construction yields an interior* trough*. The headline is therefore genuinely conditional: the sign of $\mathcal{C}(\bar{\pi})$ in* (19) *selects hump versus trough.*

Proof. Combine Lemma 7 (equal endpoints), Lemma 8 (channel (A) single-peaked under (C*)), and Hypothesis 1 (channel (B) does not overturn this). The converse follows by reversing the inequality in (C*) in Lemma 8. See Appendix B.13 for the closed-form posterior-variance backbone and Appendix B.14 for the C^1 cutoff map and the computable residual bound. □

Remark (what is proved versus what is conditional). Endpoint symmetry (Lemma 7) and the channel-(A) sign under (C*) (Lemma 8) are proved. The condition (C*) is an explicit primitive that selects hump versus trough and is verified at the baseline; Hypothesis 1 (channel (B) does not overturn the shape) is corroborated numerically in Section 5, not asserted as a theorem. The baseline calibration confirms a well-behaved single-peaked hump (Figure H.2).

Appendix F sharpens this division in both directions. In one direction, Theorem 3 bounds the GE channel by an *inversion-free* quantity $\bar{B}(\kappa)$ computable from a single equilibrium solve (the contraction modulus of Assumption (A6) replaces the unknown path derivative), and certifies single-peakedness, up to ε-localization of the peak, as a *theorem* on the interior region where $\bar{B}$ is dominated: strictly monotone on either side of a small ball around the channel-(A) peak, global maximum interior to the ball — at the baseline, $\kappa \in [0.35, 0.825]$ under the inversion-free bound and $[0.30, 0.85]$ under the exact implicit-function bound, comfortably containing $\kappa^\dagger$ (Corollary 3, Figure H.14). In the other direction, Proposition 17 certifies that Hypothesis 1 *fails* at the high-σ_ξ robustness cells: channel (A) remains single-peaked there, yet the full-equilibrium profile is a trough, so the GE channel genuinely overturns the shape and no calibration-free version of the hypothesis can be true. The hump-versus-trough boundary is itself economic — channel (A)'s amplitude scales with the sensitivity of bidder entry to expected resistance, so GE feedback dominates exactly where entry is premia-insensitive (Remark 20).

The economic logic is as follows. At very low liquidity ($\kappa \to 0$), order flow is nearly perfectly revealing. The market infers the blockholder's action, fully pricing the expected engagement into the share price and deterring marginal takeover bids. Although the blockholder still benefits fundamentally from engagement, this bid deterrence suppresses the overall minority gains. At very high liquidity ($\kappa \to 1$), noise trading dominates, posteriors converge to the unconditional prior across all states, and prices flatten. The blockholder can no longer extract adverse-selection rents from informed trading, so the expected return to engagement falls below its private cost for all signal realizations. In this upper limit, $\Delta^{\text{act}}(\kappa) \to 0$. Between these extremes, an interior κ optimally balances the cover provided by noise traders with the survival of engagement incentives.

Between these extremes, an intermediate level of liquidity maximizes minority gains. The blockholder retains enough information advantage to profit from engagement, noise traders provide enough cover for informed trading, and the market's Bayesian inference sustains a meaningful activism premium. The turning point $\kappa^\dagger$ marks the liquidity level at which the marginal benefit of additional noise trading (higher bid incidence, better cover for informed orders) is exactly offset by the marginal cost (weaker inference, diluted engagement incentives).

Section 5 complements this theory with numerical analysis. In the baseline calibration, $\Delta^{\min}(\kappa)$ is hump-shaped (Figure H.2) and the activism component varies smoothly with κ (Figure H.3).

Disclosure Attenuation. The activism-driven component $\Delta^{\text{act}}(\kappa)$ can be decomposed by disclosure status. In disclosed states ($D = 1$), engagement is known with certainty, so the activism premium $\tilde{m} - m_0$ is constant and there is no inference channel. In nondisclosed states ($D = 0$), engagement must be inferred from order flow, making the premium κ-sensitive through the Bayesian formulas in Proposition 2.

This implies that the liquidity sensitivity of $\Delta^{\text{act}}(\kappa)$ comes entirely from the $D = 0$ component. Shifting probability mass toward disclosed activism (larger ω_P) attenuates this inference-driven sensitivity, as more takeovers occur in states where engagement is observed rather than inferred. The following proposition formalizes this observation.

This result has direct policy implications: shifting more activism from hidden to observable (for example, by lowering disclosure thresholds) attenuates the sensitivity of takeover premia to market liquidity.

Proposition 6 (Disclosure Attenuation (Partial Equilibrium))**.** *Hold the blockholder's strategy constant: fix the cutoffs (k_1, k_0, k_D) and hence the action probabilities, and vary κ only through its effect on inference and pricing (a partial equilibrium exercise). Decompose the activism-driven premium as*

$$\Delta^{\text{act}}(\kappa) = \underbrace{(\tilde{m} - m_0)\omega_P \cdot \bar{p}_1}_{\textit{disclosed component}} + \underbrace{(\tilde{m} - m_0)\mathbb{E}\big[\pi(X,0) \cdot p(X,0) \mid D = 0\big]\mathbb{P}(D = 0)}_{\textit{inferred component}},$$

where $\bar{p}_1 \equiv p(P^*(X,1),1)$ is the (constant) bid probability on the disclosed branch. Only the inferred component depends on κ through the posteriors $\pi(X,0)$, and it carries the weight $\mathbb{P}(D = 0) = 1 - \omega_P$. Consequently

$$\Big|\frac{\partial \Delta^{\text{act}}}{\partial \kappa}\Big| = (1 - \omega_P)\,(\tilde{m} - m_0)\,\Big|\frac{\partial}{\partial \kappa}\,\mathbb{E}\big[\pi(X,0)\,p(X,0) \mid D = 0\big]\Big| :$$

the liquidity sensitivity is proportional to the undisclosed mass $1 - \omega_P$. Holding the conditional sensitivity of the inferred component fixed, *shifting probability mass toward disclosed engagement (larger ω_P) attenuates the liquidity sensitivity of the activism-driven premium.*

Proof sketch. In disclosed states, $\pi(X, 1) = 1$ and all pricing objects are κ-invariant (Appendix B.3), so the disclosed component contributes zero to $\partial\Delta^{\text{act}}/\partial\kappa$, and the display follows by differentiating the decomposition. The qualifier is substantive: a configuration change that raises

ω_P moves k_D, which also alters the conditional term $\mathbb{E}[\pi(X,0)p(X,0) \mid D=0]$; the proposition signs only the weight channel, which is the mechanical and dominant effect in the calibrated region (Figure H.6). $\square$

5 Comparative Statics and Numerical Analysis

This section combines analytical comparative statics with numerical analysis. Table G.3 in Appendix G summarizes the baseline parameterization; Table G.1 in Appendix G reports equilibrium outcomes.

5.1 Baseline Calibration and Equilibrium

The baseline parameters satisfy the sufficient condition for Assumption (A5): $\delta/\sigma_\xi = 0.95/0.40 = 2.375 < 1/\phi(0) \approx 2.507$. Under this parameterization, the Hold region collapses ($k_0 = k_1$), so the blockholder's strategy reduces to Exit, Quiet Voice, and Public Voice. Figure H.1 in Appendix H depicts the resulting cutoff structure.

Three features of the equilibrium stand out:

- **Disclosed states have maximal activism premium:** $\bar{m}(X,1) = \tilde{m}$ for all $X \in \{0,1,2\}$, since $D=1$ reveals $a=1$ with certainty.
- **Nondisclosed states show inference:** The posterior $\pi(X,0)$ is highest when order flow is most diagnostic of the quiet-engagement action. With no stake acquisition in $D=0$ states, $X=1$ reveals $q=0$ and hence $\pi(1,0) = \omega_Q/(\omega_H + \omega_Q)$.
- **Disclosure deters bids:** For fixed X, $p(X,1) < p(X,0)$ because the bidder knows it must pay the full activism premium when $D=1$.

Figure H.4 plots equilibrium prices across all (X,D) states. Conditional on disclosure ($D=1$), order flow reflects only noise given Public Voice, so $P(X,1)$ is constant across X. The disclosed price level exceeds comparable nondisclosed states because the market knows engagement has occurred rather than inferring it.

5.2 Nonmonotonicity and Decomposition

Figure H.2 displays the headline comparative static: a nonmonotone relation between liquidity and expected minority takeover gains. As κ rises from low levels, $\Delta^{\min}(\kappa)$ initially increases because bid incidence grows. At higher liquidity, activism-related premia contribute less, and $\Delta^{\min}(\kappa)$ declines. The vertical dashed line marks the interior turning point $\kappa^\dagger$.

Figure H.3 decomposes these gains into the baseline component $m_0 \cdot \mathbb{P}(\text{bid})$ and the activism component $\Delta^{\text{act}}(\kappa)$. Both components vary smoothly with liquidity, and the hump shape arises from the interaction between bid incidence and the weight placed on activism-related premia.

Figure H.5 traces how equilibrium cutoffs shift with liquidity. The ordering $k_1 \le k_0 \le k_D$ is preserved throughout, though regions can collapse (e.g., $k_0 = k_1$).

5.3 Activism Frictions

Sensitivity to engagement frictions. The model's general equilibrium structure precludes simple closed forms, but analytical results pin down the direction of effects within each regime. Higher engagement costs shrink the voice regions; larger expected engagement benefits and takeover premia expand them.

Figures H.7 and H.8 confirm robustness to key parameters.

Engagement cost C_0. Raising C_0 shrinks the voice regions and lowers Δ^{act}, but it can raise bid incidence by reducing expected resistance. The net effect on $\Delta^{\min}$ is therefore ambiguous, and the turning point in κ shifts across cost levels.

Premium wedge $(m_1 - m_0)$. Increasing the wedge raises the takeover premium conditional on bidding but can deter bids. In the calibration, expected minority takeover gains peak at an intermediate wedge.

5.4 Liquidity Effects

How does liquidity affect the market's inference about activism? Holding the blockholder's strategy fixed, the within-regime effects are as follows. Fix the equilibrium cutoffs (k_1, k_0, k_D) and hence the action probabilities $(\omega_E, \omega_H, \omega_Q, \omega_P)$. The posteriors in Proposition 2 satisfy:

(a) In nondisclosed states, $\partial\pi(1,0)/\partial\kappa = 0$ and $\partial\pi(-1,0)/\partial\kappa \geq 0$;

(b) $\partial\pi(0,0)/\partial\kappa \leq 0$;

(c) $\pi(-2,0) = 0$; in disclosed states, $\pi(X,1) = 1$ for all feasible X.

Consequently, the inferred premium wedge $\bar{m}(X,0) = m_0 + (\tilde{m} - m_0)\pi(X,0)$ is increasing in κ for $X = -1$, decreasing in κ for $X = 0$, and invariant for $X = 1$, while disclosed states have $\bar{m}(X,1) = \tilde{m}$. (Proof: see Appendix B.7.)

The intuition runs as follows. At $X = 1$, the buy order fully reveals that the blockholder did not sell, pinning down the posterior regardless of κ. At $X = -1$, higher liquidity makes a sell by the noise trader more likely, so the market revises upward the probability that the blockholder engaged quietly rather than exited. At $X = 0$, the opposite logic applies: higher liquidity makes the zero-order-flow event more likely under exit, reducing the inferred engagement probability.

Across-equilibrium liquidity effects. Beyond these within-regime inference effects, changing liquidity also shifts the blockholder's incentives through adverse selection and moves equilibrium cutoffs and prices jointly. Because these objects adjust endogenously with κ, the full cross-equilibrium patterns are characterized in Figure H.5.

In disclosed states ($D = 1$), $\pi(X,1) = 1$ and $\bar{m}(X,1) = \tilde{m}$, so liquidity has no inference content conditional on disclosure. In nondisclosed states ($D = 0$), $\pi(X,0)$ varies with κ through the Bayesian updating formulas in Proposition 2, making the $D = 0$ component of $\Delta^{\text{act}}(\kappa)$ liquidity-sensitive. This asymmetry between disclosed and nondisclosed states drives the model's core predictions.

Figure H.6 isolates the disclosure-inference channel. I fix the blockholder's cutoff strategy at the baseline equilibrium ($\kappa = 0.5$) and vary liquidity κ only through the noise distribution in order flow, then compare the resulting $\Delta^{\mathrm{act}}(\kappa)$ under threshold disclosure to a counterfactual without disclosure. Removing disclosure amplifies the role of inference and makes $\Delta^{\mathrm{act}}(\kappa)$ more sensitive to κ; threshold disclosure attenuates this sensitivity by shifting probability mass toward states with observed engagement ($D = 1$).

5.5 Takeover Environment

How do the takeover environment parameters shape bid incidence? Fix any reached information set (X, D) and treat the inferred premium wedge $\bar{m}(X, D)$ as given. The bid probability function $p(X, D)$ defined in (1) satisfies:

(a) **(Synergies)** $\partial p(X, D)/\partial \bar{S} > 0$: higher synergies increase bid incidence uniformly.

(b) **(Entry cost)** $\partial p(X, D)/\partial K < 0$: higher bidding costs reduce bid incidence uniformly.

(c) **(Moderate feedback via bidder heterogeneity)** For each reached information set (X, D),

$$\frac{\partial p(X, D)}{\partial P(X, D)} = -\frac{1}{\sigma_\xi}\,\phi\!\left(\frac{\bar{m}(X, D) + K - \bar{S} + P(X, D)}{\sigma_\xi}\right) < 0.$$

In particular, the maximal slope satisfies $\sup_P \left|\partial p/\partial P\right| = \phi(0)/\sigma_\xi$, so a larger σ_ξ attenuates the price-to-entry feedback and relaxes the regularity condition in Assumption (A5).

(Proof: see Appendix B.8.)

5.6 Additional Sensitivity Analysis

I extend the sensitivity analysis to three additional structural parameters: the engagement success probability ρ, the bidder synergy volatility σ_ξ, and the discount factor δ.

First, I analyze the sensitivity to the engagement success probability $\rho \in \{0.5, 0.7, 0.9\}$ (Figure H.9). The baseline calibration utilizes $\rho = 0.9$, which is economically defensible because it represents the success rate conditional on the blockholder choosing to engage. An activist who selectively engages only when her private signal is highly favorable should plausibly succeed at a high rate. Nevertheless, lowering ρ mechanically shrinks the expected standalone improvement $\tilde{\Delta}$ and the expected takeover premium wedge $\tilde{m} - m_0$. As ρ falls, the hump-shaped curve of $\Delta^{\min}(\kappa)$ flattens significantly, reflecting the diminished payoff to both quiet and public engagement, without altering its qualitative shape. The vertical scale of the nonmonotonicity compresses, but the peak persists.

Second, I vary the bidder heterogeneity $\sigma_\xi \in \{0.25, 0.40, 0.60\}$ (Figure H.10). The parameter σ_ξ governs the variance of the bidder's private synergy shock. Higher σ_ξ increases average bid incidence because the right tail of bidder valuations grows. Importantly, it also serves a theoretical function: raising σ_ξ attenuates the price-to-entry feedback loop, cleanly relaxing the regularity condition defined in Assumption (A5). The numerical sweeps show that while higher σ_ξ shifts the entire $\Delta^{\min}(\kappa)$ curve upward (due to a higher baseline bid probability), the hump shape generated by the order-flow inference channel remains robust.

Finally, I test the sensitivity to the discount factor $\delta \in \{0.85, 0.90, 0.95\}$ (Figure H.11). Lowering δ mechanically reduces equilibrium prices through the pricing equation $P = \delta[(1-p)\hat{V} + p(P + \bar{m})]$, which in turn increases bid incidence as the bidder faces lower acquisition costs. Although lower δ simultaneously weakens the blockholder's incentive to bear the upfront engagement cost C_0, the bid-probability channel dominates: expected minority gains shift upward as δ falls. Moreover, the hump shape progressively flattens, because the inference channel matters less when bids are already likely. This interacts with the regularity condition $\delta/\sigma_\xi < 1/\phi(0)$: lower δ actively relaxes the price-feedback constraint.

6 Testable Implications

The model generates several predictions regarding how liquidity, disclosure regimes, and takeover premia interact. These predictions differentiate the framework from purely microstructure or purely governance-focused approaches.

1. **Cross-country disclosure attenuation.** The core theoretical result (Proposition 6) establishes that lowering the disclosure threshold shifts activism from inferred to observable, which dampens the sensitivity of premia to liquidity. *Prediction:* The sensitivity of takeover premia to stock liquidity is lower in jurisdictions with strict disclosure regimes (e.g., the 3% U.K. threshold) than in those with loose regimes (e.g., the 5% U.S. threshold). *Empirical strategy:* Regress M&A takeover premia on a liquidity proxy (e.g., Amihud illiquidity) interacted with a U.K./U.S. country indicator, controlling for firm fundamentals. Data can be sourced from SDC Platinum matched with CRSP/Compustat and LSE data. Identification requires controlling for baseline cross-country M&A differences.

2. **Order-flow inference and bid prediction.** In nondisclosed states, the market infers engagement structurally from order flow. In the baseline equilibrium, $\pi(0,0) > \pi(-1,0)$ because zero order flow is more likely generated by the Hold or Quiet Voice actions. *Prediction:* Zero abnormal volume predicts a higher latent probability of activism, and therefore higher subsequent bid rates, compared to moderate institutional selling. *Empirical strategy:* Sort target firms by pre-bid abnormal volume. Test whether firms exhibiting zero abnormal volume have higher subsequent M&A bid rates than firms with moderate negative volume, avoiding confounding public announcements.

3. **Nonmonotone liquidity effect on premia.** The model dictates that the inference channel collapses at both extremes of liquidity, generating a hump-shaped relationship (Proposition 5). *Prediction:* Across firms, the relationship between market liquidity and expected minority takeover gains is nonmonotone (hump-shaped). *Empirical strategy:* Bin target firms by Amihud illiquidity quintiles and plot average expected takeover premia. The model predicts an interior peak at moderate liquidity quintiles. The challenge is ensuring liquidity is not purely proxying for firm size.

4. **Activism premium in price response to disclosure.** Because $\pi(X,1) = 1 > \pi(X,0)$ for all nondisclosed states, the expected activism premium is strictly higher in disclosed states. *Prediction:* An activist crossing the regulatory threshold triggers a discrete price jump that isolates the gap between inferred engagement (embedded in pre-filing prices) and known engagement. *Empirical strategy:* Conduct a standard 13D event study. The model predicts

the abnormal return upon filing is increasing in the prevailing market illiquidity, because illiquid stocks had lower pre-filing inferred engagement ($\pi(X, 0)$). Data can be sourced from EDGAR 13D filings matched with CRSP.

5. **Bid deterrence under disclosure.** Because Public Voice mandates $D = 1$, the bidder internalizes the maximum required premium wedge, $\bar{m}(X, 1) = \tilde{m}$. *Prediction:* Publicly disclosed activist campaigns deter takeover bids more strongly than quiet engagement, but those bids that do occur feature higher premia. *Empirical strategy:* Match 13D filers to M&A bids using SharkRepellent and SDC. Test whether the hazard rate of receiving a bid is lower for 13D targets compared to a matched sample of 13F (quiet) targets, conditional on the eventual premium being higher.

7 Welfare Analysis

This section carries the two positive results—the liquidity hump and the disclosure-attenuation mechanism—into a *normative* register, asking when they translate into planner statements about total surplus and minority welfare once prices and premia are netted as pure transfers. I treat welfare as a genuine planner problem rather than as a corollary of the positive analysis. Section 7.1 defines total surplus, justifies the present-value normalization under which prices and premia are pure transfers, and defends $W_{\min} = \Delta^{\min}$ as the takeover-premium welfare object through the free-rider logic of Grossman and Hart (1980). Section 7.2 states two liquidity programs—one for total surplus, one for the minority—and signs the wedge $\kappa^* - \kappa^\dagger$ from primitives by an envelope argument, quarantining the genuinely unsigned general-equilibrium cutoff-shift terms into a named hypothesis. Section 7.3 studies a planner who sets the disclosure threshold and characterizes underdisclosure. Section 7.4 supplies the first-best benchmark and the cutoff ordering.

Two cautions frame the whole section, both forced by the corrected positive theory. First, minority value creation does *not* collapse at high liquidity: as $\kappa \to 0$ and as $\kappa \to 1$ the realized non-disclosed posterior converges to the *same* two-point law, so $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$ (Lemma 7; strict positivity of the common limit is a numerical fact at the calibration, ≈ 0.067). The interior hump in $\Delta^{\min}$—and in W—is therefore not a collapse phenomenon but a *conditional* curvature property, which can invert to a trough off the baseline calibration. Second, nothing below presumes that more liquidity or more disclosure raises efficiency; each comparative static is signed from the netted surplus, and where a residual price term is genuinely unsigned the conclusion is stated conditionally on an explicit primitive hypothesis.

7.1 Welfare Objects and Transfer Netting

Fix an equilibrium with cutoffs $k = (k_1, k_0, k_D)$ and induced action probabilities $(\omega_E, \omega_H, \omega_Q, \omega_P)$. With expectations taken over (v, s, z, ξ) under the equilibrium strategies, define the three party surpluses

$$W_{\min}(\kappa) \equiv \Delta^{\min}(\kappa) = \mathbb{E}\big[\, m^R(a)\, \mathbf{1}\{\text{bid}\}\,\big], \tag{23}$$

$$W_B(\kappa) \equiv \mathbb{E}\big[\, U(q^*, a^* \mid s)\,\big] = \mathbb{E}\big[- q^* P(X, D) + \delta\, h(q^*)\, Y - a^* C(s)\big], \tag{24}$$

$$W_{\text{bid}}(\kappa) \equiv \mathbb{E}\big[\max(\Pi_B, 0)\big] = \mathbb{E}\big[\, \mathbf{1}\{\text{bid}\}\, \big(\bar{S} - P(X, D) + \xi - \bar{m}(X, D) - K\big)\big], \tag{25}$$

and total surplus $W(\kappa) = W_{\min}(\kappa) + W_B(\kappa) + W_{\text{bid}}(\kappa)$. In (24), $h(q^*) = 1 + q^*$ is the blockholder's *post-trade share holding*: starting from one share, $h(-1) = 0$ under Exit (she sells her share), $h(0) = 1$ under Hold and Quiet Voice, and $h(+1) = 2$ under Public Voice (she buys one share). This matches the implementation, in which the continuation value is scaled by h and the Public-Voice terminal carries the two-share terms (`compute_expected_payoff`).[10]

Lemma 9 (Transfer netting and the present-value form of W)**.** *Measure all $t = 1$ and $t = 2$ flows in common present-value units, i.e. set $\delta = 1$.*[11] *Then the market price $P(X, D)$ and the realized offer premium $m^R(a)$ enter W only through differences that cancel, and*

$$W(\kappa) = \mathbb{E}\Big[\mathbf{1}\{\text{bid}\}\big(\bar{S} + \xi - K - v - a\,\tilde{\Delta}\big) + v + a\,\tilde{\Delta} - a\,C(s)\Big]. \tag{26}$$

Proof. The firm has a unit mass of shares: the blockholder's pre-trade share and a continuum of atomistic float holders. Three transfers move cash among the parties, and each nets to zero at the firm level.

(i) The secondary trade at $t = 1$. The blockholder submits q^* and pays $-q^*P(X, D)$; the counterparties (the noise trader and the competitive market maker, who absorbs net order flow at the break-even price) receive $+q^*P(X, D)$. The market maker prices at $P = \delta\,\mathbb{E}[Y \mid X, D]$ (eq. (2)), so it earns zero expected profit conditional on (X, D), and the noise trader transacts at that same competitive price. Hence $\mathbb{E}[-q^*P] + \mathbb{E}[q^*P] = 0$: the trade redistributes shares (changing h from 1 to $1 + q^*$) but adds nothing to the social sum.

(ii) The takeover transfer at $t = 2$. In a consummated bid the raider buys the float and the blockholder's h shares at a per-share consideration $b = P + m^R(a)$. The blockholder receives $h\,b$, entering W_B through the $\delta\,h\,Y$ term (with $Y = P + m^R(a)$ on a bid); the atomistic float, of mass $1 - h$, receives $(1 - h)\,b$, of which the premium part $(1 - h)\,m^R(a)$ is exactly $W_{\min}$ on the float's share and the h-share premium $h\,m^R(a)$ accrues to the blockholder. The raider pays the *whole* unit mass: its outlay is $1 \cdot b = P + m^R(a)$ per share, which enters Π_B as $-P - \bar{m}$ (with $\mathbb{E}[m^R(a) \mid X, D] = \bar{m}(X, D)$). Adding target receipts and raider outlay over the unit share mass, $h\,b + (1 - h)\,b - 1 \cdot b = 0$: every dollar of price and premium the target side receives is a dollar the raider pays. Thus P and $\bar{m}$ cancel between $W_B + W_{\min}$ and W_{bid}.

(iii) What survives is the real resource flow. A consummated bid replaces the standalone continuation $v + a\,\tilde{\Delta}$ by the raider's valuation $\bar{S} + \xi$ net of the bid cost K; absent a bid the firm is worth $v + a\,\tilde{\Delta}$; engagement consumes $a\,C(s)$ regardless. Collecting, $W = \mathbb{E}[\mathbf{1}\{\text{bid}\}(\bar{S} + \xi - K) + (1 - \mathbf{1}\{\text{bid}\})(v + a\tilde{\Delta}) - a\,C(s)]$, which rearranges to (26). The price P has vanished, confirming it is a pure transfer. $\square$

Equation (26) is the planner's maximand; it coincides with the surplus expression computed by `numerical.model.compute_welfare`, whose component decomposition $W = W_{\min} + W_{\text{bid}} + W_B$

[10] The symbol h here is the integer share count, $h \in \{0, 1, 2\}$; it is unrelated to the comparative-statics integrand $h(\pi) = \pi\,p(\pi)$ used in the hump analysis of Section 5. The two never co-occur, but we flag the clash.

[11] *Present-value normalization.* The discount factor δ converts $t = 2$ flows into $t = 1$ units. Putting all dates in common units ($\delta = 1$) is a units choice that affects no comparative static: every flow is scaled identically, and the equilibrium cutoffs (4)–(6) and posteriors (Proposition 2) are homogeneous of degree zero in that common scale. The price P enters W only through the netting below, so it drops out under any $\delta \le 1$; with $\delta < 1$ the same identity holds after discounting the real $t = 2$ flows, again without P-dependence.

reproduces (26) to machine precision at every converged κ (the sum equals the price-free total with difference 0), an exact numerical check of the netting in Lemma 9.

Why $W_{\min}$ is the takeover-premium welfare object. Dispersed minority shareholders are atomistic. By the free-rider logic of Grossman and Hart (1980), an atomistic holder tenders only if offered at least the post-takeover per-share value, so the bidder must cede essentially the full incremental value to the float; the realized *premium* is what accrues to minority holders. That realized premium per share is $m^R(a)$, paid only on a consummated bid, so ex ante minority welfare from control is $\mathbb{E}[m^R(a)\,\mathbf{1}\{\text{bid}\}] = \Delta^{\min}(\kappa)$, exactly (23). It is a distributional slice of W, not W itself.

Planner versus rent recipient. A planner maximizing total surplus is distinct from the float maximizing its control rent. Levit et al. (2024) stress, in a model of trading and shareholder voting, that the secondary market reallocates control rights to traders whose private objectives need not coincide with value maximization; the governance arrangement that best serves one constituency need not be socially efficient. The present section makes the analogous point in a liquidity-and-disclosure setting: the liquidity (or disclosure threshold) maximizing the float's control rent $W_{\min}$ need not maximize W, because a higher inferred activism premium deters marginal but value-creating bids. Proposition 7 isolates the exact sign of the divergence.

7.2 The Liquidity Planner: κ^* versus $\kappa^\dagger$

Define the two programs

$$\kappa^* \in \arg\max_{\kappa\in[0,1]} W(\kappa), \qquad \kappa^\dagger \in \arg\max_{\kappa\in[0,1]} W_{\min}(\kappa). \tag{27}$$

Both maximands are continuous on the compact $[0,1]$, so maximizers exist by Weierstrass. On the interior, write $k(\kappa) = (k_1(\kappa), k_0(\kappa), k_D(\kappa))$ for the equilibrium cutoffs, differentiable in κ where the equilibrium correspondence is smooth (maintained on the calibrated region; see the Numerical Regularity below).

The blockholder privately chooses (q, a) given the price and posterior *schedules*; the equilibrium cutoffs are her own indifference points (eqs. (4)–(6)). The next lemma is the crux of the correction: the envelope theorem kills only the part of the cutoff response that runs through her *own* margin, not the part that runs through the equilibrium schedules she takes as given. Dropping the latter—as the refuted draft did—is an error.

Lemma 10 (The W_B cutoff-shift term does not vanish by envelope)**.** *On the interior where the cutoffs are differentiable in κ,*

$$\frac{dW_B}{d\kappa} = \underbrace{\left.\frac{\partial W_B}{\partial \kappa}\right|_k}_{\textit{direct}} + \underbrace{\sum_{i\in\{1,0,D\}} \frac{\partial W_B}{\partial k_i}\frac{dk_i}{d\kappa}}_{\textit{cutoff-shift channel}}, \tag{28}$$

and the cutoff-shift channel is generically nonzero and of indeterminate sign. The envelope theorem annihilates only the part of $\partial W_B/\partial k_i$ that routes through the blockholder's own *indifferent-type* *relocation; it does* not *annihilate the part that routes through the equilibrium price and posterior schedules $P(\,\cdot\,;k)$, $\pi(\,\cdot\,;k)$, which she takes as given.*

Proof. Write $W_B(\kappa) = \mathbb{E}_s\, V\big(s, k;\, P(\,\cdot\,; k^{\text{eq}}(\kappa)), \pi(\,\cdot\,; k^{\text{eq}}(\kappa)), \kappa\big)$, where V is her payoff functional, k is her own choice of action boundaries, and the price/posterior schedules are evaluated at the equilibrium population cutoffs $k^{\text{eq}}(\kappa)$, which an atomistic blockholder does not control. At her optimum she is indifferent between adjacent actions at each k_i (eqs. (4)–(6)), so U is continuous across the boundary and the own-choice relocation of the measure-zero indifferent type has zero first-order value, $(\partial V/\partial k)(dk/d\kappa) = 0$. But V also depends on κ through the schedule arguments $P(\,\cdot\,; k^{\text{eq}}(\kappa))$, $\pi(\,\cdot\,; k^{\text{eq}}(\kappa))$, which are not her choice variables; her first-order condition says nothing about them. Their total derivative

$$\frac{\partial V}{\partial P}\,\frac{\partial P(\,\cdot\,;k)}{\partial k}\,\frac{dk^{\text{eq}}}{d\kappa} + \frac{\partial V}{\partial \pi}\,\frac{\partial \pi(\,\cdot\,;k)}{\partial k}\,\frac{dk^{\text{eq}}}{d\kappa} \tag{29}$$

is generically nonzero whenever $\partial V/\partial P \neq 0$ or $\partial V/\partial \pi \neq 0$, and it is the cutoff-shift channel in (28). For Quiet and Public Voice $\partial V/\partial P \neq 0$: the payoff carries the price terms $-q^* P$ and, on a bid, $\delta\, h\,(P + \tilde{m})$ (with the $h{=}2$ term under Public Voice), so a movement of the price *schedule*—not her location on it—changes V at first order. Hence the cutoff-shift channel does not vanish. □

Lemma 11 (The cutoff-shift terms in $W_{\min}, W_{\text{bid}}$ do not vanish)**.** *For $F \in \{W_{\min}, W_{\text{bid}}\}$, $\frac{dF}{d\kappa} = \frac{\partial F}{\partial \kappa}\big|_k + \sum_i \frac{\partial F}{\partial k_i}\frac{dk_i}{d\kappa}$, and the second sum is generically nonzero and of indeterminate sign.*

Proof. $W_{\min}$ and W_{bid} are equilibrium objects evaluated at the blockholder's optimum, not maximized over k, so no envelope theorem applies and the chain rule retains every term. The sensitivities $dk_i/d\kappa$ solve the implicit differentiation of the equilibrium system (4)–(6) with the pricing fixed point (3); their signs are not pinned by the Standing Assumptions. For instance, $dk_D/d\kappa$ aggregates the inference response (Proposition 2: $\partial\pi(-1,0)/\partial\kappa \geq 0$, $\partial\pi(0,0)/\partial\kappa \leq 0$) with the price feedback, which push the public/quiet margin in opposite directions. □

Proposition 7 (Liquidity wedge: sign of $\kappa^* - \kappa^\dagger$)**.** *Suppose W and $W_{\min}$ are single-peaked in κ* (a Numerical Regularity at the calibration, not a theorem; see below)*, and let $\kappa^\dagger$ be the interior regular maximizer of $W_{\min}$ ($W'_{\min}(\kappa^\dagger) = 0$, $W''_{\min}(\kappa^\dagger) < 0$). Define the* liquidity-surplus bundle

$$\mathcal{B}(\kappa) \equiv \underbrace{\left.\frac{\partial W_B}{\partial \kappa}\right|_k}_{\textit{direct trading rent}} + \underbrace{\sum_i \frac{\partial W_B}{\partial k_i}\frac{dk_i}{d\kappa}}_{W_B\ \textit{cutoff-shift (Lem. 10)}} + \underbrace{\frac{dW_{\text{bid}}}{d\kappa}}_{\textit{bidder (Lem. 11)}}. \tag{30}$$

Then $W'(\kappa^\dagger) = \mathcal{B}(\kappa^\dagger)$, so

$$\operatorname{sgn}\big(\kappa^* - \kappa^\dagger\big) = \operatorname{sgn}\, \mathcal{B}(\kappa^\dagger). \tag{31}$$

Under the primitive condition

$$(\text{W}^*) \qquad \mathcal{B}(\kappa^\dagger) > 0 \quad (\textit{resp.}\ < 0), \tag{32}$$

$\kappa^* > \kappa^\dagger$ (resp. $\kappa^* < \kappa^\dagger$); with $\mathcal{B}(\kappa^\dagger) = 0$, $\kappa^* = \kappa^\dagger$.

Proof. By Lemmas 10–11, $W' = W'_{\min} + dW_B/d\kappa + dW_{\text{bid}}/d\kappa$ with *every* term retained. At the interior regular maximizer $W'_{\min}(\kappa^\dagger) = 0$, so $W'(\kappa^\dagger) = dW_B/d\kappa + dW_{\text{bid}}/d\kappa = \mathcal{B}(\kappa^\dagger)$, giving (31). Under the first branch of (W*) the gradient is strictly positive, so W is strictly increasing at $\kappa^\dagger$; with W single-peaked its maximizer κ^* lies strictly to the right. The reverse and equality cases follow identically from the sign of $\mathcal{B}(\kappa^\dagger)$. □

Primitive reading of (W*). The three surviving pieces have transparent economics. $\partial W_B/\partial\kappa|_k$ is the marginal trading-rent effect: more noise camouflages the informed blockholder, raising adverse-selection rents and loosening the voice participation margin. The W_B cutoff-shift term is the genuinely unsigned general-equilibrium channel of Lemma 10: as $k(\kappa)$ moves, the price/posterior schedule the blockholder faces moves, and she does not internalize it. $dW_{\text{bid}}/d\kappa$ combines a price channel (higher κ tends to lower the average pre-bid P, raising $\Pi_B = \bar{S} - P + \xi - \bar{m} - K$) with its own unsigned cutoff-shift sum. (W*) states that the *sum* is positive at $\kappa^\dagger$. When it holds, the minority planner stops short of the surplus optimum because extra liquidity, while eroding the inferred premium the float extracts, expands the value-creating set of consummated takeovers—the free-rider tension of Grossman and Hart (1980). We do *not* have a primitive monotone comparative static that signs $\mathcal{B}$, and we say so: (W*) is a calibrated, stated hypothesis, not a theorem. It would be a mistake to delete the W_B cutoff-shift term by an envelope appeal—that is exactly the refuted move, ruled out by Lemma 10.

Numerical illustration (verified). At the baseline calibration both objects are interior-peaked. The paper's own solver (residual $\leq 10^{-5}$ on the interior) gives the minority optimum $\kappa^\dagger \approx 0.59$ with $\Delta^{\min}(\kappa^\dagger) \approx 0.078$, and the surplus optimum κ^* at essentially the same liquidity, so $\mathcal{B}(\kappa^\dagger) \approx 0$: at baseline the equality case of Proposition 7 obtains, the surplus and minority objectives being closely aligned because the same interior liquidity that best sustains the inferred activism premium also best balances the camouflage and value-creation channels in W_B and W_{bid}. Endpoint symmetry holds on the valid near-ends, $\Delta^{\min}(0.05) \approx \Delta^{\min}(0.95) \approx 0.067$ (Lemma 7), so value creation does *not* vanish at the extremes. The alignment is calibration-specific: the $\Delta^{\min}$ profile flips from a hump to a trough in a non-trivial corner of $(\sigma_\xi, \bar{S})$-space, so both the existence of an interior $\kappa^\dagger$ and the sign of the wedge depend on primitives. The proposition is therefore stated two-sidedly and the baseline orientation is reported, not assumed.

Numerical Regularity (single peak, single-valuedness, smoothness). On the calibrated grid the multi-start solver returns a single fixed point at each κ, and the maps $\kappa \mapsto k(\kappa)$, $\kappa \mapsto W(\kappa)$, $\kappa \mapsto W_{\min}(\kappa)$ are continuous and single-peaked. The single-peakedness invoked in Proposition 7 is this numerical regularity, not a theorem; uniqueness rests on the contraction hypothesis of Assumption (A6), verified numerically as in the equilibrium section.

7.3 The Disclosure-Threshold Planner

Now hold κ fixed and let the planner influence the disclosure threshold, which the blockholder's strategy turns into the equilibrium cutoff k_D. Crossing k_D flips the disclosure flag. Bids arrive on *both* branches of the model (eq. (1) assigns interior bid probability to every reached cell), but at very different rates and prices: disclosure *deters* bids — at the baseline the disclosed bid probability is $p_1^* = 0.013$ against an average $\bar{p}_Q \approx 0.35$ on the quiet branch — while raising the threshold a successful bid must clear. The marginal type therefore undergoes a discrete switch between two different bid lotteries even though the planner's objective varies smoothly with k_D.

The blockholder's k_D, derived. By the monotone best-response structure (Lemmas 4 and 5) the difference $U_P(s) - U_Q(s)$ is strictly increasing in s, so a unique k_D solves the indifference condition (6), $U_Q(k_D) = U_P(k_D)$. Her value as a function of the threshold is

$$\Pi^{\text{block}}(k_D) = \int_{k_0}^{k_D} U_Q(s)\, dF_s(s) + \int_{k_D}^{\infty} U_P(s)\, dF_s(s) + \Pi^{\text{block}}_{<k_0}, \tag{33}$$

with the Exit/Hold contribution $\Pi^{\text{block}}_{<k_0}$ independent of k_D. Since the integrand is continuous and $U_P - U_Q$ single-crosses zero at k_D, Π^{block} is C^1 on the interior with

$$\frac{d\Pi^{\text{block}}}{dk_D} = \big[U_Q(k_D) - U_P(k_D)\big]\, f_s(k_D) = 0 \tag{34}$$

at the equilibrium threshold, by (6). Two points make this a derivation, not an invocation. (a) An interior smooth optimum exists: by single-crossing $U_P - U_Q < 0$ below k_D and > 0 above, so Π^{block} has a unique interior stationary point that maximizes over admissible thresholds (the corners $k_D \downarrow k_0$ and $k_D \to \infty$ are dominated under interior-cutoff Assumption (A1)). (b) Equation (34) is the blockholder's own envelope condition on her own action margin—the indifference condition times the density—so it holds with content, unlike a blanket envelope appeal.

Planner's objective in k_D. Write $W(k_D)$ for total surplus (26) as a function of the threshold (with κ and the lower cutoffs fixed). Decompose

$$\frac{dW}{dk_D} = \underbrace{f_s(k_D)\big[\Sigma_Q(k_D) - \Sigma_P(k_D)\big]}_{\text{(i) reclassification of the marginal type}} + \underbrace{\left.\frac{\partial W}{\partial k_D}\right|_{\text{infra}}}_{\text{(ii) GE price/entry on infra-marginal types}}, \tag{35}$$

where $\Sigma_Q(s), \Sigma_P(s)$ are the *social* surpluses of a type-s blockholder under Quiet and Public Voice, read from (26); neither contains P (it netted out in Lemma 9), so the reclassification term is in real units.

Lemma 12 (The reclassification jump is a bid-screening term)**.** *Assume the bidder synergy shock is mean-zero Gaussian, $\xi \sim N(0, \sigma_\xi^2)$, with $\bar{S}$ the synergy* net of $\mathbb{E}[\xi] = 0$ *(so $\bar{S}$ does not double-count the truncation terms below). Under both Quiet and Public Voice the blockholder engages ($a = 1$), so the standalone improvement $\tilde{\Delta}$ and the cost $C(k_D)$ are common to Σ_Q, Σ_P and cancel in the difference. What does not cancel is the* bid lottery*, which differs across the two branches:*

$$\Sigma_P(k_D) - \Sigma_Q(k_D) = \big(p_1^* - \bar{p}_Q\big)\Big(\bar{S} - K - \mathbb{E}[v \mid s{=}k_D] - \tilde{\Delta}\Big) + \sigma_\xi\Big(\phi(T_1) - \mathbb{E}_z\, \phi(T_{X,0})\Big), \tag{36}$$

where $p_1^* = 1 - \Phi(T_1)$ is the disclosed-branch bid probability with $T_1 = (\tilde{m} + K - \bar{S} + P^*(\cdot, 1))/\sigma_\xi$ (eq. (1) with $\pi = 1$, $\bar{m}(\cdot, 1) = \tilde{m}$), and $\bar{p}_Q = \mathbb{E}_z\, p(X, 0)$, $T_{X,0}$ are the quiet-branch analogues: a

Quiet type generates $X = z$ with the noise law, and each cell $(X, 0)$ carries its own bid probability and threshold.

Proof. By (26) an engaging type contributes $v + \tilde{\Delta} - C(s)$ with no bid and $\bar{S} + \xi - K - C(s)$ with a consummated bid; prices and premia net out (Lemma 9). Under Quiet Voice ($q = 0$, $D = 0$) the order flow is $X = z$ and a bid arrives at cell $(X, 0)$ with probability $p(X, 0)$; hence

$$\Sigma_Q = v + \tilde{\Delta} - C(k_D) + \mathbb{E}_z\Big[p(X,0)\big(\bar{S} - K - v - \tilde{\Delta}\big) + \sigma_\xi\, \phi(T_{X,0})\Big],$$

using the mean-zero Gaussian truncation identity $\mathbb{E}[\xi \mid \xi \geq \sigma_\xi T]\,(1 - \Phi(T)) = \sigma_\xi \phi(T)$ cell by cell. Under Public Voice ($D = 1$) the disclosed branch is order-flow invariant (Appendix B.3), so the same computation gives $\Sigma_P = v + \tilde{\Delta} - C(k_D) + p_1^*(\bar{S} - K - v - \tilde{\Delta}) + \sigma_\xi\, \phi(T_1)$. Subtracting, with $v = \mathbb{E}[v \mid s{=}k_D]$, gives (36).

The planner objective $W(k_D)$ is itself *smooth* in k_D: the term $f_s(k_D)[\Sigma_Q - \Sigma_P]$ in (35) is the ordinary continuous-cutoff Leibniz boundary term. What changes *discretely* across the threshold is the marginal type's bid *lottery*: crossing k_D flips D, switching the type from the frequent, low-threshold quiet-branch bids ($\bar{p}_Q$) to the rare, high-threshold disclosed bids ($p_1^* < \bar{p}_Q$: deterrence), so $\Sigma_P - \Sigma_Q \neq 0$ even as the cutoff varies continuously. At the baseline the jump is positive ($\approx +0.064$): the bids that disclosure deters are value-*destroying* at the margin (the marginal type's standalone value $\mathbb{E}[v \mid k_D] + \tilde{\Delta} \approx 1.86$ exceeds the raider's mean creation $\bar{S} - K = 1.29$, contributing $+0.19$), and this screening gain exceeds the foregone ξ-selection (-0.12). Disclosure is socially valuable for the marginal type because it *screens out* low-quality control transactions, not because it summons them. $\square$

Proposition 8 (Disclosure wedge)**.** *Let k_D^{eq} solve* (6) *and $k_D^\star \in \arg\max_{k_D} W(k_D)$. Define the infra-marginal general-equilibrium term*

$$G(k_D) \equiv \frac{\partial W}{\partial k_D}\bigg|_{\text{infra}} = \sum_{(X,D)} \frac{\partial W}{\partial p(X,D)}\,\frac{\partial p(X,D)}{\partial k_D}, \qquad \frac{\partial p(X,D)}{\partial k_D} = -\frac{1}{\sigma_\xi}\,\phi(T_{X,D})\,\frac{\partial P(X,D)}{\partial k_D}, \quad (37)$$

so $\operatorname{sgn} G = -\operatorname{sgn}(\partial P/\partial k_D) \cdot \operatorname{sgn}(\partial W/\partial p)$*, which is endogenous and not assumed. Suppose $W(k_D)$ is single-peaked* (Numerical Regularity) *and, at k_D^{eq}:* (i) *the marginal-type jump is positive, $\Sigma_P(k_D^{\text{eq}}) - \Sigma_Q(k_D^{\text{eq}}) > 0$; and* (ii) *the infra-marginal term is non-negative, $G(k_D^{\text{eq}}) \geq 0$. Then $dW/dk_D|_{k_D^{\text{eq}}} < 0$ and $k_D^\star < k_D^{\text{eq}}$: the equilibrium discloses too little (underdisclosure). If (i) holds but (ii) is reversed strongly enough, or if (i) fails, the conclusion is suspended.*

Proof. By the blockholder's optimality (34), perturbing k_D from k_D^{eq} has zero first-order effect on Π^{block}, so the social derivative is governed entirely by the externalities she ignores. In (35) the reclassification term is $f_s(k_D)[\Sigma_Q - \Sigma_P] = -f_s(k_D)[\Sigma_P - \Sigma_Q] < 0$ by (i): raising k_D pushes the marginal type out of Public Voice into Quiet Voice, destroying the real takeover surplus of Lemma 12. The infra-marginal term is $G(k_D^{\text{eq}}) \geq 0$ by (ii). The sum is strictly negative, so $dW/dk_D|_{k_D^{\text{eq}}} < 0$; with W single-peaked the maximizer lies strictly to the left, $k_D^\star < k_D^{\text{eq}}$, enlarging the disclosing region $\{s \geq k_D\}$. $\square$

Why the price term is left unsigned. From $p = 1 - \Phi((\bar{m} + K - \bar{S} + P)/\sigma_\xi)$ (eq. (1)), $\partial p/\partial k_D = -\sigma_\xi^{-1}\phi(\cdot)\,\partial P/\partial k_D$, so $\operatorname{sgn}(\partial p/\partial k_D) = -\operatorname{sgn}(\partial P/\partial k_D)$. The sign of $\partial P/\partial k_D$ is general-equilibrium endogenous: raising k_D shrinks the public-voice/buy region, alters the $D=0$ posterior split $\pi(1,0) = \omega_Q/(\omega_H + \omega_Q)$ and the disclosed price, and feeds through the pricing fixed point (3). I therefore do not assert $\partial p/\partial k_D < 0$; the whole effect is quarantined in $G(k_D)$ and its sign *carried as hypothesis (ii) inside the proposition statement.* The directional content of Proposition 8 comes from the reclassification jump (i), signed from primitives in Lemma 12.

Reading (i) from primitives. By (36), with deterrence ($\bar{p}_Q > p_1^*$, the baseline configuration) the jump is positive iff

$$\big(\bar{p}_Q - p_1^*\big)\Big(\mathbb{E}[v \mid s{=}k_D^{\mathrm{eq}}] + \tilde{\Delta} - (\bar{S} - K)\Big) \;>\; \sigma_\xi\Big(\mathbb{E}_z\,\phi(T_{X,0}) - \phi(T_1)\Big), \tag{38}$$

i.e. when the value destroyed by the quiet-branch bids that disclosure deters — the marginal type's standalone value $\mathbb{E}[v \mid k_D]+\tilde{\Delta}$ net of the raider's mean creation $\bar{S}-K$, weighted by the deterred mass — exceeds the foregone ξ-selection. At the baseline both sides are computable from `prices.csv`: 0.19 against 0.12, so (38) holds. It fails when the marginal control transaction is value-*creating* ($\bar{S} - K$ large relative to the standalone value), in which case deterring bids destroys surplus, the jump turns negative, and the conclusion is correctly suspended.

Numerical support (partial probe, verified). A full general-equilibrium planner over k_D requires re-solving the lower cutoffs and the pricing fixed point as k_D moves. A *partial* probe—holding (k_1, k_0) at their equilibrium values and maximizing (26) over k_D alone—returns an interior W-maximizing threshold strictly below the equilibrium threshold at every κ examined: $(\kappa, k_D^{\mathrm{eq}}, k_D^{\star}) = (0.20,\ 2.51,\ 1.71), (0.50,\ 2.26,\ 2.16), (0.80,\ 2.40,\ 1.80)$. In every case $k_D^{\star} < k_D^{\mathrm{eq}}$, consistent with Proposition 8. **This probe corroborates only the reclassification channel (i); it cannot test hypothesis (ii).** By construction it holds (k_1, k_0) fixed and recomputes nothing through the price fixed point for the moved k_D, so it *freezes exactly the general-equilibrium feedback $G(k_D)$ that (ii) is about.* Hypothesis (ii) $G(k_D^{\mathrm{eq}}) \geq 0$ thus remains unverified numerically, and the directional conclusion rests on it; a full test requires re-solving the price fixed point at the lowered k_D, which I leave open.

7.4 First-Best Benchmark and Cutoff Ordering

The constrained problem moves only the threshold induced by the blockholder's strategy. The first-best lets the planner dictate the action pointwise.

Proposition 9 (First-best action rule)**.** *Maintaining $\xi \sim N(0, \sigma_\xi^2)$, a surplus-maximizing planner who observes s and dictates (q, a) subject only to feasibility engages quietly ($a = 1$, $D = 0$) rather than not at all iff*

$$\big(1 - \bar{p}_Q\big)\,\tilde{\Delta} - C(s) \;\geq\; 0, \tag{39}$$

(the improvement is realized only when no bid displaces the firm; premia are transfers and do not enter), and prefers public disclosure to quiet engagement iff the jump (36) *is positive. The first-best disclosure cutoff k_D^{FB} solves $\Sigma_P(k_D^{\mathrm{FB}}) = \Sigma_Q(k_D^{\mathrm{FB}})$, i.e. the zero of the jump.*

Proof. The objective (26) is additively separable across types once prices net out (Lemma 9), so the planner optimizes pointwise in s. For a quiet type, engagement raises the no-bid branch by $\tilde{\Delta}$

at cost $C(s)$ and leaves the bid-branch surplus $\bar{S}+\xi-K$ unchanged (the improvement is displaced by the control transfer; the premium is a transfer by Lemma 9 and does not enter), giving (39). Among engaging types, disclosure is preferred iff $\Sigma_P(s) \geq \Sigma_Q(s)$; by single-crossing of $\Sigma_P - \Sigma_Q$ in s — under deterrence ($\bar{p}_Q > p_1^*$) the term $\mathbb{E}[v \mid s]$ enters (36) with the positive coefficient $(\bar{p}_Q - p_1^*)\beta$, so the jump is strictly *increasing* in s — the optimum is a cutoff at the zero of the jump. $\square$

Corollary 1 (Constrained versus first-best disclosure)**.** *Suppose the jump is positive at the equilibrium threshold (hypothesis (i) of Proposition 8) and deterrence holds ($\bar{p}_Q > p_1^*$). Then:*

(a) $k_D^{\mathrm{FB}} < k_D^{\mathrm{eq}}$*: the first-best discloses strictly more than the equilibrium.*

(b) *If in addition (ii′) $G \geq 0$ on $[k_D^{\mathrm{FB}}, k_D^{\mathrm{eq}}]$ and W is single-peaked, then $k_D^{\mathrm{FB}} \leq k_D^{\star} < k_D^{\mathrm{eq}}$.*

Proof. (a) The private cutoff equates the blockholder's own Quiet/Public payoffs (eq. (6)), which omit the takeover externality on minority and bidder surplus; the first-best cutoff equates the *social* surpluses, $\Sigma_P = \Sigma_Q$. By (i) the jump $\Sigma_P - \Sigma_Q$ is strictly positive at k_D^{eq}, and since the jump is strictly *increasing* in s under deterrence (Proposition 9) its zero k_D^{FB} lies at a strictly lower signal, $k_D^{\mathrm{FB}} < k_D^{\mathrm{eq}}$.

(b) We must show the *constrained* optimum $k_D^{\star}$ lies between the two, $k_D^{\mathrm{FB}} \leq k_D^{\star} < k_D^{\mathrm{eq}}$, and reconcile the two roles of G. The constrained planner solves $dW/dk_D = 0$, i.e. $f_s(k_D)[\Sigma_P(k_D) - \Sigma_Q(k_D)] = G(k_D)$ by (35); the first-best solves $\Sigma_P - \Sigma_Q = 0$. *There is no contradiction in the direction of G: $G \geq 0$* enters the *level of the FOC*, not its sign at k_D^{eq}. At k_D^{eq} the reclassification term is strictly negative (it equals $-f_s[\Sigma_P - \Sigma_Q] < 0$ by (i)) and $G \geq 0$, so the total $dW/dk_D < 0$ and the constrained optimum lies to the *left*, below k_D^{eq} (this is Proposition 8). At k_D^{FB}, by contrast, the reclassification term vanishes ($\Sigma_P = \Sigma_Q$), so $dW/dk_D|_{k_D^{\mathrm{FB}}} = G(k_D^{\mathrm{FB}}) \geq 0$ by (ii′), which supplies non-negativity of G at this point (this is why the hypothesis is stated on the interval rather than at k_D^{eq} alone); the constrained objective is still weakly increasing at the first-best cutoff, so the constrained maximizer lies weakly to the *right* of k_D^{FB}. Combining, $k_D^{\mathrm{FB}} \leq k_D^{\star} < k_D^{\mathrm{eq}}$. The economics: the same non-negative infra-marginal price externality G pushes the constrained optimum *below* the private cutoff (relative to k_D^{eq}, where the marginal-type term dominates) yet *above* the first-best (relative to k_D^{FB}, where the marginal-type term is zero and only G remains). The directions are consistent once the FOC is evaluated at each cutoff in turn. $\square$

Scope. The disclosure ordering is conditional on the named hypotheses (i)–(ii), of which the baseline partial probe corroborates only (i) ($k_D^{\star} < k_D^{\mathrm{eq}}$, interior, at $\kappa \in \{0.2, 0.5, 0.8\}$). The liquidity wedge is approximately zero at baseline ($\kappa^* \approx \kappa^{\dagger} \approx 0.59$). The relevant $\Delta^{\min}$ and W profiles flip sign across the parameter map, so neither an unconditional non-monotone-welfare claim nor an unconditional wedge orientation is asserted; the single-peakedness used to convert signed gradients into argmax orderings is a Numerical Regularity carried inside the proposition statements.

8 Extensions

Everything in this section stress-tests the two headline results rather than adding a third. The polar full- and no-disclosure benchmarks bracket the disclosure-attenuation mechanism of Proposition 6, marking its upper and lower bounds; the noisy-rumor regime shows the same attenuation survives when partial observability arrives through media or leaks, not only through filings; and the general-equilibrium analysis shows where that attenuation ceases to carry a determinate sign once the blockholder re-optimizes her cutoffs. Each locates the boundary of the two established results rather than introducing a mechanism of its own.

Disclosure requirements for activist investors vary widely across jurisdictions, creating natural laboratories for testing the model's predictions. I consider three benchmarks that span the policy spectrum.

In the United States, Schedule 13D mandates filing within five business days of crossing a 5% beneficial ownership threshold. The United Kingdom imposes a lower 3% threshold with a tighter two-trading-day deadline under the FCA's Disclosure and Transparency Rules. The European Union's Transparency Directive sets a 5% baseline but permits member states to adopt lower thresholds; Germany, for instance, applies 3% for voting-rights notifications. At one extreme, very low thresholds approach full disclosure of activist intent; at the other, weak enforcement or broad institutional exemptions leave much activism unobserved.

Institutional background: ownership-disclosure regimes. The model's disclosure threshold τ maps to real stake-disclosure rules, which are distinct from *control* thresholds (discussed below).

- **United States — Schedule 13D.** An activist (non-passive) investor must file once beneficial ownership exceeds 5%. The SEC's 2024 modernization *shortened* the filing deadline from 10 calendar days to 5 business days, effective February 5, 2024.
- **United States — 13G and 13F.** Schedule 13G is the parallel short-form regime for passive holders crossing 5%; Form 13F is a separate quarterly position report required of institutional investment managers exercising discretion over at least $100 million in U.S.-listed equities.
- **United Kingdom — FCA DTR 5.** Issuers in scope require disclosure once a holding reaches 3%, and thereafter at each whole percentage point (1% increments).
- **European Union — Transparency Directive.** Voting-rights notification is triggered at the 5%, 10%, 15%, 20%, 25%, 30%, 50%, and 75% thresholds (member states may set additional lower thresholds).

Confounder to keep separate. The roughly 30% *mandatory-bid*/control threshold (e.g., under the U.K. City Code on Takeovers and Mergers) governs when an acquirer must extend an offer to all shareholders. It is a control-acquisition rule, not a disclosure rule, and the two should not be conflated when interpreting τ.

Throughout this section, I hold the blockholder's cutoff strategy fixed, and hence the action probabilities $\omega_E, \omega_H, \omega_Q, \omega_P$, to isolate the effect of the disclosure regime on inference and pricing.

8.1 Full Disclosure Benchmark

Some jurisdictions have considered or implemented very low disclosure thresholds (such as proposals to reduce the US Schedule 13D threshold from 5% to 2.5%) or require disclosure of activist intent regardless of stake size. In the limit, such regimes approach full disclosure.

Under full disclosure, the market observes the blockholder's action (q, a) directly. The engagement posterior is trivial: $\pi = 1$ when $a = 1$ and $\pi = 0$ when $a = 0$. The activism premium is deterministic in each state, and the inference channel that links liquidity to premia shuts down completely. Order flow reveals only the noise-trader realization z, which is uninformative about fundamentals. This benchmark marks the upper bound on how much disclosure can attenuate the liquidity-sensitivity of the activism premium.

8.2 No Disclosure Benchmark

At the other extreme, some jurisdictions have high thresholds, weak enforcement, or broad exemptions that leave much activism unobserved. The no-disclosure benchmark captures this limiting case: the market receives no disclosure signal and must infer everything from order flow alone.

Under no disclosure, the market conditions only on aggregate order flow $X \in \{-2, -1, 0, 1, 2\}$. Only the extreme realizations pin down engagement with certainty: $X = -2$ arises only from Exit ($\pi = 0$), while $X = 2$ arises only from Public Voice ($\pi = 1$). At intermediate values of X, the market mixes over all four actions with weights that depend on the noise distribution and hence on κ. Compared to the baseline, this regime makes the activism premium more liquidity-sensitive because inference operates across all states rather than being confined to nondisclosed states. This benchmark marks the lower bound on how much disclosure can attenuate the inference channel.

8.3 An Intermediate Regime: Stake Disclosure Plus Noisy Rumors

Between the polar benchmarks of full disclosure and no disclosure lies a realistic intermediate case: baseline stake-triggered disclosure augmented by a noisy public signal about quiet engagement. This regime captures modern information environments, such as the United Kingdom, where a lower 3% threshold is augmented by active financial media (L. Fang and Peress, 2009), or markets where analyst coverage, Bloomberg terminal leaks, and quarterly 13F filings offer noisy, delayed signals about activist positions. Furthermore, the rumor structure serves as a tractable proxy for wolf pack activism, where information leaks through a network of secondary blockholders.

To formalize this, I augment the baseline model with a binary public rumor signal $R \in \{0, 1\}$ that fires in nondisclosed states ($D = 0$). If the blockholder engages ($a = 1$), the rumor fires with true-positive probability η_1. If she does not engage ($a = 0$), the rumor fires with false-positive probability η_0, where $\eta_0 < \eta_1$. In disclosed states ($D = 1$), the rumor is irrelevant because engagement is already known.

Applying Bayes' rule, the updated posterior in the nondisclosed state becomes:

$$\pi(X, 0, R) = \frac{\omega_Q \mathbb{P}(X|q = 0)\mathbb{P}(R|a = 1)}{\omega_E \mathbb{P}(X|q = -1)\mathbb{P}(R|a = 0) + \mathbb{P}(X|q = 0)(\omega_H \mathbb{P}(R|a = 0) + \omega_Q \mathbb{P}(R|a = 1))}. \tag{40}$$

When the rumor fires ($R = 1$), the market updates toward engagement; when it remains silent ($R = 0$), the market updates away.

Numerical analysis of this regime (Figure H.12) reveals that as the rumor becomes more precise (i.e., as $\eta_1 - \eta_0$ increases), the $\Delta^{\min}(\kappa)$ curve systematically flattens. Comparing a baseline model ($\eta_1 = \eta_0$), a moderate rumor regime ($\eta_1 = 0.75, \eta_0 = 0.25$), and a highly precise regime ($\eta_1 = 0.95, \eta_0 = 0.05$), the activism premium becomes progressively less sensitive to liquidity. In the limit as $\eta_1 \to 1$ and $\eta_0 \to 0$, the regime converges to the full-disclosure benchmark within nondisclosed states. This confirms the robustness of the core theoretical mechanism: any friction or institutional feature that shifts engagement from inferred to observable intrinsically attenuates the sensitivity of takeover premia to secondary market liquidity. Appendix B.15 provides the posterior formulas for general (η_0, η_1), and Table G.2 in Appendix G reports a numerical comparison.

8.4 General Equilibrium Disclosure Effects

Proposition 6 established a partial-equilibrium result: holding the blockholder's strategy fixed, stricter disclosure attenuates the liquidity sensitivity of premia by increasing transparency. However, in general equilibrium, altering the regulatory disclosure threshold fundamentally changes the blockholder's incentive to engage in the first place.

Proposition 10 (General Equilibrium Disclosure Trade-off)**.** *Let τ parameterize the strictness of the disclosure regime, and let the equilibrium region probabilities $\omega_j(\tau)$, $j \in \{E, H, Q, P\}$, depend on τ through the blockholder's cutoffs. The total derivative of the activism-driven premium decomposes* exactly*, by the chain rule, into a direct (transparency) channel and an indirect (deterrence) channel operating through the equilibrium region masses:*

$$\frac{d\Delta^{\text{act}}}{d\tau} = \underbrace{\frac{\partial \Delta^{\text{act}}}{\partial \tau}\bigg|_{\omega}}_{\textit{Transparency effect}} + \underbrace{\sum_{j \in \{E,H,Q,P\}} \frac{\partial \Delta^{\text{act}}}{\partial \omega_j} \frac{d\omega_j}{d\tau}}_{\textit{Deterrence effect}}. \tag{41}$$

The decomposition itself is definitional (chain rule). The channel signs are carried as conditional readings, not theorems: the transparency effect—the partial derivative holding the equilibrium masses fixed—is nonnegative in the calibrated region*, where shifting mass to observable states ($D = 1$) removes the inference discount (the general-equilibrium counterpart of the weight channel in Proposition 6); the deterrence channel collects the indirect terms—a stricter threshold forces earlier disclosure, raising bid deterrence and lowering the expected return to quiet engagement, which contracts the engagement mass $\omega_Q + \omega_P$—and is nonpositive* whenever *the premium is increasing in the engagement mass and $d(\omega_Q + \omega_P)/d\tau \leq 0$. The* net *sign is indeterminate and governed by primitives. As with the liquidity comparative static, the indirect (cutoff-shift) terms admit no general closed-form sign—they require inverting the blockholder's cutoff Jacobian—so the trade-off is stated as an exact decomposition with conditionally signed channels rather than a signed net effect; no formal proof of the channel signs is claimed beyond the stated conditions.*

The net welfare impact on minority shareholders is therefore ambiguous and depends on primitives. Under sufficient conditions where engagement costs C_0 are low and liquidity κ is moderate, the blockholder's constraint is slack. The transparency effect dominates, and strict disclosure unambiguously aids minority shareholders. Conversely, if C_0 is high, the blockholder is at the margin of indifference. Imposing strict disclosure destroys the Quiet Voice region without proportionally expanding Public Voice. In this regime, the deterrence effect dominates, and strict disclosure paradoxically harms minority shareholders by deterring the very activism that generates takeover premia.

9 Conclusion

This paper develops a unified theoretical framework demonstrating that secondary-market liquidity, activism disclosure, and takeover premia are jointly determined in equilibrium. By embedding an informed blockholder's choice among Exit, Quiet Voice, and Public Voice into a setting with order-flow inference and takeover feedback, the model isolates a novel pricing mechanism: an activism premium that is either directly observed through regulatory disclosure or inferred by a competitive market maker. The central finding is a disclosure-attenuation result: by moving the activism premium from an inferred to a directly observable basis, stricter disclosure makes takeover premia less sensitive to market liquidity. As a supporting comparative static, liquidity exerts a nonmonotone, hump-shaped effect on the premium captured by minority shareholders. Rising noise-trading intensity lowers adverse-selection costs and spurs bid incidence, but simultaneously erodes the informativeness of order flow, dampening the inference channel that sustains engagement incentives. An interior, premium-maximizing liquidity level $\kappa^{\dagger}$ thus strikes the optimal balance for minority shareholders.

The analysis provides several distinct, testable implications. Most notably, threshold-based disclosure policy acts as an implicit governor on the liquidity-premia relationship. Lowering the disclosure threshold increases the share of activism that is directly observable, thereby attenuating the inference channel. Consequently, the model predicts that the sensitivity of takeover premia to market liquidity is structurally lower in strict-disclosure jurisdictions (such as the United Kingdom) than in permissive jurisdictions (such as the United States). The model further highlights a critical general equilibrium trade-off for policymakers: while stricter thresholds increase market transparency, they simultaneously reduce the expected returns to quiet engagement, potentially deterring activism altogether. Furthermore, the welfare analysis reveals that policies protecting minority shareholders by targeting optimal liquidity may diverge from the objective of maximizing total social surplus.

Extending the framework to dynamic settings with multiple trading rounds would capture the gradual position-building that characterizes real-world activist campaigns. Introducing multiple blockholders or endogenous information acquisition would illuminate how collective action and rumor networks, such as the wolf pack dynamics discussed in Section 8.3, reshape the inference channel. Each of these extensions preserves the core insight developed here: liquidity, disclosure, and corporate governance are inextricably linked, and evaluating any one requires a framework that internalizes their joint equilibrium feedback.

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A Notation

Symbol	Meaning
Fundamentals and information	
v	Standalone fundamental at $t=0$, $v \sim \mathcal{N}(\mu, \sigma_v^2)$.
s	Blockholder signal, $s = v + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
$\mu, \sigma_v^2, \sigma_\varepsilon^2, \sigma_s^2$	Prior mean/variances; $\sigma_s^2 \equiv \sigma_v^2 + \sigma_\varepsilon^2$.
β	Posterior weight in $\hat{v}(s) = \mathbb{E}[v \mid s] = \mu + \beta(s-\mu)$; $\beta \equiv \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2)$.
$\hat{v}(s)$	Blockholder posterior mean of v given s.
Φ, ϕ	Standard normal CDF and PDF.
$\lambda_L(\alpha), \lambda_U(\alpha)$	Inverse Mills ratios: $\lambda_L(\alpha) = \phi(\alpha)/\Phi(\alpha)$ and $\lambda_U(\alpha) = \phi(\alpha)/(1-\Phi(\alpha))$.
Trading, liquidity, disclosure	
q	Blockholder order at $t=1$, $q \in \{-1, 0, +1\}$.
z	Noise-trader order, $z \in \{-1, 0, +1\}$ with $\mathbb{P}(z=0) = 1-\kappa$ and $\mathbb{P}(z=\pm 1) = \kappa/2$.
κ	Noise-trading intensity (liquidity).
X	Total order flow observed by the market maker, $X = q + z$.
D	Disclosure indicator for threshold-crossing stake acquisition, $D = 1 \iff (q = +1)$.
$P(X, D)$	Competitive price at $t=1$ conditional on (X, D).
Engagement (voice)	
a	Engagement choice at $t=1$, $a \in \{0, 1\}$.
$C(s), C_0, \chi$	Engagement cost at $t=1$ (private to the blockholder); $C(s) = C_0 \exp\left(-\chi \frac{s-\mu}{\sigma_s}\right)$.
ρ	Engagement success probability.
$\Delta, \tilde{\Delta}$	Standalone improvement if engagement succeeds; $\tilde{\Delta} \equiv \rho\Delta$ is the expected improvement under $a=1$.
m_0, m_1	Takeover premia above P: baseline m_0 (no successful engagement) and $m_1 > m_0$ (successful engagement).
$\tilde{m}$	Expected premium under engagement, $\tilde{m} \equiv m_0 + \rho(m_1 - m_0)$.
$\pi(X, D)$	Posterior engagement probability, $\pi(X, D) \equiv \mathbb{P}(a = 1 \mid X, D)$.
$\bar{m}(X, D)$	Expected premium wedge conditional on (X, D), $\bar{m}(X, D) = \mathbb{E}[m^R(a) \mid X, D] = m_0 + (\tilde{m} - m_0)\pi(X, D)$.
Takeover and payoffs	
$\bar{S}$	Baseline bidder synergy (incremental value from acquisition).
K	Fixed bidding / entry cost.
ξ, σ_ξ^2	Bidder's private synergy shock, $\xi \sim \mathcal{N}(0, \sigma_\xi^2)$.
$\Pi_B(X, D)$	Bidder expected net deal surplus conditional on (P, D) (on-path shorthand (X, D)).
$p(X, D)$	Bid probability conditional on (P, D); on-path shorthand for $p(P(X, D), D)$.
$m^R(a)$	Realized premium wedge conditional on engagement: $m^R(a) = m_0 + a(\tilde{m} - m_0)$.
$b(X, D, a)$	Per-share offer if a bid occurs, $b(X, D, a) = P(X, D) + m^R(a)$.
τ	Time between pricing/entry and settlement in $\delta = \exp(-r\tau)$.
δ	Discount factor between pricing/entry and settlement, $\delta = \exp(-r\tau)$ (or $1/(1+r)$).
Y	Terminal per-share payoff (takeover vs. standalone).
$U(q, a \mid s)$	Blockholder expected utility at $t=1$ conditional on s and action (q, a).
$\Delta^{\min}(\kappa)$	Expected takeover premium earned by a representative minority share, $\mathbb{E}[m^R(a) \cdot \mathbf{1}\{\text{bid}\}]$.
$\Delta^{\text{act}}(\kappa)$	Activism-driven component of minority takeover gains, $\mathbb{E}[(m^R(a) - m_0) \cdot \mathbf{1}\{\text{bid}\}]$.
Equilibrium thresholds and regions	
k_1, k_0, k_D	Signal cutoffs for Exit, Hold, Quiet Voice (hidden engagement), and Public Voice (disclosed engagement).
α_i	Standardized cutoffs $\alpha_i \equiv (k_i - \mu)/\sigma_s$ for $i \in \{1, 0, D\}$.
$\omega_E, \omega_H, \omega_Q, \omega_P$	Ex ante probabilities of Exit / Hold / Quiet Voice / Public Voice regions.

Table 1: Notation summary (quick reference).

Label	Content	Location
(A1)	Interior cutoffs (nondegeneracy)	Engagement Technology (§3)
(A2)	Engagement cost decreasing in signal	Engagement Technology (§3)
(A3)	Premium wedge: $\tilde{m} > m_0$	Engagement Technology (§3)
(A4)	Interior bid probability	Bidder Entry (§3)
(A5)	Unique pricing fixed point	Price Formation (§3)
(A6)	Contraction (uniqueness)	Equilibrium Concept (§3)
(A7)	Price schedule separates order flow	Bidder Entry (§3)

Table 2: Summary of standing assumptions (A1)–(A7).

B Proofs and Derivations

B.1 Proof of Proposition 1

Proof. Write the four feasible actions as

$$E \equiv (-1, 0), \quad H \equiv (0, 0), \quad Q \equiv (0, 1), \quad P \equiv (+1, 1),$$

corresponding to Exit, Hold, Quiet Voice, and Public Voice.

Step 1: Best responses are characterized by (at most) three cutoffs. Fix any candidate pricing rule $P(\cdot, \cdot)$ and bidder-entry probabilities $p(\cdot, \cdot)$ at each information set (X, D). For an action (q, a) with holdings $h = 1 + q$ and disclosure $D = \mathbf{1}\{q = +1\}$, the blockholder's $t = 1$ payoff conditional on signal s is

$$U(q, a \mid s) = \mathbb{E}\Big[-q \cdot P(X, D) + \delta h \cdot Y - a \cdot C(s) \,\Big|\, s, q, a\Big],$$

where $X = q + z$ and $Y = \mathbf{1}\{\text{bid}\} \cdot (P(X, D) + m^R(a)) + (1 - \mathbf{1}\{\text{bid}\}) \cdot (v + a\tilde{\Delta})$. Since the bid event depends only on ξ conditional on (X, D), $\mathbb{E}[\mathbf{1}\{\text{bid}\} \mid X, D] = p(X, D)$, and risk neutrality implies $\mathbb{E}[v \mid s] = \hat{v}(s)$. Taking expectations over (v, ξ, z) conditional on s yields

$$U(q, a \mid s) = \mathbb{E}_z\Big[-q \cdot P(X, D) + \delta h\big(p(X, D) \cdot (P(X, D) + m^R(a)) \\ + (1 - p(X, D)) \cdot (\hat{v}(s) + a\tilde{\Delta})\big)\Big] - a \cdot C(s).$$

Define

$$B_{q,a} \equiv \delta h \cdot \mathbb{E}_z[1 - p(X, D)], \qquad X = q + z,$$

and collect all remaining terms (which do not depend on s) into $A_{q,a}$. Then

$$U(q, a \mid s) = A_{q,a} + B_{q,a}\, \hat{v}(s) - a \cdot C(s). \tag{42}$$

Because $\hat{v}(s) = \mu + \beta(s - \mu)$ is strictly increasing in s, and $C(s)$ is weakly decreasing by Assumption (A2), the payoff differences that determine the adjacent-action indifference cutoffs are single-crossing in s. In particular:

- **Exit vs. Hold.** $U_H(s) - U_E = A_H - A_E + B_H \hat{v}(s)$ is strictly increasing in s (since $B_H > 0$ for

$h = 1$). Hence there is at most one cutoff k_1 solving $U_E = U_H(k_1)$, and whenever it exists it separates Exit from Hold.

- **Hold vs. Quiet Voice.** Since H and Q share $(q, D, h) = (0, 0, 1)$, their $\hat{v}(s)$ coefficients are the same, and

$$U_Q(s) - U_H(s) = \delta\,\mathbb{E}_z\Big[p(X,0)\cdot(\tilde{m} - m_0) + (1 - p(X,0))\cdot\tilde{\Delta}\Big] - C(s), \tag{43}$$

  which is weakly increasing in s (strictly increasing when $\chi > 0$). Thus there is at most one cutoff k_0 solving $U_H(k_0) = U_Q(k_0)$, and whenever it exists it separates Hold from Quiet Voice.

- **Quiet vs. Public Voice.** Since Q and P both have $a = 1$, the cost term cancels, and $U_P(s) - U_Q(s) = (A_P - A_Q) + (B_P - B_Q)\hat{v}(s)$ is affine in $\hat{v}(s)$ and hence single-crossing. The coefficients B_Q and B_P are generically distinct because Q and P differ in holdings ($h = 1$ versus $h = 2$) and in disclosure status ($D = 0$ versus $D = 1$), which enter the expected bid probabilities in $B_{q,a}$. Moreover, $B_P - B_Q > 0$: although the purely algebraic relation $h_P = 2 > 1 = h_Q$ does not guarantee $2(1 - p_P) > 1 - p_Q$ for arbitrary bid probabilities, in equilibrium, Public Voice ($D = 1$) forces the buyer to internalize the full premium $\tilde{m}$. This maximal premium strongly deters takeover bids relative to the nondisclosed ($D = 0$) branch, ensuring $p(X, 1)$ is sufficiently lower than $p(X, 0)$ so that the coefficient on $\hat{v}(s)$ is strictly larger for Public Voice, making $U_P - U_Q$ strictly increasing in s. Thus there is at most one cutoff k_D solving $U_Q(k_D) = U_P(k_D)$, and whenever it exists it separates Quiet Voice from Public Voice.

Under Assumption (A1) (interiority/nondegeneracy of regions), the relevant indifference cutoffs exist and are (weakly) ordered, yielding a best response with thresholds $k_1 \le k_0 \le k_D$ as stated in Proposition 1.[12]

Step 2: Given cutoffs, beliefs and prices are pinned down. Fix a cutoff vector (k_1, k_0, k_D). This pins down the ex ante region probabilities $(\omega_E, \omega_H, \omega_Q, \omega_P)$ and hence (by Bayes' rule) the posteriors $\pi(X, D)$ in Proposition 2. Given these posteriors and conditional means $\mathbb{E}[v \mid X, D]$, the competitive pricing rule is defined by $P(X, D) = \delta\,\mathbb{E}[Y \mid X, D]$ and is characterized by the fixed point (3). Assumption (A5) ensures that for each reached information set (X, D) this fixed point has a unique solution $P^*(X, D)$.

Step 3: Fixed point over cutoff vectors. Define the cutoff mapping $T : (k_1, k_0, k_D) \mapsto (k_1', k_0', k_D')$ by: (i) compute $(\omega_E, \omega_H, \omega_Q, \omega_P)$ from (k_1, k_0, k_D); (ii) compute posteriors $\pi(X, D)$; (iii) solve for prices $P^*(X, D)$; and (iv) define (k_1', k_0', k_D') as the (unique) solutions to the indifference conditions (4)–(6) induced by these objects (using the single-crossing properties from Step 1). By construction and Assumption (A5), each step is continuous, so T is continuous.

Assumption (A6) states that T is a contraction on a suitable nonempty compact subset $\Theta \subset \mathbb{R}^3$ (with $k_1 \le k_0 \le k_D$) under the sup norm. Since $(\Theta, \|\cdot\|_\infty)$ is complete, the Banach fixed-point theorem implies that T has a unique fixed point $(k_1, k_0, k_D) \in \Theta$ and that iterating T converges to it. At this fixed point, the blockholder's cutoff strategy is optimal given the induced prices, beliefs are Bayes-consistent, and the market maker's pricing rule is competitive, which together constitute a Perfect Bayesian Equilibrium. The disclosure outcomes follow from the rule $D = 1 \iff (q = +1)$. □

[12] The weak inequalities allow for limiting cases in which some regions collapse (e.g., $k_0 = k_1$).

B.2 Proof of Proposition 2

Proof. For disclosed states, $D = 1$ occurs if and only if $q = +1$ by the disclosure rule. Since the only feasible action with $q = +1$ is Public Voice, $\pi(X, 1) = \mathbb{P}(a = 1 \mid X, 1) = 1$ for all feasible $X \in \{0, 1, 2\}$.

For nondisclosed states, $D = 0$ rules out Public Voice. The remaining actions consistent with $D = 0$ are Exit $E \equiv (-1, 0)$, Hold $H \equiv (0, 0)$, and Quiet Voice $Q \equiv (0, 1)$, which occur with ex ante probabilities $(\omega_E, \omega_H, \omega_Q)$.

Write the noise trade probabilities as $\mathbb{P}(z = 0) = 1 - \kappa$ and $\mathbb{P}(z = \pm 1) = \kappa/2$. Conditional on each action, $X = q + z$ has the following support and probabilities:

	$X = q - 1$	$X = q$	$X = q + 1$
$q = -1$ (E)	$\kappa/2$	$1 - \kappa$	$\kappa/2$
$q = 0$ (H, Q)	$\kappa/2$	$1 - \kappa$	$\kappa/2$
$q = +1$ (P)	$\kappa/2$	$1 - \kappa$	$\kappa/2$

Note that Hold H and Quiet Voice Q generate identical order-flow distributions since both have $q = 0$. Applying Bayes' rule, for any X with $D = 0$,

$$\pi(X, 0) = \mathbb{P}(a = 1 \mid X, 0) = \mathbb{P}(Q \mid X, 0) = \frac{\omega_Q \, \mathbb{P}(X \mid Q)}{\omega_E \, \mathbb{P}(X \mid E) + (\omega_H + \omega_Q) \, \mathbb{P}(X \mid q{=}0)}.$$

If $X = -2$, then X can only be generated by Exit, so $\pi(-2, 0) = 0$.

If $X = 1$, then $X = 1$ requires $q = 0$ and $z = +1$, so conditioning on $D = 0$ implies

$$\pi(1, 0) = \frac{\omega_Q}{\omega_H + \omega_Q}.$$

If $X = -1$, then $\mathbb{P}(X = -1 \mid q{=}0) = \mathbb{P}(z = -1) = \kappa/2$ and $\mathbb{P}(X = -1 \mid E) = \mathbb{P}(z = 0) = 1 - \kappa$, giving

$$\pi(-1, 0) = \frac{\omega_Q(\kappa/2)}{(\omega_H + \omega_Q)(\kappa/2) + \omega_E(1 - \kappa)}.$$

If $X = 0$, then $\mathbb{P}(X = 0 \mid q{=}0) = \mathbb{P}(z = 0) = 1 - \kappa$ and $\mathbb{P}(X = 0 \mid E) = \mathbb{P}(z = 1) = \kappa/2$, so

$$\pi(0, 0) = \frac{\omega_Q(1 - \kappa)}{(\omega_H + \omega_Q)(1 - \kappa) + \omega_E(\kappa/2)}.$$

□

B.3 Disclosed-Branch Invariance

Proof. Consider any equilibrium under the Standing Assumptions and suppose the disclosed branch is reached (i.e., $\mathbb{P}(D = 1) > 0$). By the feasible action set and the disclosure rule, $D = 1$ occurs if and only if the blockholder chooses Public Voice $P \equiv (+1, 1)$, hence $a = 1$ almost surely on $\{D = 1\}$ and $X = 1 + z$ with $z \in \{-1, 0, 1\}$.

Since z is independent of (v, s, ξ), conditioning on $(X, D) = (x, 1)$ reveals only the noise realization $z = x - 1$ and does not change beliefs about v beyond what is already implied by $D = 1$.

Formally, for any Borel set $B \subset \mathbb{R}$ and any $x \in \{0, 1, 2\}$,

$$\mathbb{P}(v \in B \mid X = x, D = 1) = \mathbb{P}(v \in B \mid D = 1).$$

In particular, $\pi(x, 1) = \mathbb{P}(a = 1 \mid X = x, D = 1) = 1$ and $\mathbb{E}[v \mid X = x, D = 1] = \mathbb{E}[v \mid D = 1] = \mu_P$ for all $x \in \{0, 1, 2\}$.

Therefore, for each $x \in \{0, 1, 2\}$ the pricing fixed point (3) on the disclosed branch has the same right-hand side: it depends on (X, D) only through $\hat{V}(X, D)$ and $\bar{m}(X, D)$, both of which are constant when $D = 1$, and through $p(P, 1)$, which depends on (P, D) but not on X separately. By Assumption (A5), this fixed point admits a unique solution, so $P^*(x, 1)$ is the same for all $x \in \{0, 1, 2\}$. The bidder's bid probability satisfies $p(x, 1) = p(P^*(x, 1), 1)$ and is therefore also constant across x on the disclosed branch. □

B.4 Derivation of Conditional Means

Lemma 13 (Truncated normal expectations). *Let $S \sim \mathcal{N}(\mu, \sigma^2)$ and define standardized thresholds $\alpha = (a - \mu)/\sigma$ and $\gamma = (b - \mu)/\sigma$. Then*

$$\mathbb{E}[S \mid S < a] = \mu - \sigma \frac{\phi(\alpha)}{\Phi(\alpha)},$$
$$\mathbb{E}[S \mid a \le S < b] = \mu + \sigma \frac{\phi(\alpha) - \phi(\gamma)}{\Phi(\gamma) - \Phi(\alpha)},$$
$$\mathbb{E}[S \mid S \ge b] = \mu + \sigma \frac{\phi(\gamma)}{1 - \Phi(\gamma)}.$$

Proof. These are standard formulas for the mean of a truncated normal distribution. For completeness, let $Z \sim \mathcal{N}(0, 1)$ and note that $\mathbb{E}[S \mid \cdot] = \mu + \sigma\, \mathbb{E}[Z \mid \cdot]$. For the left tail,

$$\mathbb{E}[Z \mid Z < \alpha] = \frac{\int_{-\infty}^{\alpha} z\phi(z)\, dz}{\Phi(\alpha)} = \frac{-\phi(\alpha)}{\Phi(\alpha)},$$

where the last equality follows from $\frac{d}{dz}\phi(z) = -z\phi(z)$. The right tail follows similarly. For an interval $[\alpha, \gamma)$,

$$\mathbb{E}[Z \mid \alpha \le Z < \gamma] = \frac{\int_{\alpha}^{\gamma} z\phi(z)\, dz}{\Phi(\gamma) - \Phi(\alpha)} = \frac{\phi(\alpha) - \phi(\gamma)}{\Phi(\gamma) - \Phi(\alpha)}.$$

□

Derivation of $\mu_E, \mu_H, \mu_Q, \mu_P$. By iterated expectations,

$$\mathbb{E}[v \mid s \in \mathcal{S}] = \mathbb{E}[\mathbb{E}[v \mid s] \mid s \in \mathcal{S}] = \mathbb{E}[\hat{v}(s) \mid s \in \mathcal{S}],$$

for any measurable set $\mathcal{S}$. Since $\hat{v}(s) = \mu + \beta(s - \mu)$,

$$\mathbb{E}[v \mid s \in \mathcal{S}] = \mu + \beta\big(\mathbb{E}[s \mid s \in \mathcal{S}] - \mu\big).$$

Applying Lemma 13 to $s \sim \mathcal{N}(\mu, \sigma_s^2)$ with the relevant truncation regions $\mathcal{S}$ yields the expressions for $\mu_E, \mu_H, \mu_Q, \mu_P$ in the main text. In particular:

- $\mu_E = \mathbb{E}[v \mid s < k_1]$ uses the left-tail formula with $\alpha = (k_1 - \mu)/\sigma_s$.
- $\mu_H = \mathbb{E}[v \mid k_1 \leq s < k_0]$ uses the interval formula with $\alpha = (k_1 - \mu)/\sigma_s$ and $\gamma = (k_0 - \mu)/\sigma_s$.
- $\mu_Q = \mathbb{E}[v \mid k_0 \leq s < k_D]$ uses the interval formula with $\alpha = (k_0 - \mu)/\sigma_s$ and $\gamma = (k_D - \mu)/\sigma_s$.
- $\mu_P = \mathbb{E}[v \mid s \geq k_D]$ uses the right-tail formula with $\alpha = (k_D - \mu)/\sigma_s$.

□

Derivation of $\mathbb{E}[v \mid X, D]$. For $D = 1$, only Public Voice is feasible, so $\mathbb{E}[v \mid X, 1] = \mu_P$ for all compatible $X \in \{0, 1, 2\}$.

For $D = 0$, Exit, Hold, and Quiet Voice are feasible. Since $X = q + z$ with z independent of (v, s), the conditional distribution of v given (X, D) is obtained by mixing the action-specific conditional means using Bayes weights proportional to the ex ante action probabilities and the likelihoods $\mathbb{P}(X \mid q)$. Note that Hold (H) and Quiet Voice (Q) both have $q = 0$ and hence generate identical order-flow distributions. Using Bayes' rule and the conditional means (μ_E, μ_H, μ_Q) gives:

- $X = -2$ implies Exit, hence $\mathbb{E}[v \mid -2, 0] = \mu_E$.
- $X = 1$ implies $q = 0$ and $z = +1$, so

$$\mathbb{E}[v \mid 1, 0] = \frac{\omega_H \mu_H + \omega_Q \mu_Q}{\omega_H + \omega_Q}.$$

- $X = -1$ mixes Exit ($z = 0$), Hold ($z = -1$), and Quiet Voice ($z = -1$). Writing the Bayes weights as

$$w_E = \frac{\omega_E(1-\kappa)}{\mathcal{D}_{-1}}, \quad w_H = \frac{\omega_H(\kappa/2)}{\mathcal{D}_{-1}}, \quad w_Q = \frac{\omega_Q(\kappa/2)}{\mathcal{D}_{-1}},$$

with $\mathcal{D}_{-1} = (\omega_H + \omega_Q)(\kappa/2) + \omega_E(1-\kappa)$, yields $\mathbb{E}[v \mid -1, 0] = w_E\mu_E + w_H\mu_H + w_Q\mu_Q$.
- $X = 0$ mixes Quiet Voice/Hold with $z = 0$ and Exit with $z = 1$. Writing the Bayes weights as

$$w_E = \frac{\omega_E(\kappa/2)}{\mathcal{D}_0}, \quad w_H = \frac{\omega_H(1-\kappa)}{\mathcal{D}_0}, \quad w_Q = \frac{\omega_Q(1-\kappa)}{\mathcal{D}_0},$$

with $\mathcal{D}_0 = (\omega_H + \omega_Q)(1-\kappa) + \omega_E(\kappa/2)$, yields $\mathbb{E}[v \mid 0, 0] = w_E\mu_E + w_H\mu_H + w_Q\mu_Q$.

□

B.5 Proof of Proposition 3

Proof. The market maker sets $P(X, D) = \delta\,\mathbb{E}[Y \mid X, D]$. Conditional on (X, D), the price $P(X, D)$ and the expected premium wedge $\bar{m}(X, D)$ are constants. The bid indicator $\mathbf{1}\{\text{bid}\}$ is a measurable function of $(P(X, D), D, \xi)$, and the bidder shock ξ is independent of (v, a) and of (q, z). Hence $\mathbf{1}\{\text{bid}\}$ is conditionally independent of (v, a) given (X, D), and $\mathbb{E}[\mathbf{1}\{\text{bid}\} \mid X, D] = p(X, D)$ with $p(X, D)$ understood as $p(P(X, D), D)$.

In the takeover payoff, the realized premium wedge is $m^R(a) = m_0 + a(\tilde{m} - m_0)$. Using conditional independence,

$$\mathbb{E}[(1 - \mathbf{1}\{\text{bid}\}) \cdot v \mid X, D] = (1 - p(X, D))\mathbb{E}[v \mid X, D],$$

and

$$\mathbb{E}[(1-\mathbf{1}\{\text{bid}\})\cdot a \mid X, D] = (1-p(X,D))\mathbb{E}[a \mid X, D] = (1-p(X,D))\pi(X,D).$$

Taking expectations of Y conditional on (X, D) yields

$$\mathbb{E}[Y \mid X, D] = p(X,D)\cdot(P(X,D)+\bar{m}(X,D)) + (1-p(X,D))\cdot\big(\mathbb{E}[v \mid X, D] + \tilde{\Delta}\pi(X,D)\big),$$

and multiplying by δ gives the stated decomposition. □

B.6 Proof of Bid Probability Monotonicity

Proof. Let

$$T(X,D) \equiv \frac{\bar{m}(X,D) + K - \bar{S} + P(X,D)}{\sigma_\xi}.$$

Then $p(X,D) = 1 - \Phi(T(X,D))$.

For the first claim (higher prices deter bids), holding (X, D) fixed and differentiating with respect to $P(X, D)$ yields

$$\frac{\partial p(X,D)}{\partial P(X,D)} = -\frac{1}{\sigma_\xi}\,\phi(T(X,D)) < 0,$$

since $\phi > 0$ and $\sigma_\xi > 0$.

For the second claim (higher inferred engagement deters bids when $\tilde{m} > m_0$), note that $\bar{m}(X,D) = m_0 + (\tilde{m} - m_0)\pi(X,D)$, so $\partial m/\partial\pi = \tilde{m} - m_0$. Hence

$$\frac{\partial p(X,D)}{\partial \pi(X,D)} = -\phi(T(X,D))\cdot\frac{\tilde{m}-m_0}{\sigma_\xi} < 0$$

whenever $\tilde{m} > m_0$. □

B.7 Proof of Within-Regime Liquidity Effects

Proof. Fix $(\omega_E, \omega_H, \omega_Q)$ and write $p_0 \equiv 1-\kappa$ and $p_1 \equiv \kappa/2$. For any reached nondisclosed information set (so denominators are nonzero), Proposition 2 gives:

$$\pi(1,0) = \frac{\omega_Q}{\omega_H+\omega_Q}, \qquad \pi(-1,0) = \frac{\omega_Q p_1}{(\omega_H+\omega_Q)p_1+\omega_E p_0}, \qquad \pi(0,0) = \frac{\omega_Q p_0}{(\omega_H+\omega_Q)p_0+\omega_E p_1}.$$

The first expression does not involve κ, hence $\partial\pi(1,0)/\partial\kappa = 0$.

For $X = -1$, differentiating with respect to κ yields

$$\frac{\partial\pi(-1,0)}{\partial\kappa} = \frac{(\omega_Q/2)\omega_E}{\big((\omega_H+\omega_Q)(\kappa/2)+\omega_E(1-\kappa)\big)^2} \geq 0,$$

with strict inequality whenever $\omega_E\omega_Q > 0$.

For $X = 0$, differentiating yields

$$\frac{\partial\pi(0,0)}{\partial\kappa} = -\frac{\omega_Q\omega_E/2}{\big((\omega_H+\omega_Q)(1-\kappa)+\omega_E(\kappa/2)\big)^2} \leq 0,$$

again strict whenever $\omega_E\omega_Q > 0$.

Finally, $\pi(-2,0) = 0$ and $\pi(X,1) = 1$ follow from Proposition 2. The claims for $\bar{m}(X,D) = m_0 + (\tilde{m}-m_0)\pi(X,D)$ follow because m is affine in π with slope $\tilde{m}-m_0 > 0$ under Assumption (A3). □

B.8 Proof of Takeover Comparative Statics

Proof. Recall the bid probability function

$$p(P,D) = 1 - \Phi\left(\frac{\bar{m}(P,D) + K - \bar{S} + P}{\sigma_\xi}\right).$$

Holding (P,D) fixed, $\partial p/\partial \bar{S} > 0$ and $\partial p/\partial K < 0$ follow immediately from $\Phi' = \phi > 0$.

For the price derivative at any reached information set (X,D), set $T(X,D) \equiv (\bar{m}(X,D) + K - \bar{S} + P(X,D))/\sigma_\xi$, so $p(X,D) = 1 - \Phi(T(X,D))$. Differentiating with respect to $P(X,D)$ yields

$$\frac{\partial p(X,D)}{\partial P(X,D)} = -\phi(T(X,D)) \cdot \frac{1}{\sigma_\xi} < 0.$$

Since ϕ is maximized at zero, $\sup_P |\partial p/\partial P| = \phi(0)/\sigma_\xi$. □

B.9 Proof of Equilibrium Cutoff Equations

Proof. Define the action-specific expected payoffs by evaluating $U(q, a \mid s)$ for the four actions in Proposition 1:

$$\begin{aligned} U_E &\equiv U(-1,0 \mid s), \quad U_H(s) \equiv U(0,0 \mid s), \quad U_Q(s) \equiv U(0,1 \mid s), \\ U_P(s) &\equiv U(+1,1 \mid s). \end{aligned}$$

In a cutoff strategy equilibrium with thresholds $k_1 < k_0 < k_D$, the blockholder is indifferent between adjacent actions at the boundary signals where the action switches. The three indifference conditions are:

- At $s = k_1$: Exit vs. Hold, i.e., $U_E = U_H$.
- At $s = k_0$: Hold vs. Quiet Voice, i.e., $U_H = U_Q(k_0)$.
- At $s = k_D$: Quiet Voice vs. Public Voice, i.e., $U_Q(k_D) = U_P(k_D)$.

These are exactly the conditions (4)–(6). □

B.10 Proof of Minority Takeover Gains Decomposition

Proof. By definition,

$$\Delta^{\min}(\kappa) = \mathbb{E}\big[m^R(a) \cdot \mathbf{1}\{\text{bid}\}\big].$$

Conditional on (X,D), the bid indicator $\mathbf{1}\{\text{bid}\}$ is a measurable function of $(P(X,D), D, \xi)$, and ξ is independent of (v,a). Hence $\mathbf{1}\{\text{bid}\}$ is conditionally independent of (v,a) given (X,D), so

$$\mathbb{E}\big[m^R(a) \cdot \mathbf{1}\{\text{bid}\} \mid X, D\big] = \mathbb{E}[m^R(a) \mid X,D] \cdot \mathbb{E}[\mathbf{1}\{\text{bid}\} \mid X,D] = \bar{m}(X,D) \cdot p(X,D).$$

Taking expectations over (X, D) yields $\Delta^{\min}(\kappa) = \mathbb{E}[\bar{m}(X, D) \cdot \mathbf{1}\{\text{bid}\}]$.

Using $\bar{m}(X, D) = m_0 + (\tilde{m} - m_0)\pi(X, D)$ and linearity of expectation,

$$\begin{aligned} \Delta^{\min}(\kappa) &= \mathbb{E}\big[(m_0 + (\tilde{m} - m_0)\pi(X, D)) \cdot \mathbf{1}\{\text{bid}\}\big] \\ &= m_0 \cdot \mathbb{E}[\mathbf{1}\{\text{bid}\}] + (\tilde{m} - m_0) \cdot \mathbb{E}\big[\pi(X, D) \cdot \mathbf{1}\{\text{bid}\}\big], \end{aligned}$$

which is the claimed decomposition with $\mathbb{P}(\text{bid}) = \mathbb{E}[\mathbf{1}\{\text{bid}\}]$. □

B.11 Existence: self-map on the cutoff polytope

Existence is proved in the main text by exhibiting a Brouwer self-map on the cutoff polytope Θ of (7): Lemma 2 brackets every best-response cutoff in a fixed interval $[\underline{s}, \bar{s}]$ independent of the conjecture, and Lemma 3 shows the best-response map T is continuous and maps Θ into itself (ordering preservation follows from the slope ordering $B(E) < B(H) = B(Q) < B(P)$ established in Lemmas 4 and 5). Brouwer's theorem then yields the fixed point of Proposition 4. The fixed point may lie on a collapse face, recovering the baseline Hold collapse.

B.12 Uniqueness: numerical regularity

Global uniqueness is not claimed as a theorem; see Remark 3. Assumption (A6) (contraction of T) would deliver it, but $\|DT\|_\infty < 1$ cannot be derived from primitives without further structure on the cross-cutoff Jacobian, an obstacle common to discrete order-flow models with price feedback. Following the methodological precedent of Edmans, Goldstein, et al. (2015), we verify uniqueness numerically: iterating T from arbitrary ordered starting points in Θ converges to a single fixed point in every exercise reported in this paper, and a 30-seed multistart at $\kappa \in \{0.2, 0.5, 0.8\}$ converges in all cases to one fixed point per κ, confirming that the spectral radius is strictly below one at the calibration.

B.13 Proof of the corrected minority-gain results (Lemmas 7, 8; Proposition 5): the realized $D = 0$ law, posterior variance, and curvature of h

Fix the action masses $(\omega_E, \omega_H, \omega_Q, \omega_P)$. Disclosure is stake-triggered, $D = \mathbf{1}\{q = +1\}$, so $\{D = 0\} = \{\text{Exit, Hold, Quiet Voice}\}$. Write

$$A := \omega_E, \qquad B := \omega_H + \omega_Q, \qquad Q := \omega_Q, \qquad \bar{\pi} := \frac{\omega_Q}{\omega_H + \omega_Q} = \frac{Q}{B},$$

and, with $p_0 := 1 - \kappa$, $p_1 := \kappa/2$,

$$T_- := A\,p_0 + B\,p_1, \qquad T_0 := A\,p_1 + B\,p_0.$$

Enumerating the four $D = 0$ order-flow cells (consistent with Proposition 2) gives Table 3.

Tower identity. Summing the realized atoms,

$$\mathbb{E}\big[\pi\,\mathbf{1}\{D{=}0\}\big] = \sum_X \mathbb{P}(X, D{=}0)\,\pi(X, 0) = \mathbb{P}(\text{Quiet Voice}, D{=}0) = \omega_Q = Q, \tag{44}$$

X	contributing (q,z) (mass)	pooled mass at X	$\pi(X,0)$
-2	$(-1,-1)$: Ap_1	Ap_1	0
-1	$(-1,0)$: Ap_0; $(0,-1)$: Bp_1	T_-	$\dfrac{Q\,p_1}{T_-}$
0	$(-1,+1)$: Ap_1; $(0,0)$: Bp_0	T_0	$\dfrac{Q\,p_0}{T_0}$
$+1$	$(0,+1)$: Bp_1	Bp_1	$\dfrac{Q}{B}=\bar{\pi}$

Table 3: The no-disclosure ($D=0$) order-flow cells, pooled mass, and realized engagement posterior $\pi(X,0)$. The realized law is supported on at most four points $\{0,\pi(-1,0),\pi(0,0),\bar{\pi}\}$; the ends $0,\bar{\pi}$ are the chord endpoints, $\pi(-1,0),\pi(0,0)$ the interior atoms. Only $\{0,\bar{\pi}\}$ survive the $\kappa\to 0^+,1^-$ limits, which underlies Lemma 7.

exact for every κ (Quiet-Voice types never disclose), and $\mathbb{P}(D{=}0)=A(2p_1{+}p_0){+}B(2p_1{+}p_0)=A{+}B$ since $2p_1+p_0=1$. Hence the conditional mean $\mathbb{E}[\pi\mid D{=}0]=Q/(A+B)$ is κ-invariant *at fixed masses*; in equilibrium the masses move with κ, so the conditional mean drifts while the unnormalized moment (44) stays fixed.

Lemma 14 (Closed-form posterior variance on $\{D=0\}$: strictly U-shaped)**.** *Fix the masses (A,B,Q) with $A,B,Q>0$, $Q\le B$. Conditional on $\{D=0\}$ and at fixed masses, the second moment is*

$$\mathbb{E}\big[\pi^2\mathbf{1}\{D{=}0\}\big]=\frac{Q^2p_1^2}{T_-}+\frac{Q^2p_0^2}{T_0}+\frac{Q^2p_1}{B},$$

and, writing $f(\kappa):=\mathbb{E}[\pi^2\mathbf{1}\{D{=}0\}]-\frac{Q^2}{B}$,

$$f(\kappa)\;=\;-\,\frac{A\,Q^2\,(A+B)\,\kappa\,(2-\kappa)\,(1-\kappa)}{4\,B\,T_-\,T_0}\;<\;0\quad \textit{on } (0,1),\qquad f(0)=f(1)=0. \tag{45}$$

In the variable $x:=\kappa/2\in[0,\frac{1}{2}]$, with $g(x):=\mathbb{E}[\pi^2\mathbf{1}\{D{=}0\}]/Q^2$, the second moment is strictly convex,

$$g''(x)\;=\;\frac{2A^2}{T_-(x)^3}+\frac{2A^2}{T_0(x)^3}\;>\;0, \tag{46}$$

so the conditional variance $\mathrm{Var}[\pi\mid D{=}0]$ *(an increasing-affine image of g, since the mean is κ-invariant) is strictly convex with* $\mathrm{Var}(0)=\mathrm{Var}(1)$ *and a* unique *interior minimizer; its endpoint slopes are $V'(0)=-Q^2/(2B^2)<0$ and $V'(1)=+Q^2/B^2>0$.*

Proof. The second moment is $\sum_X\mathbb{P}(X,D{=}0)\,\pi(X,0)^2$: the $X{=}-2$ term vanishes and the other three give the stated sum. At $\kappa=0$ ($p_1=0$) and $\kappa=1$ ($p_0=0$) both reduce to Q^2/B, so $f(0)=f(1)=0$; placing the three terms over the common denominator $4BT_-T_0$ and reducing yields (45), every factor of which is signed on $(0,1)$, so $f<0$ strictly there. For convexity, substitute $x=\kappa/2$, so $T_-=A+(B-2A)x$ and $T_0=B+(A-2B)x$ are affine; the elementary identity $\frac{d^2}{dx^2}[x^2/T]=2\alpha^2/T^3$ for $T=\alpha+\beta x$ (here $\alpha=A$) applied termwise gives (46), strictly positive since $A>0$, $T_-,T_0>0$. A strictly convex function with equal endpoints has a unique interior minimizer; the endpoint slopes follow from $f'(0),f'(1)$ in (45). (For $A=B$ the minimizer is $\kappa = 2 - \sqrt{2} \approx 0.586$, matching the equilibrium hump peak $\kappa^\dagger \approx 0.59$.) The qualifier "at fixed masses" is essential: in equilibrium the cutoffs, hence (A, B, Q), move with κ, and the moving-mass variance need not be monotone on each side. $\square$

Lemma 15 (Curvature of $h = \pi p$ and the condition (C**))**.** *With* $T(\pi) = (\bar{m}(\pi) + K - \bar{S} + P(\pi))/\sigma_\xi$, $p(\pi) = 1 - \Phi(T(\pi))$, *and the price locally affine on the realized hull* ($P'' \approx 0$, *so* $T'' \approx 0$),

$$h''(\pi) = 2p'(\pi) + \pi\, p''(\pi) = \phi\big(T(\pi)\big)\, T'(\pi) \left[\, \pi\, T(\pi)\, T'(\pi) - 2\,\right]. \tag{47}$$

Since $\phi(T) > 0$ *and* $T' > 0$, h *is concave on all of* $[0, \bar{\pi}]$ *((C**)) iff* $\max_{\pi \in [0,\bar{\pi}]} \pi\, T(\pi)\, T'(\pi) \leq 2$, *with slack* $\mathrm{slack} := 2 - \max_{[0,\bar{\pi}]} \pi T T'$; (C**) *reversed (h convex) iff* $\pi T T' \geq 2$ *on* $[0, \bar{\pi}]$. *Full-interval concavity implies the chord condition* $\mathcal{C}(\bar{\pi}) < 0$ *of* (19)*, which is the operative diagnostic for the two-point realized law. At the baseline,* $\max_{[0,\bar{\pi}]} \pi T T' = 0.42$ *and* $\mathrm{slack} = +1.58 > 0$, *so* (C**) *holds with strict slack (the slack is conservative: the true feedback* $P'(\pi) \geq 0$ *only lowers* $\pi T T'$*).*

Proof. $h = \pi p \Rightarrow h'' = 2p' + \pi p''$; with $p' = -\phi(T) T'$ and $p'' = \phi(T)[T T'^2 - T'']$ and $T'' = 0$, $h'' = \phi(T) T' [\pi T T' - 2]$, giving the sign equivalence. The Jensen bridge of Lemma 8 compares h to its chord at the realized atoms; full-interval concavity makes the chord defect $h - \ell \geq 0$ at every atom, so $\mathbb{E}_\kappa[h]$ exceeds the common endpoint value for every interior κ (an interior global maximum of channel (A)). The baseline numbers are computed from the model's own price fixed point. $\square$

B.14 Proof of the general-equilibrium transfer (Hypothesis 1): the cutoff map is C^1 and the residual is computably bounded

Lemma 16 (C^1 equilibrium map and a computable bound on the GE residual)**.** *Under Assumption* (A6) *(the equilibrium map is a local contraction with invertible Jacobian) the cutoff map* $\kappa \mapsto k(\kappa) = (k_1, k_0, k_D)(\kappa)$ *is* C^1 *on the interior, by the implicit function theorem applied to the equilibrium system* $F(k, \kappa) = 0$ *with* $\det D_k F \neq 0$. *Writing* $\Delta^{\min}(\kappa) = \Phi\big(k(\kappa), \kappa\big)$,

$$\frac{\mathrm{d}\Delta^{\min}}{\mathrm{d}\kappa} = \underbrace{\frac{\partial \Phi}{\partial \kappa}}_{E'(\kappa)\ \textit{(channel A)}} + \underbrace{\sum_i \frac{\partial \Phi}{\partial k_i} \frac{\mathrm{d}k_i}{\mathrm{d}\kappa}}_{B(\kappa)\ \textit{(channel B)}}, \qquad \frac{\mathrm{d}k}{\mathrm{d}\kappa} = -\,(D_k F)^{-1}\, \partial_\kappa F, \tag{48}$$

and $B(\kappa)$ *is bounded by* $|B(\kappa)| \leq \big(\sum_i |\partial\Phi/\partial k_i|\big)\, \|\mathrm{d}k/\mathrm{d}\kappa\|_\infty$, *each factor an explicit equilibrium object. If, on a sub-interval* $[\underline{\kappa}, \overline{\kappa}] \ni \kappa^\dagger$ *with channel-(A) amplitude* $A^* = \max_{[\underline{\kappa},\overline{\kappa}]} E - \frac{1}{2}(E(\underline{\kappa}) + E(\overline{\kappa}))$, $\int_{\underline{\kappa}}^{\kappa^\dagger} |B| < A^*$ *and* $\int_{\kappa^\dagger}^{\overline{\kappa}} |B| < A^*$, *then* $\Delta^{\min}$ *inherits an interior maximum on* $[\underline{\kappa}, \overline{\kappa}]$.

Proof. C^1-ness and (48) are the chain rule plus the IFT under Assumption (A6); the bound is the triangle inequality with the IFT comparative static. Integrating (48) from $\underline{\kappa}$ to $\kappa \leq \kappa^\dagger$ gives $\Delta^{\min}(\kappa) - \Delta^{\min}(\underline{\kappa}) \geq (E(\kappa) - E(\underline{\kappa})) - \int_{\underline{\kappa}}^{\kappa} |B|$; at $\kappa^\dagger$ the first term is $\geq A^*$ while the residual integral is $< A^*$, so $\Delta^{\min}(\kappa^\dagger) > \Delta^{\min}(\underline{\kappa})$, and symmetrically $> \Delta^{\min}(\overline{\kappa})$; the maximum is interior. $\square$

Remark 5 (Status of the GE step)**.** *That the GE channel is genuinely first-order is confirmed numerically: at the peak* $\mathrm{d}k/\mathrm{d}\kappa \approx (-0.0599, -0.0599, +0.0982)$ *for* (k_1, k_0, k_D), *so* $B \not\equiv 0$ *and the envelope caveat is real. The sub-interval transfer inequality is verified, not assumed, on the region where the script reports the residual integral below the channel-(A) amplitude; there the* *equilibrium hump is a theorem. Establishing it on the* entire *interior is the numerically verified regularity behind Hypothesis 1; it is not claimed analytically everywhere, because near the exact endpoints both E' and B blow up (grid-edge degeneracy). Equilibrium uniqueness is the contraction property of Assumption* (A6)*, verified numerically (multistart solves return a single fixed point at every tested κ), not derived in closed form. Appendix F upgrades this lemma's bound: the quantified contraction modulus yields an inversion-free bound on $B(\kappa)$ requiring no path derivative, a certified single-peakedness region at the baseline, and a certified counterexample showing the hypothesis fails globally.*

B.15 Derivations for Section 8

(ND) No disclosure. Under the no-disclosure regime $S_{\mathrm{ND}} = \emptyset$, the market conditions only on $X = q + z$ with $\mathbb{P}(z = 0) = p_0 = 1 - \kappa$ and $\mathbb{P}(z = \pm 1) = p_1 = \kappa/2$. Writing $\mathbb{P}(X \mid q)$ for the order-flow likelihood induced by z, Bayes' rule implies

$$\pi_{\mathrm{ND}}(X) \equiv \mathbb{P}(a = 1 \mid X) = \frac{\omega_Q \mathbb{P}(X \mid q = 0) + \omega_P \mathbb{P}(X \mid q = +1)}{\omega_E \mathbb{P}(X \mid q = -1) + (\omega_H + \omega_Q)\mathbb{P}(X \mid q = 0) + \omega_P \mathbb{P}(X \mid q = +1)}.$$

Substituting $\mathbb{P}(X = q) = p_0$ and $\mathbb{P}(X = q \pm 1) = p_1$ (and zero otherwise) yields the closed-form expressions reported in Section 8.

(NR) Baseline stake disclosure plus a noisy rumor. Under $S_{\mathrm{NR}} = (D, R)$ with $D = \mathbf{1}\{q = +1\}$, the disclosed branch $D = 1$ pins down Public Voice in the baseline action set, so $\pi(X, 1, R) = 1$ for all feasible X. When $D = 0$, $q \in \{-1, 0\}$ and the rumor has hit/false-alarm rates $\mathbb{P}(R = 1 \mid q = 0, a = 1) = \eta_1$ and $\mathbb{P}(R = 1 \mid q = 0, a = 0) = \eta_0$, with the convention $\mathbb{P}(R = 1 \mid q = -1) = \eta_0$. Bayes' rule gives, for each feasible (X, R),

$$\begin{aligned}
\pi(X, 0, R) &= \mathbb{P}(Q \mid X, 0, R) \\
&= \frac{\omega_Q\, \mathbb{P}(X \mid q = 0)\mathbb{P}(R \mid Q)}{\omega_E\, \mathbb{P}(X \mid q = -1)\mathbb{P}(R \mid E) + \mathbb{P}(X \mid q = 0)\big(\omega_H\, \mathbb{P}(R \mid H) + \omega_Q\, \mathbb{P}(R \mid Q)\big)}.
\end{aligned}$$

Evaluating $\mathbb{P}(X \mid q)$ using $X = q + z$ yields the cases in Section 8: $X = -2$ is degenerate (Exit only); $X = 1$ pins down $q = 0$ so κ drops out; and $X \in \{-1, 0\}$ mixes $q = -1$ and $q = 0$ with weights proportional to (p_0, p_1), generating the displayed formulas.

C A bargaining reading of the control premia under an explicit threat-point condition

This appendix gives a generalized-Nash-bargaining reading of the bid-conditional control premia (m_0, m_1) used in the body. We are deliberate about the *epistemic status* of every claim, because the load-bearing object — the engagement-induced shift in the seller bloc's disagreement payoff — cannot be honestly derived from deeper primitives within the scope of this paper.

Honest scope statement (updated). A full microfoundation must derive the threat-point shift $d(1) - d(0) > 0$ (and hence Assumption (A3), $\widetilde{m} > m_0$) from primitives. Within this appendix we do not do that: the sign and magnitude of the shift are governed by a pivotality/appropriability coefficient λ that can only be pinned by a complete tender-game equilibrium. **That tender game is now solved in Appendix D**, which derives $\lambda = 1 - q(1-\gamma)\psi$ from the disagreement-node environment and characterizes exactly when Condition 1 holds (Theorem 2). This appendix therefore retains its original role as the *bargaining layer* — everything proved here conditional on Condition 1 now operates on a derived premise — and its honest-scope discipline applies one level down, to the institutional primitives of the tender environment listed in §D.8. Following the strategy of Edmans, Goldstein, et al. (2015) and the corporate-finance tradition of stating results under transparent primitive conditions, we instead: (i) *state* the threat-point shift as an explicit, economically interpretable condition (Condition 1); (ii) *motivate* that condition with a free-rider tender heuristic in the spirit of Grossman and Hart (1980), flagged honestly as motivation rather than proof; (iii) prove what genuinely follows from the bargaining protocol *given* the condition; and (iv) carry the one remaining numerical claim (the single-peak comparative static) as a *numerically verified regularity* at the baseline calibration, clearly labelled as such.

Two distinct primitives, kept distinct. The body uses two separate engagement primitives: the *standalone* improvement Δ (baseline $\Delta = 0.25$), which enters only through $\widetilde{\Delta} = \rho\Delta$ on the no-takeover branch, and the *premium wedge* $m_1 - m_0$ (baseline $m_1 - m_0 = 0.30 - 0.10 = 0.20$), which is its own primitive. The bargaining reading introduces an engagement value Δ_{eng} that shifts the target bloc's *sale* threat point and so drives the wedge; conceptually Δ_{eng} is the engagement improvement that survives into a sale, and it need not numerically equal the body's standalone Δ.

C.1 Environment and the bargaining problem

Condition on the takeover reaching the bargaining table, i.e. on the bidder finding it worthwhile to bid (the bid rule $\Pi_B = \bar{S} - P + \xi - \bar{m} - K \geq 0$ is taken *as given* from the body). Let $a \in \{0, 1\}$ index whether engagement has *succeeded* ($a = 1$) or not ($a = 0$) by the time control is contested. Two parties bargain over the gains from transferring control: the target's incumbent control bloc (Nash weight $\theta \in (0, 1)$) and the bidder (weight $1 - \theta$). The dispersed minority float free-rides at the negotiated price and does not sit at the bargaining table.

The surplus, and why the price P does not enter it. If a deal is struck, the value created over and above the parties' disagreement values is the takeover pie

$$G(a, \xi) \; = \; \bar{S} + \xi - K - d(a), \tag{49}$$

where $\bar{S} + \xi$ is the gross valuation the bidder realizes, K is the body's bid cost, and $d(a)$ is the target bloc's disagreement payoff. The body's bidder surplus $\Pi_B = \bar{S} - P + \xi - \bar{m} - K$ contains the market price P, whereas (49) contains *no* P; the two are consistent. In the body the bidder pays P to the *market* — a transfer to the dispersed minority float, who free-ride at the negotiated price — and pays the premium to the *bloc*. The pie (49) is split only between the bidder and the *bloc*; P is a transfer to a third party (the float) and is orthogonal to the bidder–bloc split. Hence P never enters G, and there is no inconsistency between the body's Π_B and the appendix's bidder net gain.

The disagreement payoff: stated as a condition. We use

$$d(a) \;=\; w + a\,\lambda\,\widetilde{\Delta}_{\text{eng}}, \qquad \widetilde{\Delta}_{\text{eng}} := \rho\,\Delta_{\text{eng}}, \qquad \lambda \in (0,1], \tag{50}$$

where w is the target's standalone continuation value absent engagement, $\widetilde{\Delta}_{\text{eng}} = \rho\Delta_{\text{eng}}$ is the expected engagement improvement reaching the sale, and λ is the fraction of that improvement the bloc can appropriate at the disagreement node. Equation (50) is the single load-bearing modeling object. **We adopt it as an explicit structural condition, not as a derived result** (Condition 1); the free-rider heuristic of §C.2 motivates its sign but does not constitute a proof of $\lambda > 0$ from deeper primitives.

Condition 1 (Appropriable engagement threat-point shift)**.** *At the bargaining table the target bloc's disagreement (no-sale-to-this-bidder) payoff takes the form* (50) *with a strictly positive appropriability coefficient* $\lambda \in (0,1]$ *and a strictly positive expected engagement improvement* $\widetilde{\Delta}_{\text{eng}} = \rho\Delta_{\text{eng}} > 0$*. That is, a fraction* λ *of the engagement improvement that survives to a sale is captured by the bloc as an outside option, so the engagement-induced shift in the bloc's threat point is* $d(1) - d(0) = \lambda\,\widetilde{\Delta}_{\text{eng}} > 0$*.*

The Nash split and the realized minority premium. The bidder's disagreement value is its no-deal payoff 0, so the bidder's net gain is $(1-\theta)G(a,\xi)$ and the bloc captures $d(a) + \theta G(a,\xi)$. The control premium paid is the bloc's bargained rent net of the common baseline w:

$$m^R(a,\xi) \;=\; \big[d(a) + \theta\,G(a,\xi)\big] - w \;=\; a\,\lambda\,\widetilde{\Delta}_{\text{eng}} + \theta\big(\bar{S} + \xi - K - w - a\,\lambda\,\widetilde{\Delta}_{\text{eng}}\big). \tag{51}$$

Collecting terms,

$$m^R(a,\xi) \;=\; \theta\,(\bar{S} - K - w) \;+\; \theta\,\xi \;+\; a\,(1-\theta)\,\lambda\,\widetilde{\Delta}_{\text{eng}}. \tag{52}$$

Lemma 17 (Closed-form Nash rent and participation)**.** Given *Condition 1, on the consummation set* $\{G(a,\xi) \ge 0\} = \{\xi \ge \xi^*(a)\}$ *with* $\xi^*(a) = d(a) + K - \bar{S}$*, the generalized-Nash split gives the bloc the rent* (51) *and leaves the bidder* $\Pi_A = (1-\theta)G(a,\xi) \ge 0$*. The bloc's individual rationality (*$m^R(a,\xi) \ge d(a) - w$*, with equality only at* $\xi = \xi^*(a)$*) and the bidder's incentive compatibility both hold, so* $\{\xi \ge \xi^*(a)\}$ *is the equilibrium agreement region.*

Proof. The maximand $N(x) = x^\theta (G-x)^{1-\theta}$ on $x \in [0,G]$ is strictly log-concave with first-order condition $\theta/x = (1-\theta)/(G-x)$, i.e. $x^* = \theta G$; adding back $d(a)$ gives bloc payoff $d(a) + \theta G$ and the bidder $(1-\theta)G$. On $\{G \ge 0\}$, $\theta G \ge 0$ (zero only at $G = 0$) and $(1-\theta)G \ge 0$, giving IR and participation. A rent below $d(a)$ violates the bloc's IR; offering exactly $d(a)$ is the $\theta = 0$ corner, excluded by $\theta > 0$, so the unique split is interior. The boundary $\xi^*(a)$ solves $G = 0$. (Numerically: the argmax matches θG to 6.0×10^{-10}, and the IR/IC checks register zero violations over simulated draws.) □

C.2 Motivating the threat-point condition: a free-rider tender heuristic

We now give the economic *motivation* for Condition 1. This subsection is a heuristic that makes the sign of the threat-point shift plausible; **it is not a derivation of $\lambda > 0$ from primitives**, and we do not present it as one (Remark 6 states precisely why).

Why "no sale" is not a literal status quo. "No sale to this bidder" is not a literal status quo w for the engaging bloc; it is a continuation value, which is the bloc's outside option in the sense of Binmore et al. (1986). Suppose bargaining with the incumbent bidder breaks down. At the disagreement node the target is a free-standing firm whose fundamentals have been improved by engagement to $w + a\,\Delta^*$, where Δ^* is the realized fundamental improvement at the disagreement node. A competitive fringe of potential acquirers may then mount a takeover. By the free-rider logic of Grossman and Hart (1980), atomistic shareholders tender only at a price equal to their post-takeover per-share value, so a value-improving fringe raider captures essentially none of the improvement; the dispersed float retains it. The control bloc, holding a pivotal block, is *not* a price-taker in the tender and retains its pro-rata claim on the improved fundamentals. This makes it *plausible* that the bloc's continuation value per share is

$$d(a) \;\approx\; w \;+\; a\,\Lambda\,\Delta^*, \tag{53}$$

with $\Lambda \in (0,1]$ the fraction of the fundamental improvement that survives to the bloc at the disagreement node. Writing $\lambda\,\widetilde{\Delta}_{\mathrm{eng}} := \Lambda\,\Delta^*$ identifies the coefficient of Condition 1. The heuristic delivers the *sign* ($\Lambda > 0$ whenever the bloc is pivotal and the float will not tender below value); it does *not* pin Λ, which depends on block size, tender mechanics, and the fringe's composition.

Remark 6 (Honest status: this is motivation, not a closed derivation)**.** *Equation* (53) *is a heuristic and Condition 1 is an* assumption*. A genuine derivation would solve a complete tender-game equilibrium and produce* Λ *(equivalently* λ*) as a function of primitives (block size, reservation values, the fringe's bidding protocol); that game is orthogonal to this paper's disclosure mechanism and we do not solve it. We therefore treat* $\lambda \in (0,1]$ *as a reduced-form pivotality parameter and carry it explicitly everywhere below. The substantive interpretive content — engagement enters the bloc's threat point with a strictly positive, appropriable coefficient, and a recognisable free-rider mechanism makes that sign plausible — is real; the claim that this sign is* proved *from primitives is one we do not make. The free-rider mechanism is from Grossman and Hart (1980); the outside-option reading of the threat point is from Binmore et al. (1986).*

C.3 The wedge is ξ-free; the levels are not

The engagement *wedge* — the premium object the body's κ-comparative statics use — is the within-state (common-ξ) difference of (52):

$$m^R(1,\xi) - m^R(0,\xi) \;=\; (1-\theta)\,\lambda\,\widetilde{\Delta}_{\mathrm{eng}} \;=\; (1-\theta)\,\lambda\,\rho\,\Delta_{\mathrm{eng}}. \tag{54}$$

The $\theta\,\xi$ term in (52) is common to both states and cancels; the bid cost K and baseline w cancel for the same reason. Hence the wedge is exactly ξ-free, K-free, and w-free, while the *level* $m^R(a,\xi)$ is manifestly ξ-dependent through $\theta\,\xi$.

C.4 Reduced form, folding, and the ρ non-identification

The body posits primitives (m_0, m_1) and sets $\widetilde{m} = m_0 + \rho(m_1 - m_0)$, so $\widetilde{m} - m_0 = \rho(m_1 - m_0)$. The bargaining reading pins the *wedge* via (54); the body's single-ρ fold then reads off $\widetilde{m}$ with the *same* (m_0, m_1):

$$m_1 - m_0 \;:=\; m^R(1) - m^R(0) \;=\; (1-\theta)\,\lambda\,\rho\,\Delta_{\text{eng}}, \qquad \widetilde{m} - m_0 \;=\; \rho\,(m_1 - m_0) \;=\; \rho^2(1-\theta)\,\lambda\,\Delta_{\text{eng}}. \tag{55}$$

The ρ^2 is justified by an explicit event tree (Lemma 18): one ρ is the success probability of the engagement *campaign* acting on the engagement improvement (inside $\widetilde{\Delta}_{\text{eng}} = \rho\Delta_{\text{eng}}$), the second is the body's own ρ in the fold (the premium-state arrival acting on the wedge); the two attach to *distinct independent* Bernoulli draws, so $\rho^2 = \mathbb{E}[B_1 B_2] \neq \mathbb{E}[B_1^2] = \rho$.

Proposition 11 (Reduced form is the constant-θ special case)**.** *Fix $\theta \in (0,1)$ constant and $\lambda \in (0,1]$, and assume Condition 1. Then the bargained premia* (54)–(55) *reproduce the body's reduced form: there exist (m_0, m_1) with $m_1 - m_0 = (1-\theta)\lambda\rho\Delta_{\text{eng}}$ and $\widetilde{m} = m_0 + \rho(m_1 - m_0)$. Conversely, any body calibration (m_0, m_1, ρ) with $m_1 > m_0$ is realized by a one-parameter family of primitives $\{(\theta, \lambda, \Delta_{\text{eng}}) : (1-\theta)\lambda\rho\Delta_{\text{eng}} = m_1 - m_0,\ \theta \in (0,1),\ \lambda \in (0,1]\}$. The body verbatim is the case $\lambda = 1$ with state-independent θ.*

Proof. Constancy of θ in a makes the wedge $(1-\theta)\lambda\rho\Delta_{\text{eng}}$ a single number $m_1 - m_0$; the body's fold defines $\widetilde{m}$. For the converse, given $(m_1 - m_0, \rho)$ and any $\lambda \in (0,1]$, pick any $\Delta_{\text{eng}} > (m_1 - m_0)/(\lambda\rho)$ and set $\theta = 1 - (m_1 - m_0)/(\lambda\rho\Delta_{\text{eng}}) \in (0,1)$; then $(1-\theta)\lambda\rho\Delta_{\text{eng}} = m_1 - m_0$. □

Remark 7 (Identification: only the wedge *product* is pinned)**.** *Proposition 11 is also a* non-identification statement*. The body's calibration pins the product $(1-\theta)\lambda\rho\Delta_{\text{eng}} = m_1 - m_0$ but not its factors: the bargaining weight θ is* set-identified*, not point-identified, given the wedge and ρ, and the structural comparative static in θ (Remark 9) is therefore conditional on an external pin for θ. An identifying restriction — e.g. an observable target-resistance moment such as the empirical pass rate of management defenses — would restore point identification; ρ is likewise not separately identified, so the body's $\rho = 0.9$ is a normalization on the locus, not an over-identifying restriction.*

Lemma 18 (The two ρ's attach to distinct draws)**.** *Let engagement success be $B_1 \sim \text{Bern}(\rho)$ (realizing the improvement Δ_{eng}, so $\widetilde{\Delta}_{\text{eng}} = \mathbb{E}[B_1]\Delta_{\text{eng}} = \rho\Delta_{\text{eng}}$) and let premium-state arrival be $B_2 \sim \text{Bern}(\rho)$, independent of B_1. Then the ex-ante engagement contribution to the premium is*

$$\mathbb{E}\big[(m^R(B_1,\cdot) - m_0)\cdot \mathbf{1}\{B_2\}\big] \;=\; \underbrace{\mathbb{P}(B_2{=}1)}_{=\rho}\cdot\underbrace{(1-\theta)\lambda\,\mathbb{E}[B_1]\,\Delta_{\text{eng}}}_{=(1-\theta)\lambda\rho\Delta_{\text{eng}}} \;=\; \rho^2(1-\theta)\lambda\,\Delta_{\text{eng}},$$

so the two factors of ρ are $\mathbb{P}(B_2{=}1)$ and $\mathbb{E}[B_1]$ — two distinct Bernoulli draws, not a repeated deflation of one event.

Proof. The wedge realized when the campaign succeeds is $m_1 - m_0 = (1-\theta)\lambda\widetilde{\Delta}_{\text{eng}}$, but $\widetilde{\Delta}_{\text{eng}}$ already integrates out B_1: $\widetilde{\Delta}_{\text{eng}} = \mathbb{E}[B_1]\Delta_{\text{eng}} = \rho\Delta_{\text{eng}}$. Multiplying by the fold probability $\mathbb{P}(B_2{=}1) = \rho$ (independent of B_1) gives the displayed product. Each ρ is the mean of a separate, independent Bernoulli, so the square is the product of two distinct expectations, not $\mathbb{E}[B^2]$ for a single B (indeed $\mathbb{E}[B_1 B_2] = \rho^2 \neq \mathbb{E}[B_1^2] = \rho$). □

C.5 Assumption A3 as a conditional theorem

Theorem 1 (A3 from bargaining, conditional on Condition 1)**.** Assume Condition 1. *Suppose further that the bloc is not a monopolist over the surplus ($\theta < 1$). Then the bargained premia satisfy*

$$\widetilde{m} - m_0 \;=\; \rho^2(1-\theta)\,\lambda\,\Delta_{\text{eng}} \;>\; 0, \qquad \textit{equivalently} \qquad \widetilde{m} > m_0.$$

Thus standing Assumption (A3) *holds as a* consequence *of bargaining* given *the structural Condition 1. The condition is sharp: A3 fails iff $\theta = 1$ (the bloc captures the entire pie), $\lambda = 0$ (engagement is unappropriable at the threat point), or $\rho\Delta_{\text{eng}} = 0$ (engagement is worthless).*

Proof. By (55), $\widetilde{m} - m_0 = \rho^2(1-\theta)\lambda\Delta_{\text{eng}}$. Under Condition 1 ($\lambda > 0$, $\rho\Delta_{\text{eng}} > 0$) and $\theta < 1$ each factor is strictly positive, so the product is strictly positive and $\widetilde{m} > m_0$. Sharpness: if $\theta = 1$ then $1-\theta = 0$ and the wedge (54) vanishes; if $\lambda = 0$ or $\rho\Delta_{\text{eng}} = 0$ then $\lambda\widetilde{\Delta}_{\text{eng}} = 0$ and again the wedge vanishes. □

Remark 8 (What the conditional theorem rests on)**.** *Theorem 1 is a theorem* conditional on *Condition 1; it is* not *an unconditional theorem. The economic content of A3 has moved from the reduced-form assertion "$m_1 > m_0$" to the structural condition "$d(1) - d(0) = \lambda\widetilde{\Delta}_{\text{eng}} > 0$," which the free-rider heuristic of §C.2 makes plausible but does not prove from primitives. The* sign *of the wedge holds for any $\lambda > 0$; the* magnitude *scales with λ, so any quantitative match holds λ fixed.*

Reproducing the body's numbers. The body's baseline has $m_0 = 0.10$, $m_1 = 0.30$, $\rho = 0.9$ — hence wedge $m_1 - m_0 = 0.20$ and, by the body's fold, $\widetilde{m} - m_0 = \rho(m_1 - m_0) = 0.18$ (so $\widetilde{m} = 0.28$). Three reconciliations all reproduce the wedge 0.20: (R1) $\theta = \frac{1}{2},\ \lambda = 1 \Rightarrow \Delta_{\text{eng}} = 4/9$; (R2) $\Delta_{\text{eng}} = \Delta = 0.25,\ \lambda = 1 \Rightarrow \theta = 1/9$; (R3) $\theta = \frac{1}{2},\ \lambda = \frac{1}{2} \Rightarrow \Delta_{\text{eng}} = 8/9$. All three give $(1-\theta)\lambda\rho\Delta_{\text{eng}} = 0.20$ and, through the single-ρ fold, $\widetilde{m} - m_0 = 0.18$ — they are observationally identical at the body's 0.20, illustrating Remark 7.

Remark 9 (Bargaining power enters the premium — conditional on a pin)**.** *On $\theta \in (0,1)$ the bloc captures share θ of the gross control rent while the minority benefit of engagement is $(1-\theta)\lambda\rho\Delta_{\text{eng}}$.* If θ were independently identified*, this would deliver a structural comparative static absent from the reduced form: takeover premia and the minority gain move oppositely in θ, since a more powerful bloc extracts more of the gross surplus as private rent and concedes less of the threat-shift as minority value. Because θ is only set-identified, we state this as a conditional structural prediction. The toehold/auction extension, in which the engagement stake confers an auction advantage, is in the spirit of Bulow et al. (1999).*

C.6 The level reading and the shape of $\Delta^{\min}$

The body's headline object $\Delta^{\min}(\kappa) = \mathbb{E}[m^R(a)\,\mathbf{1}\{\text{bid}\}]$ (eq. (14)) uses the *level* $m^R(a)$, not just the wedge. We show the ξ-dependence of the level in (52) does not contaminate the *shape* of $\Delta^{\min}$.

Proposition 12 (Level enters $\Delta^{\min}$ as κ-independent coefficients times analysed bid statistics)**.** *With $m^R(a,\xi) = \theta(\bar{S} - K - w) + \theta\xi + (1-\theta)\lambda a\,\widetilde{\Delta}_{\text{eng}}$ from* (52)*,*

$$\Delta^{\min}(\kappa) = \theta(\bar{S} - K - w)\,\underbrace{\mathbb{E}[\mathbf{1}\{\text{bid}\}]}_{=p(\kappa)} + \theta\,\underbrace{\mathbb{E}[\xi\,\mathbf{1}\{\text{bid}\}]}_{=\sigma_\xi\,\varphi(t(\kappa))} + (1-\theta)\lambda\,\widetilde{\Delta}_{\text{eng}}\,\underbrace{\mathbb{E}[a\,\mathbf{1}\{\text{bid}\}]}_{\textit{wedge channel}}, \tag{56}$$

where, under the body's bidding rule $\mathbf{1}\{\text{bid}\} = \mathbf{1}\{\xi \geq \sigma_\xi t(\kappa)\}$ *with* $t(\kappa) = (\bar{m} + K - \bar{S} + P(\kappa))/\sigma_\xi$ *and* $\xi \sim N(0, \sigma_\xi^2)$*, the middle term equals* $\theta\sigma_\xi\varphi(t(\kappa))$*. Hence* $\Delta^{\min}$ *is an affine combination, with* κ*-independent coefficients, of three functions of the* same *equilibrium objects analysed in the body:* $p(\kappa)$*,* $\sigma_\xi\varphi(t(\kappa))$*, and the wedge channel* $\mathbb{E}[a\mathbf{1}\{\text{bid}\}]$*.*

Proof. Take $\mathbb{E}[\,\cdot\,\mathbf{1}\{\text{bid}\}]$ of (52) for $a \in \{0, 1\}$ and use linearity. The ξ term gives $\theta\mathbb{E}[\xi\mathbf{1}\{\text{bid}\}]$; under the Gaussian ξ and the threshold rule, $\mathbb{E}[\xi\mathbf{1}\{\xi \geq \sigma_\xi t\}] = \sigma_\xi\varphi(t)$ (truncated-normal first moment, the missing-mass term vanishing because $\mathbb{E}[\xi] = 0$). The coefficients are constants in κ. $\square$

C.7 Does the single peak survive a state-dependent wedge?

The constant-θ reduced form delivers a constant wedge. We perturb it by a bounded state-dependent factor,

$$m_1 - m_0 \;\mapsto\; (1-\theta)\lambda\rho\Delta_{\text{eng}}\big(1 + \lambda_{\text{sd}}\, g(\kappa)\big), \qquad g(\kappa) = \kappa - \tfrac{1}{2}, \tag{57}$$

with $\lambda_{\text{sd}} \in \{-0.4, -0.2, 0, +0.2, +0.4\}$, and re-solve for the equilibrium cutoffs and $\Delta^{\min}(\kappa)$ with the paper's own solver.

Numerical Regularity 1 (Single peak survives a state-dependent wedge — at the baseline calibration)**.** *This is a* numerically verified regularity at the baseline calibration, not a theorem. *In the baseline, with the* constant *wedge (*$\lambda_{\text{sd}} = 0$*),* $\Delta^{\min}(\kappa)$ *is single-peaked over* $\kappa \in [0.01, 0.99]$ *with interior maximizer* $\kappa^\dagger \approx 0.59$*, peak* ≈ 0.0776*, and amplitude* ≈ 0.0107 *(*$\approx 14\%$ *of level). Under the state-dependent wedge* (57)*, the interior maximizer stays at* $\kappa^\dagger \approx 0.60$ *and the profile remains single-peaked for* every $\lambda_{\text{sd}} \in \{-0.4, -0.2, 0, +0.2, +0.4\}$*; the peak rises/falls monotonically with* λ_{sd} *from* ≈ 0.0722 *to* ≈ 0.0829*. All re-solves pass the solver's residual gate. The headline single-hump orientation is* conditional *on the chord-curvature condition of the nonmonotonicity analysis; when that condition fails the profile inverts to a trough.*

C.8 What is proved, what is conditional, and what is open

For transparency: *proved* are the closed-form Nash rent and participation (Lemma 17), the ξ-invariance of the wedge (54), the reduced-form nesting and converse (Proposition 11), the ρ^2 event tree (Lemma 18), the level-shape decomposition (Proposition 12), and A3 *conditional on* Condition 1 (Theorem 1). *Conditional / numerical:* the threat-point shift is *stated* as Condition 1 and only *motivated* by the free-rider heuristic; θ is set-identified (Remark 7); and the single interior peak is a numerical regularity (Numerical Regularity 1). *Resolved since this record was written:* the appropriability coefficient λ *is* now derived from a complete tender-game equilibrium — Appendix D solves the disagreement-node game and delivers $\lambda = 1 - q(1-\gamma)\psi$ from primitives (Corollary 2), discharging what this ledger originally listed as open; the residual open margins of that derivation (bidder–fringe competition, repeated arrival, endogenous charter terms) are catalogued in §D.8.

D The disagreement-node tender game: deriving the appropriability coefficient

This appendix replaces the *stated* threat-point condition of Appendix C (Condition 1) with a *solved game*. Appendix C was deliberate that its load-bearing object — the engagement-induced shift $d(1) - d(0) = \lambda \widetilde{\Delta}_{\text{eng}}$ in the bloc's disagreement payoff — could "only be pinned by a complete tender-game equilibrium," and carried $\lambda \in (0, 1]$ as a reduced-form pivotality parameter with the free-rider logic of Grossman and Hart (1980) offered as motivation, not proof (Remark 6). Here we specify and solve that tender game at the disagreement node. The appropriability coefficient emerges in closed form,

$$\lambda \;=\; 1 - q\,(1-\gamma)\,\psi \;\in\; [0, 1],$$

with q the equilibrium probability of a fringe raid, γ the portability of the activist's improvement under a fringe acquirer, and ψ a pivotality factor. Condition 1 becomes a corollary (its functional form is *derived*), and Assumption (A3) becomes a theorem whose failure region is characterized by primitives rather than excluded by assumption. The honest residue is stated in §D.8: what remains assumed are *institutional* primitives of the tender environment (equal-treatment offers, the standard tie-breaking selection, charter parameters), not any condition on the endogenous threat point itself.

Notation. Throughout this appendix $\alpha \in (0, 1)$ denotes the control bloc's stake and $\tau_c \in (0, 1]$ the control threshold (fraction of all shares needed for control); these are local to the tender game.[13] We write $\overline{G}(x) \equiv \mathbb{P}(S_F \geq x)$ for the survival function of the fringe synergy distribution G, assumed atomless on its support and with finite mean. We maintain throughout that

$$\alpha \;<\; \tau_c :$$

the bloc *alone* cannot deliver control (satisfied at the baseline calibration, $\alpha = 0.30 < \tau_c = 0.75$). Without it the raider's cheapest successful coalition is a bloc-only lowball at the bloc's reservation, the bloc is held at $y(a)$ in *every* raid, and $\lambda = 1$ identically — the closed form below would not apply (see §D.8).

D.1 Environment

Condition on the disagreement node of Appendix C: negotiation with the incumbent bidder has broken down, and the target is a freestanding firm. As in Appendix C, $a \in \{0, 1\}$ indexes the engagement state at the node, and we work conditional on the *realized* engagement improvement $\delta_e \geq 0$ (so $\delta_e = \Delta_{\text{eng}}$ if the campaign succeeded, $\delta_e = 0$ otherwise); the fold back to expected objects $\widetilde{\Delta}_{\text{eng}} = \rho\Delta_{\text{eng}}$ is exact and is performed in Remark 10, consistent with the two-draw event

[13] The symbol α is unrelated to the completion-probability robustness device $\alpha(P, D)$ mentioned in passing in the body; τ_c is unrelated to the discounting horizon τ. We write φ for the standard normal density (as in Appendix C) and reserve ϕ for the dilution parameter below, flagging the departure from the body's occasional use of $\phi(\cdot)$ for the density.

tree of Lemma 18. The firm's standalone per-share value at the node is

$$y(a) \;=\; w + a\,\delta_e.$$

Players. (i) A *fringe raider* with privately observed type $(c_F, S_F) \sim H \otimes G$, independent draws: $c_F \geq 0$ is the cost of mounting a tender offer, and $S_F \geq 0$ is the raider's synergy over the firm *as it stands under the raider's plan* (below). (ii) The *control bloc*, holding stake $\alpha \in (0,1)$, strategic. (iii) A dispersed *float* of measure $1-\alpha$, atomistic.

Portability. If the raider takes control, the post-takeover per-share gross value is

$$Z(a) \;=\; w + \gamma\, a\,\delta_e + S_F, \qquad \gamma \in [0,1].$$

The parameter γ is the fraction of the activist's improvement that *survives a change of control*: $\gamma = 1$ when the improvement is portable (governance fixes, balance-sheet repairs, divestitures already executed — complements to any owner), $\gamma = 0$ when the raider's plan supersedes it (the raider re-optimizes operations wholesale, so the activist's program is redundant — substitutes). Intermediate γ mixes the two.

Dilution. As in Grossman and Hart (1980), the corporate charter permits a controlling raider to dilute minority (non-tendered) shares by $\phi \geq 0$ per share; the diverted value accrues to the raider.

Timing.

1. The raider observes (c_F, S_F) and the public state a, and chooses whether to *enter* (sinking c_F); upon entry it posts a *uniform, conditional, unrestricted* tender offer at price $\hat{b}$ per share, conditional on receiving at least τ_c of the shares.

2. Float members tender atomistically; the bloc tenders strategically (all-or-none is without loss under a uniform price).

3. If the tendered mass is at least τ_c, control transfers: tendered shares receive $\hat{b}$, non-tendered shares are worth $Z(a) - \phi$. Otherwise the offer lapses and the status quo $y(a)$ prevails.

Equilibrium concept. Subgame-perfect equilibrium of the entry-offer-tender game, with the standard tie-breaking selection in the tendering subgame: shareholders who are exactly indifferent tender. This is the selection under which value-improving raids are possible at all in the atomistic limit (Grossman and Hart, 1980); Bagnoli and Lipman (1988) provide the finite-shareholder foundation in which pivotal tendering delivers this outcome as the limit of exact equilibria, and Holmström and Nalebuff (1992) characterize the mixed equilibria we do not select. We say the bloc is *pivotal* if the float alone cannot deliver control, i.e. $1 - \alpha < \tau_c$.

D.2 The tendering subgame and the free-rider floor

Lemma 19 (Optimal offers: the free-rider floor and the reservation top-up)**.** *Fix an entered raider and the state a, and define the* acquiescence region *as all of the support of S_F when the bloc is non-pivotal, and $\{S_F \geq \phi + (1-\gamma)a\delta_e\}$ when it is pivotal (Lemma 21); its complement (pivotal only) is the* blocking region*. In every subgame-perfect tendering outcome under the stated selection:*

(i) *On the acquiescence region the optimal offer is the* free-rider floor

$$\hat{b}^*(a) \;=\; Z(a) - \phi,$$

at which the entire float tenders. The raider's payoff per share it does not *acquire is the dilution ϕ, and per share it* does *acquire is $Z(a) - \hat{b}^*(a) = \phi$; its gross payoff from a successful floor raid is ϕ on the full register, independently of the tendered mass, and net of cost $\phi - c_F$.*

(ii) *On the blocking region the floor offer fails (the pivotal bloc refuses), and the raider's best successful offer is the* reservation top-up *$\hat{b} = y(a) > Z(a) - \phi$: the float strictly tenders, the indifferent bloc tenders under the selection, the raider acquires the full register and nets*

$$Z(a) - y(a) - c_F \;=\; S_F - (1-\gamma)\, a\, \delta_e - c_F,$$

undertaking the raid iff this is nonnegative. In every top-up raid the bloc is paid exactly *its reservation $y(a)$.*

Proof. An atomistic float member is never pivotal, so conditional on the raid succeeding her choice is $\hat{b}$ (tender) against $Z(a) - \phi$ (retain a diluted share); conditional on failure the conditional offer lapses and both choices give $y(a)$. She therefore tenders iff $\hat{b} \geq Z(a) - \phi$, with the selection resolving indifference toward tendering.

(i) On the acquiescence region the floor succeeds: the float tenders, and (pivotal case) the bloc tenders by Lemma 21(ii). Any $\hat{b} < Z(a) - \phi$ loses the entire float, and with $\alpha < \tau_c$ bloc shares alone cannot deliver control, so the raid fails. Any $\hat{b} > Z(a) - \phi$ keeps the success event unchanged while strictly raising the price on every tendered share, hence is dominated. At the floor: each acquired share costs $Z(a) - \phi$ and is worth $Z(a)$, margin ϕ; each non-acquired share is diluted by ϕ, accruing to the raider; summing over the unit register gives ϕ gross, $\phi - c_F$ net, independent of the tendered mass and of a.

(ii) On the blocking region $y(a) > Z(a) - \phi$. Offers $\hat{b} \in [Z(a) - \phi,\, y(a))$ attract the float but not the bloc, delivering mass $1 - \alpha < \tau_c$: failure. The top-up $\hat{b} = y(a)$ exceeds the float's reservation (strict tendering) and meets the bloc's (indifference resolved toward tendering): mass one, success. Raising $\hat{b}$ above $y(a)$ only pays more on every share. At $\hat{b} = y(a)$ the raider buys the whole register at $y(a)$ per share, each worth $Z(a)$ under its control, with no minority left to dilute: net payoff $Z(a) - y(a) - c_F = S_F - (1-\gamma)a\delta_e - c_F$. Entry on the blocking region therefore occurs exactly when this is nonnegative; the bloc, tendering at its reservation, receives $y(a)$. □

The floor is the free-rider result: the raider's synergy S_F passes *entirely* to the shareholders through the offer price; the raider's only rent on floor raids is the charter dilution ϕ. Top-up raids exist precisely because a reluctant pivotal bloc can be bought out at its outside option — but for that very reason they transfer *no* surplus to the bloc, which is what keeps the threat point below clean of them.

D.3 Entry

Lemma 20 (Floor raids are dilution-financed and state-blind)**.** *Floor raids — raids at the offer $\hat{b}^*(a) = Z(a) - \phi$, which are the only raids in which the bloc earns more than its reservation —*

occur iff $c_F \leq \phi$ (and, if the bloc is pivotal, the acquiescence event of Lemma 21 holds). The floor-feasibility event $\{c_F \leq \phi\}$ has probability

$$q \;\equiv\; H(\phi) \;\in\; [0,1],$$

independent of the engagement state a and of the synergy draw S_F. Top-up raids (Lemma 19(ii)) occur on a state-dependent set — engagement shrinks *it — but pay the bloc exactly $y(a)$ and therefore do not enter the threat-point shift.*

Proof. By Lemma 19(i) a floor raid pays $\phi - c_F$ regardless of (a, S_F) and of the tendered mass; not entering pays 0 (the offer is conditional). On the acquiescence region the floor is the raider's best offer, so floor raids occur exactly on $\{c_F \leq \phi\}$ intersected with acquiescence, and $\mathbb{P}(c_F \leq \phi) = H(\phi)$ with independence from (a, S_F) immediate. On the blocking region the optimal offer is the top-up of Lemma 19(ii), profitable on $\{S_F - (1-\gamma)a\delta_e \geq c_F\}$ — a set decreasing in $a\delta_e$ — in which the bloc receives its reservation $y(a)$ exactly. ☐

The economically important part is that the *bloc-surplus-relevant* raid margin is state-blind: on floor raids the raider's profit is $\phi - c_F$ whatever the engagement state, so engagement affects the bloc's takeover option only through its own willingness to part with an improved firm — not through raider selection. Total raid incidence *is* state-dependent (engagement deters some top-up raids), but those raids hold the bloc at its outside option and so contribute nothing to the shift $d(1) - d(0)$. This is what keeps the threat-point shift below clean of entry-selection terms.

D.4 The bloc at the node: blocking and acquiescence

Lemma 21 (Blocking rule)**.** *Consider an entered raider offering the floor $\hat{b}^*(a) = Z(a) - \phi$.*

(i) ***Non-pivotal bloc*** *($1 - \alpha \geq \tau_c$): the raid succeeds for every S_F; the bloc's per-share payoff is $Z(a) - \phi$ whether it tenders or retains.*

(ii) ***Pivotal bloc*** *($1 - \alpha < \tau_c$): the raid succeeds iff the bloc tenders, and the bloc tenders iff*

$$\hat{b}^*(a) \;\geq\; y(a) \quad\iff\quad S_F \;\geq\; \phi + (1-\gamma)\, a\, \delta_e.$$

On the acquiescence set the bloc's per-share payoff is $Z(a) - \phi$; off it, the floor *raid fails, and the bloc receives $y(a)$ — either because no raid occurs, or because the raider switches to the reservation top-up of Lemma 19(ii), which pays the bloc exactly $y(a)$.*

Proof. (i) With $1 - \alpha \geq \tau_c$ the float alone delivers control (Lemma 19(i): all of it tenders at the floor). The bloc tendering yields $\hat{b}^* = Z(a) - \phi$; retaining yields the diluted value $Z(a) - \phi$. Identical. (ii) With $1 - \alpha < \tau_c$ the success event requires bloc shares. If the bloc tenders at the floor, it receives $\hat{b}^*(a)$; if it refuses, the conditional floor offer lapses and it retains a share worth $y(a)$ (or is bought out at $y(a)$ in a top-up raid). It tenders at the floor iff $Z(a) - \phi \geq y(a)$, i.e. $w + \gamma a\delta_e + S_F - \phi \geq w + a\delta_e$, i.e. $S_F \geq \phi + (1-\gamma)a\delta_e$. (Indifference, an event of G-measure zero, is resolved toward tendering.) Off the acquiescence set the raider's optimal response is characterized by Lemma 19(ii): the top-up succeeds when profitable, and in it the bloc — tendering at its reservation under the selection — receives exactly $y(a)$. In every branch off the acquiescence set the bloc's payoff is therefore $y(a)$. ☐

D.5 The threat point and the appropriability coefficient

Proposition 13 (The threat-point shift, and λ in closed form)**.** *Let $d(a)$ be the bloc's per-share equilibrium continuation value at the disagreement node, conditional on the realized improvement δ_e. Then*

$$\begin{aligned} \textit{non-pivotal:}\quad d(a) \;&=\; w + a\delta_e + q\left(\mathbb{E}[S_F] - \phi - (1-\gamma)a\delta_e\right), &(58)\\ \textit{pivotal:}\quad d(a) \;&=\; w + a\delta_e + q\,\mathbb{E}\big[(S_F - \phi - (1-\gamma)a\delta_e)^+\big]. &(59)\end{aligned}$$

In both regimes the engagement-induced shift is linear in the realized improvement,

$$d(1) - d(0) \;=\; \lambda\,\delta_e, \qquad \boxed{\lambda \;=\; 1 - q\,(1-\gamma)\,\psi \;\in\; [0,1],}$$

where $\psi = 1$ in the non-pivotal regime and, in the pivotal regime with $\gamma < 1$ and $\delta_e > 0$,

$$\psi \;=\; \frac{1}{(1-\gamma)\,\delta_e}\int_0^{(1-\gamma)\delta_e} \overline{G}(\phi + t)\,dt \;\in\; [0,1]$$

(the average probability, along the forgone-improvement interval, that the fringe synergy clears the bloc's reservation); when $\gamma = 1$ or $\delta_e = 0$ the ψ-term is multiplied by zero and $\lambda = 1$. Moreover $\psi \le 1$ implies $\lambda^{\text{pivotal}} \ge \lambda^{\text{non-pivotal}}$ at common (q,γ): pivotality protects the threat point.

Proof. Non-pivotal. By Lemmas 20–21(i), with probability q a raider enters and the raid succeeds for every S_F, paying the bloc $Z(a) - \phi = w + \gamma a\delta_e + S_F - \phi$; with probability $1-q$ the status quo $y(a) = w + a\delta_e$ prevails. Taking expectations over S_F (independent of entry) gives

$$d(a) = (1-q)(w + a\delta_e) + q\,(w + \gamma a\delta_e + \mathbb{E}[S_F] - \phi),$$

which rearranges to (58). Differencing: $d(1) - d(0) = (1-q)\delta_e + q\gamma\delta_e = \delta_e\,[1 - q(1-\gamma)]$.

Pivotal. By Lemmas 19–21, the bloc earns more than its reservation only in *floor* raids, which occur on $\{c_F \le \phi\} \cap \{S_F \ge \phi + (1-\gamma)a\delta_e\}$ and pay it $Z(a) - \phi$; in every other branch — no entry, failed floor, or a top-up raid that buys the bloc out at its outside option — the bloc has exactly $y(a)$ (the outside-option principle: marginal raids extract the bloc's full surplus). Hence

$$d(a) = y(a) + q\,\mathbb{E}\big[\big(Z(a) - \phi - y(a)\big)\,\mathbf{1}\{S_F \ge \phi + (1-\gamma)a\delta_e\}\big] = y(a) + q\,\mathbb{E}\big[(S_F - \phi - (1-\gamma)a\delta_e)^+\big],$$

using $Z(a) - \phi - y(a) = S_F - \phi - (1-\gamma)a\delta_e$, which is exactly the positive part on the acquiescence set. This is (59). Differencing, with $X \equiv S_F - \phi$ and $c \equiv (1-\gamma)\delta_e \ge 0$:

$$d(1) - d(0) = \delta_e + q\left(\mathbb{E}[(X-c)^+] - \mathbb{E}[X^+]\right) = \delta_e - q\int_0^c \mathbb{P}(X \ge t)\,dt,$$

where the last step is the layer-cake identity $\mathbb{E}[X^+] - \mathbb{E}[(X-c)^+] = \int_0^c \mathbb{P}(X \ge t)\,dt$ (Fubini on $\mathbb{E}\int_0^\infty \mathbf{1}\{t \le X\}dt$, valid for any X with $\mathbb{E}|X| < \infty$ and $c \ge 0$). Substituting $\mathbb{P}(X \ge t) = \overline{G}(\phi + t)$ and factoring out $c = (1-\gamma)\delta_e$ yields $d(1) - d(0) = \delta_e[1 - q(1-\gamma)\psi]$ with ψ as displayed. Bounds: $0 \le \overline{G} \le 1$ gives $\psi \in [0,1]$, and $q, (1-\gamma), \psi \in [0,1]$ give $\lambda \in [0,1]$. The regime comparison is immediate from $\psi \le 1$. □

Corollary 2 (Condition BG's form is derived, with a microfounded level)**.** *Define $w' \equiv d(0)$ from* (58)–(59) *(the standalone continuation value absent engagement, now inclusive of the fringe option value) and λ as in Proposition 13. Then the bloc's disagreement payoff satisfies* exactly

$$d(a) \;=\; w' + a\,\lambda\,\delta_e,$$

i.e. equation (50) *of Condition 1 holds with $\lambda = 1 - q(1-\gamma)\psi$ an equilibrium object. Every downstream result of Appendix C that takes Condition 1 as input (Lemma 17, the wedge* (54)*, Proposition 11, Theorem 1) now operates on a derived premise, with one caveat: $\lambda > 0$ is no longer guaranteed by assumption — it holds exactly on the region characterized in Theorem 2, and the wedge formula extends continuously to $\lambda = 0$.*

Proof. Linearity of $d(a)$ in a with slope $\lambda\delta_e$ is the content of Proposition 13; matching coefficients against (50) (with δ_e in place of $\widetilde{\Delta}_{\text{eng}}$ pre-fold; see Remark 10) gives the display. The downstream results of Appendix C use only the functional form and $\lambda \geq 0$; inspection of their proofs shows none uses $\lambda > 0$ except where flagged. □

Remark 10 (The fold back to $\widetilde{\Delta}_{\text{eng}}$ is exact)**.** *Proposition 13 conditions on the realized improvement $\delta_e \in \{0, \Delta_{\text{eng}}\}$. Taking expectations over the campaign-success draw $B_1 \sim \mathrm{Bern}(\rho)$ of Lemma 18: the shift is* 0 *when $B_1 = 0$ and $\lambda(\Delta_{\text{eng}})\,\Delta_{\text{eng}}$ when $B_1 = 1$, so the expected shift is $\rho\,\lambda\,\Delta_{\text{eng}} = \lambda\,\widetilde{\Delta}_{\text{eng}}$ with $\lambda \equiv \lambda(\Delta_{\text{eng}})$ evaluated at the realized improvement. No approximation is involved, and the second ρ of the body's fold (premium-state arrival B_2) composes exactly as in Lemma 18, preserving $\widetilde{m} - m_0 = \rho^2(1-\theta)\lambda\Delta_{\text{eng}}$. Note one refinement relative to the reduced form: $\lambda(\cdot)$ depends on the* size *of the realized improvement in the pivotal regime (through ψ), a state-dependence the constant-λ reduced form suppresses; quantitative statements therefore hold λ at $\lambda(\Delta_{\text{eng}})$.*

D.6 Assumption (A3) from primitives

Theorem 2 (A3 holds except on a characterized boundary)**.** *Maintain $\theta < 1$ and $\rho\,\Delta_{\text{eng}} > 0$, and let $\lambda = 1 - q(1-\gamma)\psi$ from Proposition 13. Then the bargained premia of Appendix C satisfy*

$$\widetilde{m} - m_0 \;=\; \rho^2(1-\theta)\,\lambda\,\Delta_{\text{eng}} \;>\; 0 \qquad \Longleftrightarrow \qquad \lambda > 0,$$

and $\lambda = 0$ — equivalently, Assumption (A3) *fails — if and only if* all three *of the following hold:*

(F1) *fringe raids are* certain*: $q = H(\phi) = 1$;*

(F2) *the activist's improvement is* fully superseded *by the raider's plan: $\gamma = 0$;*

(F3) *blocking is* unavailable or worthless*: the bloc is non-pivotal ($1 - \alpha \geq \tau_c$),* or *it is pivotal but the fringe synergy clears the bloc's reservation with probability one, $\mathbb{P}\big(S_F \geq \phi + \Delta_{\text{eng}}\big) = 1$.*

In particular each of the following alone implies $\lambda > 0$: some raids fail to materialize ($q < 1$); any portability ($\gamma > 0$); a pivotal bloc facing a fringe whose synergy falls short of $\phi + \Delta_{\text{eng}}$ with positive probability.

Proof. The equivalence $\widetilde{m} - m_0 > 0 \Leftrightarrow \lambda > 0$ is Theorem 1's display with λ substituted from Corollary 2, the other factors being positive by hypothesis. It remains to characterize $\lambda = 0$, i.e.

$q(1-\gamma)\psi = 1$. Since each factor lies in $[0,1]$, the product equals one iff every factor equals one. $q = 1$ is (F1); $1-\gamma = 1$ is (F2). Given $\gamma = 0$: in the non-pivotal regime $\psi \equiv 1$, which is the first disjunct of (F3); in the pivotal regime $\psi = 1$ iff $\int_0^{\delta_e} \overline{G}(\phi+t)\,dt = \delta_e$ iff $\overline{G}(\phi+t) = 1$ for a.e. $t \in (0,\delta_e)$; since $\overline{G}$ is nonincreasing and right-continuous this holds iff $\overline{G}(\phi+\delta_e^-) = 1$, i.e. $\mathbb{P}(S_F \geq \phi+\delta_e) = 1$ for $\delta_e = \Delta_{\text{eng}}$ by atomlessness of G — the second disjunct of (F3). The "in particular" statement negates each factor. $\square$

Remark 11 (Comparative statics of appropriability)**.** *From $\lambda = 1 - q(1-\gamma)\psi$ with $q = H(\phi)$: (i) λ is nonincreasing in fringe-market strength: $\partial\lambda/\partial q = -(1-\gamma)\psi \leq 0$; a hotter takeover market* erodes *the activist's bargaining-relevant appropriability even as it raises the level of the bloc's outside option. (ii) λ is nondecreasing in portability: in the pivotal regime $(1-\gamma)\psi = \frac{1}{\delta_e}\int_0^{(1-\gamma)\delta_e} \overline{G}(\phi+t)\,dt$, whose γ-derivative is $-\overline{G}(\phi+(1-\gamma)\delta_e) \leq 0$, so $\partial\lambda/\partial\gamma = q\,\overline{G}(\phi+(1-\gamma)\delta_e) \geq 0$; in the non-pivotal regime $\partial\lambda/\partial\gamma = q \geq 0$. (iii) Pivotality is protective and discontinuous: as α crosses $1-\tau_c$ from below, ψ drops from 1 to the integral average, so λ jumps* up *by $q(1-\gamma)\big(1-\psi\big) \geq 0$ — a testable stake-size threshold effect. (iv) Dilution ϕ is two-edged in the pivotal regime: $\partial\lambda/\partial\phi = -(1-\gamma)\psi\,H'(\phi) + q\,\frac{1}{\delta_e}\big[\overline{G}(\phi) - \overline{G}(\phi+(1-\gamma)\delta_e)\big] \cdot$ (sign depends on H' vs the G-mass on $(\phi, \phi+(1-\gamma)\delta_e]$): easier dilution invites more raids ($q\uparrow$, eroding λ) but also lowers the floor relative to the bloc's reservation, thinning the acquiescence set ($\psi\downarrow$, protecting λ). We leave ϕ unsigned and verify both directions numerically.*

D.7 Measured premia: rationalizing the estimated negative activism–premium effect

Celentano and Levine (2025) estimate that bid premia are 13.7% (5.2pp) lower when an activist is present in takeover negotiations, relative to a no-activist counterfactual.[14] At face value a negative activism–premium relation contradicts $m_1 > m_0$. The model says otherwise: the empirical object is not the wedge. Bid premia are measured against a market price that *already capitalizes* activism beliefs — the body's Proposition 3 prices in both the expected standalone improvement $\widetilde{\Delta}\,\pi(X,D)$ and the activism premium $(\widetilde{m}-m_0)\pi(X,D)$ — so activism moves the *denominator* as well as the numerator. When appropriability is low, the denominator wins.

Proposition 14 (Measured-premium reversal under low appropriability)**.** *Fix the blockholder's cutoffs and an information state (X,D) as in Proposition 6, and let $P(\pi)$ denote the pricing fixed point of equation* (2) *viewed as a function of the engagement posterior $\pi = \pi(X,D)$, with $\bar{m}(\pi) = m_0 + (\widetilde{m}-m_0)\pi$ and bid probability $p(P,\pi)$ from* (1)*. Define the* measured premium

$$M(\pi) \;\equiv\; \frac{\bar{m}(\pi)}{P(\pi)},$$

the expected offer over the prevailing market price, net of one. Maintain Assumption (A5a) *and*

[14] In their structural model the channel is bargaining: the activist lowers the board's entrenchment cost, which lowers the price the board demands. The mechanism developed here is complementary and differently indexed — see Remark 12. Albuquerque et al. (2022) contain no takeover-premium estimate; their role in this section is different and essential: they quantify the *capitalization* of activism in prices at disclosure (a 6.34% announcement return, roughly three-quarters of it anticipated treatment), which is precisely the denominator effect the proposition below exploits.

$m_0 > 0$. *Then*

$$\operatorname{sgn} M'(\pi) \;=\; \operatorname{sgn}\Big[(\widetilde{m} - m_0)\, P(\pi) \;-\; \bar{m}(\pi)\, P'(\pi)\Big],$$

and in the $\lambda \to 0$ *limit* $P'(\pi) > 0$ *whenever* $\widetilde{\Delta} > 0$.[15] *Since* $\widetilde{m} - m_0 = \rho^2(1-\theta)\lambda\Delta_{\mathrm{eng}} \to 0$ *as* $\lambda \to 0$ *while* $\bar{m}(\pi) \to m_0 > 0$ *and*

$$P'(\pi) \;\longrightarrow\; \frac{\delta\,(1-p)\,\widetilde{\Delta}}{1 - \delta p - \delta\,\frac{\partial p}{\partial P}\,\big(P + m_0 - \mathbb{E}[v \mid X, D] - \widetilde{\Delta}\pi\big)} \;>\; 0 \qquad (\lambda \to 0),$$

there exists $\lambda_{\mathrm{crit}} > 0$ *such that for all* $\lambda < \lambda_{\mathrm{crit}}$ *the measured premium is* strictly decreasing *in activism evidence,* $M'(\pi) < 0$ *for all interior* π *— even though* $m_1 \geq m_0$ *throughout. Activism then lowers measured bid premia as Celentano and Levine (2025) estimate, through the price channel, with the wedge's sign intact.*

Proof. Differentiate $M = \bar{m}/P$: $M' = [(\widetilde{m} - m_0)P - \bar{m}P']/P^2$, giving the sign display. For P': the fixed point is $P = \delta[\, p(P,\pi)\,(P + \bar{m}(\pi)) + (1 - p(P,\pi))\,(\mathbb{E}[v \mid X, D] + \widetilde{\Delta}\pi)\,]$. The implicit function theorem applies: the denominator $1 - \delta p - \delta\, p_P\,(P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi)$ is bounded below by $1 - \delta\big[p + |P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi|\, \varphi(0)/\sigma_\xi\big] > 0$ under the contraction bound of Assumption (A5a) (see also Remark 2 in the body: at the baseline calibration the conservative sufficient form of A5a is not met, so the numerical λ_{crit} below rests on the solver's observed contraction rather than the maintained bound), and it gives

$$P'(\pi) = \frac{\delta\big[p\,(\widetilde{m} - m_0) + (1-p)\,\widetilde{\Delta} + p_\pi\,\big(P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi\big)\big]}{1 - \delta p - \delta\, p_P\,\big(P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi\big)}.$$

As $\lambda \to 0$: $\widetilde{m} - m_0 \to 0$ directly, and $p_\pi = -\varphi(\cdot)\,(\widetilde{m} - m_0)/\sigma_\xi \to 0$ since π enters p only through $\bar{m}$; the displayed limit follows, and it is strictly positive for $\widetilde{\Delta} > 0$, $p < 1$. All equilibrium objects at fixed cutoffs are continuous in λ (they are compositions of Φ, φ, and the fixed point, which is continuous by A5 and the uniform contraction). Hence $M' \to -m_0 P'/P^2 < 0$ uniformly on compact interior π-sets, and continuity delivers $\lambda_{\mathrm{crit}} > 0$. □

Remark 12 (Composition reinforces the reversal; what would distinguish the stories)**.** *Two further forces push the same way in the cross-section. First, deterrence:* $\partial p/\partial \pi < 0$ *when* $\widetilde{m} > m_0$ *(body, §Bid Incidence), so high-*π *states select toward deals that clear a higher hurdle — fewer, not richer, marginal deals. Second, heterogeneity in* λ*: by Theorem 2, campaigns at targets whose natural acquirers are* superseding *(strategic buyers re-optimizing wholesale:* $\gamma \approx 0$*) in hot deal markets (*$q \approx 1$*) carry* $\lambda \approx 0$ *and hence near-zero wedges; if such targets are over-represented among consummated takeovers, the estimated average effect tilts negative. The model's distinguishing prediction — against Celentano and Levine (2025)'s board-bargaining channel, in which the premium effect moves only with activist presence: the activism effect on premia should be* less negative *(i) where improvements are portable (financial acquirers, asset redeployability), (ii) where the activist's stake is pivotal relative to control thresholds, and (iii) in cold fringe markets — because these are*

[15] At general λ the sign of P' is ambiguous: the deterrence term $p_\pi\,(P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi)$ can dominate $(1-p)\widetilde{\Delta}$ when the wedge is large relative to $\widetilde{\Delta}$, so the clause is claimed only in the limit, which is all the conclusion below requires. A sufficient condition at general λ is $p\,(\widetilde{m} - m_0) + (1-p)\,\widetilde{\Delta} > \varphi(t)\,(\widetilde{m} - m_0)\,(P + \bar{m} - \mathbb{E}[v|X,D] - \widetilde{\Delta}\pi)/\sigma_\xi$.

exactly the high-λ cells. Cross-sections in (q, γ, ψ) separate the two mechanisms for the same average sign.

Remark 13 (Empirical content of the primitives)**.** *Unlike the reduced-form λ, the derived coefficient is indexable: q by fringe M&A intensity in the target's industry-year; γ by the strategic-vs-financial composition of likely acquirers and asset redeployability; ϕ by charter/state anti-takeover provisions governing freeze-outs and squeeze-outs; pivotality by the bloc's stake against control thresholds. This converts Remark 7's non-identification (only the product $(1-\theta)\lambda\rho\Delta_{\text{eng}}$ pinned) into an over-identifying restriction set: variation in $(q, \gamma, \phi, \alpha)$ should move estimated wedges with the signs of Remark 11.*

D.8 What is solved, what is institutional, what remains open

Solved (no residual condition on endogenous objects): the tendering subgame and free-rider floor (Lemma 19); entry (Lemma 20); the blocking rule (Lemma 21); the threat point, $\lambda = 1 - q(1-\gamma)\psi$, and the exact recovery of Condition 1's form (Proposition 13, Corollary 2); A3's iff boundary (Theorem 2); the fold (Remark 10); the measured-premium reversal at fixed cutoffs (Proposition 14).

Institutional primitives (transparent and discussed, in the tradition of taking the tender institution as given): uniform equal-treatment conditional offers (discriminatory offers would let the raider pay the pivotal bloc exactly its reservation in *every* raid while holding the float at the floor; the bloc would then be held at $y(a)$ throughout, giving $\lambda = 1$ — its maximum. Equal treatment thus yields the *smallest* λ and is conservative for A3's failure region; it is also the empirically relevant institution under Williams-Act-type equal treatment); the maintained domain restriction $\alpha < \tau_c$ (if the bloc alone delivers control, the raider's bloc-only lowball holds it at reservation in every raid and $\lambda = 1$ identically — A3 then holds trivially, so the restriction is conservative as well); the Grossman and Hart (1980) indifference-tendering selection, with Bagnoli and Lipman (1988) as its finite-shareholder foundation and Holmström and Nalebuff (1992)'s mixed equilibria not selected; exogenous charter objects (ϕ, τ_c); one-shot fringe arrival with type $(c_F, S_F) \sim H \otimes G$ independent; the engagement state observable at the node.

Open (stated plainly): competition between the incumbent bidder and the fringe (a takeover auction in the spirit of Bulow et al. (1999), in which the engagement stake may also confer a toehold advantage); repeated fringe arrival and option-value timing at the disagreement node; endogenous design of (ϕ, τ_c); and the economic identification of the second fold draw B_2 of Lemma 18 ("premium-state arrival") as an event distinct from campaign success — under a single-event reading the wedge folds once, $\widetilde{m} - m_0 = \rho(1-\theta)\lambda\Delta_{\text{eng}}$, and the calibrated Δ_{eng} shifts from ≈ 0.516 to ≈ 0.46 with all signs and the architecture unchanged. None of these touches the *sign* architecture: each would alter the level of $d(a)$ and potentially the magnitude of λ, but the linear-in-a form of Corollary 2 and the boundary logic of Theorem 2 (raids must be certain, superseding, and unblockable to kill the wedge) are robust to them in the obvious directions.

E Robustness to the Decreasing-Cost Assumption (A2)

This appendix isolates the role of Assumption (A2) (the engagement cost $C(s) = C_0 \exp(-\chi(s - \mu)/\sigma_s)$ is *decreasing* in the signal, $\chi > 0$). Throughout we work with the realized $D = 0$ posterior support which, under the Standing Assumptions, is the two-point set $\{0, \bar{\pi}\}$ with $\bar{\pi} \equiv \pi(1,0) = \omega_Q/(\omega_H + \omega_Q)$ (Proposition 2), where $\omega_H, \omega_Q \geq 0$ are the unconditional Hold and Quiet masses.

Primitive payoff gaps. Fix the price schedule $P(\cdot)$ and write, for a blockholder with signal s,

$$
\begin{aligned}
G_{EH}(s) &:= \delta\, \mathbb{E}_z[\, Y_H - Y_E \mid s\,] && \text{(Hold over Exit)},\\
G_{HQ}(s) &:= \delta\, \mathbb{E}_z[\, Y_Q - Y_H \mid s\,] - C(s) && \text{(Quiet Voice over Hold)},\\
G_{QP}(s) &:= \delta\, \mathbb{E}_z[\, Y_P - Y_Q \mid s\,] && \text{(Public Voice over Quiet Voice)},
\end{aligned}
$$

where Y_a is the blockholder's gross continuation payoff under action $a \in \{E, H, Q, P\}$ and $C(s)$ is the engagement cost. The cost cancels in G_{QP} (two engagement actions) but not in G_{HQ}. The cutoffs solve $G_{EH}(k_1) = 0$, $G_{HQ}(k_0) = 0$, $G_{QP}(k_D) = 0$ when the cells are nonempty.

The cost-sensitivity wedge. Because C enters only G_{HQ}, A2 acts only on the Hold/Quiet margin. Define the gross Quiet-over-Hold slope and the net voice-surplus slope,

$$
\Lambda(s) := \delta\, \frac{\mathrm{d}}{\mathrm{d}s}\, \mathbb{E}_z[\, Y_Q - Y_H \mid s\,], \qquad W(s) := \Lambda(s) - C'(s). \tag{60}
$$

At fixed prices $\Lambda(s) \equiv 0$: Hold and Quiet share the trade ($q = 0$) and the disclosure flag ($D = 0$), so $Y_Q - Y_H$ is the engagement lottery, whose value is a cell-level object independent of the individual signal (eq. (43)). The wedge therefore reduces to $W(s) = -C'(s)$. Under A2, $C'(s) = -(\chi/\sigma_s)\, C(s) < 0$, so $W(s) > 0$: the decreasing cost *is* the single-crossing force on this margin. The substantive question is how the architecture behaves when C' is flat or positive.

E.1 The ordering proposition

Proposition 15 (The exact role of A2 on the Hold/Quiet margin)**.** *Suppose* (A1)*,* (A3)*–*(A7) *hold and $P(\cdot)$ is fixed. Let $\Lambda(s)$ be as in* (60)*; at fixed prices $\Lambda(s) \equiv 0$ (eq.* (43)*: the gross Hold/Quiet gap is signal-free). Then:*

(a) ***(Threshold).*** *If $W(s) = \Lambda(s) - C'(s) > 0$ for all s in the Quiet cell, then G_{HQ} is strictly increasing there, so it crosses zero at most once; with the strict monotonicity of G_{EH} and G_{QP} (*(A4)*,* (A5)*) this yields the threshold structure Exit/Hold/Quiet/Public.*

(b) ***(Placement of k_0).*** *If in addition the Quiet cell is nonempty and $G_{HQ}(k_1) < 0 < G_{HQ}(k_D)$, then the unique zero k_0 of G_{HQ} lies strictly in (k_1, k_D), so $k_1 < k_0 < k_D$.*

(c) ***(Exactness).*** *Since $\Lambda \equiv 0$, the threshold condition $W = \Lambda - C' > 0$ is exactly $C'(s) < 0$: at fixed prices, A2 is* necessary and sufficient *for strict single-crossing on the Hold/Quiet margin. With a flat cost ($C' \equiv 0$) the gap G_{HQ} is constant in s — no single crossing — and the equilibrium sits on the two-cutoff collapse face (the body's interiority footnote); with $C' > 0$ on a nondegenerate set the margin can invert (Definition 2). What this appendix delivers is therefore not dispensability of A2 but its* exact location*: A2 acts on one margin only, and the wedge $W(s) = -C'(s)$ makes the margin transparent and falsifiable.*

Proof. *(a)* By the slope structure of (8) ($B(H) - B(E) > 0$ and $\eta > 0$ under Lemma 4's condition), G_{EH} and G_{QP} are strictly monotone in s, hence each has at most one zero. For G_{HQ}, $G'_{HQ}(s) = \Lambda(s) - C'(s) = W(s)$, the wedge (60); the threshold hypothesis gives $W(s) > 0$, so G_{HQ} is strictly increasing and crosses zero at most once. Single-crossing of all three gaps delivers the ordered partition. *(b)* Since G_{HQ} is strictly increasing and continuous on $[k_1, k_D]$, the intermediate value theorem applied to the boundary signs yields a unique $k_0 \in (k_1, k_D)$. *(c)* Immediate: $C'(s) < \Lambda(s)$ is exactly $W(s) > 0$; under A2, $C' < 0 \leq \Lambda$, so the inequality is slack and the ordering holds for all $\chi \geq 0$ and for mildly increasing cost provided $C'(s) < \Lambda(s)$ does not fail. □

What survives flat cost ($\chi = 0$). Setting $\chi = 0$ gives $C(s) \equiv C_0$, $C' \equiv 0$, so $W \equiv \Lambda \equiv 0$: the Hold/Quiet gap is constant in s and there is *no* single crossing on this margin — consistent with the body's interiority footnote, the equilibrium collapses to the two-cutoff case (the Hold and Quiet cells are not separately identified). The Exit/engage and Quiet/Public margins, existence, and the comparative-statics machinery go through unchanged, since neither uses A2 beyond the Hold/Quiet split.

E.2 When the ordering can invert: a cost taxonomy

Definition 2 (Cost regimes)**.** *Call the engagement cost*

- ***persuasion-type*** *if $C'(s) < \Lambda(s)$ for all s in the Quiet cell — at fixed prices $\Lambda \equiv 0$, so this is exactly $C' < 0$, i.e. A2: the marginal cost of a louder voice falls with the strength of the case. The ordering of Proposition 15 holds.*
- ***entrenchment-type*** *if $C'(s) \geq \Lambda(s) \equiv 0$ on a nondegenerate subset: the cost of engaging is flat or* increases *in the signal (e.g. better-informed blockholders face harsher managerial entrenchment, litigation, or reputational retaliation precisely when their case is strong). Here G_{HQ} can be non-monotone and the Hold/Quiet ordering may invert locally, producing a non-threshold (possibly disconnected) Quiet cell.*

Remark 14 (Falsifiable mechanism for decreasing cost)**.** *A2 carries a refutable prediction. Decreasing $C(s)$ says that higher private conviction lowers the marginal cost of engagement: a blockholder who privately observes a strong value case (i) needs less costly information production to substantiate it, and (ii) faces a lower probability of a value-destroying proxy loss. Two testable cross-sectional implications follow: (1) conditional on engaging, campaign intensity/cost (advisory fees, proxy-solicitation spend, duration) should be* decreasing *in ex-post realized value surprises; and (2) the sensitivity of engagement to the signal should be* stronger *(steeper k_0 response) where information is cheap to verify. The entrenchment-type alternative reverses prediction (1). The two regimes are therefore empirically separable, and A2 is falsifiable.*

E.3 The minority-gains object: one integrand, on the chord

The headline object is the minority takeover gain (eq. (14)) $\Delta^{\min}(\kappa) = \mathbb{E}[m^R(a)\,\mathbf{1}\{\text{bid}\}]$. On the realized $D = 0$ event the relevant per-state contribution is the inferred premium wedge times the bid probability. Writing the wedge as a function of the success posterior π via $\bar{m}(\pi) = m_0 + (\tilde{m} - m_0)\pi$, define

$$g(\pi) := \bar{m}(\pi)\,p(\pi), \qquad p(\pi) = 1 - \Phi\Big(\tfrac{\bar{m}(\pi)+K-\bar{S}+P(\pi)}{\sigma_\xi}\Big), \tag{61}$$

where $p(\pi)$ is the draft's bid probability (1) evaluated at the price $P(\pi)$ consistent with that posterior. Throughout this subsection g is the only object whose curvature we use.

Remark 15 (Affine baseline and the bid-threshold argument)**.** *Since $\bar{m}(\pi) = m_0 + \nu\pi$ is affine with slope $\nu := \tilde{m} - m_0 > 0$, the curvature of g inherits a single multiplicative factor. To exhibit the mechanism in closed form we adopt the* value-reflecting *price schedule $P(\pi) = \bar{m}(\pi)$; the qualitative conclusions are unchanged for any C^2 schedule. With $P(\pi) = \bar{m}(\pi)$ the standardized bid-threshold argument carries* two *copies of $\bar{m}(\pi)$, so, writing $a_0 := K - \bar{S}$,*

$$a(\pi) = \frac{\bar{m}(\pi) + K - \bar{S} + P(\pi)}{\sigma_\xi} = \frac{2\,\bar{m}(\pi) + a_0}{\sigma_\xi}, \qquad \frac{\mathrm{d}a}{\mathrm{d}\pi} = \frac{2\nu}{\sigma_\xi} =: b. \tag{62}$$

Remark 16 (Calibration fact: the threshold argument is negative)**.** *At the baseline ($m_0 = 0.10$, $\tilde{m} = 0.28$, so $\nu = 0.18$; $K = 0.15$, $\bar{S} = 1.44$, $\sigma_\xi = 0.40$) we have $a_0 = K - \bar{S} = -1.29$, while $2\,\bar{m}(\pi) \in [0.20, 0.56]$ as π ranges over $[0,1]$. Hence the numerator $2\bar{m}(\pi) + a_0 \in [-1.09, -0.73]$ is* strictly negative for every *π, so $a(\pi) < 0$ throughout, and this remains true for every σ_ξ in the calibration family $\{0.10, \ldots, 0.60\}$. This single sign fact governs the rest of the appendix.*

Lemma 22 (Two-point representation of the $D = 0$ block)**.** *Let $\Phi^{D0}(\kappa)$ denote the $D = 0$ contribution to $\Delta^{\min}$. Under the two-point law $\{0, \bar{\pi}\}$, the $\pi = 0$ atom carries the* Exit *mass and the $\bar{\pi}$ atom the pooled $q = 0$ mass: $\Phi^{D0}(\kappa) = \omega_E\, g(0) + (\omega_H + \omega_Q)\, g(\bar{\pi})$, and the full minority object is $\Delta^{\min} = \Phi^{D0} + R(\kappa)$ with R the $D = 1$ (Public Voice) block. Separately, the tower identity $\mathbb{E}[\pi(a)\,\mathbf{1}\{D = 0\}] = \omega_Q$ holds, where $\pi(a) = \mathbf{1}\{a = 1\}$; it certifies the* mass *ω_Q but not the integrand value $g(\bar{\pi})$, and in general $(\omega_H + \omega_Q)\, g(\bar{\pi}) \neq \omega_Q$.*

Proof. At the κ limits the realized $D = 0$ cells separate by trade direction: the Exit trade ($q = -1$) produces the $\pi = 0$ atom with mass ω_E, while Hold and Quiet ($q = 0$) pool at the $\bar{\pi} = \omega_Q/(\omega_H + \omega_Q)$ atom with mass $\omega_H + \omega_Q$ (Lemma 25; Table G.1's cell map). The mass-weighted sum of g gives Φ^{D0}. By the tower rule $\mathbb{E}[\pi(a)\,\mathbf{1}\{D = 0\}] = \mathbb{P}(\text{Quiet}) = \omega_Q$ — consistent, since $(\omega_H + \omega_Q)\,\bar{\pi} = \omega_Q$. The premium object and the mass do not coincide: $g(\bar{\pi}) = \bar{m}(\bar{\pi})\,p(\bar{\pi}) \le \bar{m}(\bar{\pi}) < 1$ at the calibration. □

Curvature of g across the realized chord. Because the realized $D = 0$ law is the two-point law $\{0, \bar{\pi}\}$, the object governing the sign of the partial-equilibrium contribution's interior shape is the curvature of g *across the chord* $[0, \bar{\pi}]$, not pointwise curvature. Define the chord second difference $\mathcal{C}(\bar{\pi}) := g(0) - 2g(\bar{\pi}/2) + g(\bar{\pi})$. For $g \in C^2[0, \bar{\pi}]$ there is $\zeta \in (0, \bar{\pi})$ with $\mathcal{C}(\bar{\pi}) = \frac{1}{4}\bar{\pi}^2 g''(\zeta)$, so we use only the one-directional implication

$$g \text{ strictly concave on } [0, \bar{\pi}] \implies \mathcal{C}(\bar{\pi}) < 0. \tag{63}$$

The converse fails in general, so we never assert "$\mathcal{C} < 0 \iff g'' < 0$ somewhere."

Lemma 23 (Finite, signed curvature of g on $[0,1]$)**.** *Under (61) with $\bar{m}(\pi) = m_0 + \nu\pi$ and $P(\pi) = \bar{m}(\pi)$, $g \in C^\infty[0,1]$ and, with $a(\pi)$ and $b = 2\nu/\sigma_\xi$ as in (62),*

$$g''(\pi) = \bar{m}(\pi)\,p''(\pi) + 2\,\nu\,p'(\pi) = \varphi(a(\pi))\,b\,\big[\bar{m}(\pi)\,a(\pi)\,b - 2\nu\big], \tag{64}$$

where φ is the standard normal density and $p' = -\varphi(a)b$, $p'' = \varphi(a)\,a\,b^2$. In particular g'' is finite for every $\pi \in [0,1]$, including $\pi = 1$.

Proof. $\bar{m}$ is affine and $p(\pi) = 1 - \Phi(a(\pi))$ is C^∞ with a affine; hence $g = \bar{m}\,p$ is C^∞. By the product rule $g'' = \bar{m}\,p'' + 2\nu p'$ (as $\bar{m}'' = 0$). Substituting p', p'' gives (64). Every factor is bounded on $[0,1]$, so g'' is finite throughout. □

E.4 A partial-equilibrium curvature condition and why it does not adjudicate the peak

Lemma 24 (Curvature condition for g)**.** *Maintain the hypotheses of Lemma 23 and define the dimensionless signal-leverage term $T(\pi) := \bar{m}(\pi)\,a(\pi)$. Then, since $\varphi(a) > 0$ and $b > 0$,*

$$g''(\pi) < 0 \iff \bar{m}(\pi)\,a(\pi)\,b - 2\nu < 0 \iff \frac{T(\pi)}{\sigma_\xi} < 1, \tag{65}$$

using $b/(2\nu) = 1/\sigma_\xi$. This is an exact equivalence. By (63), (65) holding on all of $[0,\bar{\pi}]$ implies the chord second difference $\mathcal{C}(\bar{\pi}) < 0$.

Proof. From (64), $\operatorname{sgn} g'' = \operatorname{sgn}(\bar{m}\,a\,b - 2\nu)$; dividing by $2\nu > 0$ and using $T = \bar{m}\,a$, $b/(2\nu) = 1/\sigma_\xi$ gives $T(\pi)/\sigma_\xi < 1$. □

Remark 17 (The curvature condition is satisfied everywhere at the calibration)**.** *By Remark 16, $a(\pi) < 0$ for every π at the baseline and every σ_ξ in the calibration family. Hence $T(\pi) = \bar{m}(\pi)\,a(\pi) < 0$, so $T(\pi)/\sigma_\xi < 0 < 1$ for every π and every $\sigma_\xi > 0$. By Lemma 24, g is strictly concave on $[0,1]$ at every point of the calibration family—including the $\sigma_\xi = 0.60$ cells that the numerics certify as* troughs*. The curvature condition* cannot fail *anywhere in $\{(\sigma_\xi, \bar{S}) : \sigma_\xi \in [0.10, 0.60],\ \bar{S} \in [1.24, 1.64]\}$, and so cannot distinguish a hump from a trough there. It is non-vacuous only where $2\,\bar{m}(\pi) \geq \bar{S} - K$, i.e. $\bar{S} \lesssim 0.71$, disjoint from the paper's $\bar{S} \in [1.24, 1.64]$.* ***Consequently the hump/trough orientation in the paper's calibration is produced entirely by the general-equilibrium cutoff-shift channel of the next subsection, not by the curvature of g.***

E.5 Endpoint symmetry, interior peak, and the GE channel

Lemma 25 (Endpoint mass symmetry from the noise law)**.** *Let order flow be $X = q + z$ with $q \in \{-1, 0, +1\}$ and noise $z \in \{-1, 0, 1\}$, $\mathbb{P}(z{=}0) = 1 - \kappa$, $\mathbb{P}(z{=}\pm 1) = \kappa/2$, and $D = \mathbf{1}\{q = +1\}$. Fix the equilibrium cutoffs (k_1, k_0, k_D). Then in both limits $\kappa \to 0^+$ and $\kappa \to 1^-$ the realized $D = 0$ engagement posterior is the* same *two-point law $\{0, \bar{\pi}\}$, with the $\pi = 0$ atom carrying the Exit mass ω_E, the $\pi = \bar{\pi}$ atom carrying the entire $q{=}0$ mass $\omega_H + \omega_Q$, and $\bar{\pi} = \omega_Q/(\omega_H + \omega_Q)$.*

Proof. $\kappa \to 0^+$. $\mathbb{P}(z{=}0) \to 1$, so $X = q$ a.s.; on $D = 0$ ($q \in \{-1, 0\}$), $X = -1$ identifies Exit ($\pi = 0$) and $X = 0$ identifies the $q{=}0$ block, within which Hold and Quiet are indistinguishable, giving $\bar{\pi} = \omega_Q/(\omega_H + \omega_Q)$. $\kappa \to 1^-$. $\mathbb{P}(z{=}\pm 1) \to \frac{1}{2}$. The conditional supports are $q = -1 : \{-2, 0\}$, $q = 0 : \{-1, +1\}$, $q = +1 : \{0, +2\}$. On $D = 0$ we condition on $q \neq +1$, removing the $q = +1$ branch; the remaining supports $\{-2, 0\}$ and $\{-1, +1\}$ are disjoint, so X again separates Exit from the $q{=}0$ block, within which Hold and Quiet remain indistinguishable. The common $\kappa/2$ factor cancels in the posterior ratio, leaving $\bar{\pi} = \omega_Q/(\omega_H + \omega_Q)$ as at $\kappa \to 0^+$. Both limits deliver the identical two-point law and mass map. □

Proposition 16 (Endpoint symmetry and an interior maximizer)**.** *Maintain the persuasion regime of Definition 2, so Proposition 15 delivers the four-action ordering for every* $\chi \geq 0$*. Suppose* $\Delta^{\min}$ *is continuous on* $[0,1]$ *and the cutoffs* (k_1, k_0, k_D) *converge to a common triple as* $\kappa \to 0^+$ *and* $\kappa \to 1^-$*. Then:* (i) $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$*;* (ii) *if there exists interior* $\kappa_0 \in (0,1)$ *with* $\Delta^{\min}(\kappa_0) > \Delta^{\min}(0^+)$*, then* $\Delta^{\min}$ *attains its maximum at an interior* $\kappa^\dagger \in (0,1)$*.*

Proof. (i) By Lemma 25 the realized $D = 0$ law is the same two-point set at both limits with masses identical functions of the cutoffs; by Lemma 22, Φ^{D0} is then a fixed function of $(\omega_H, \omega_Q, \bar{\pi})$. The same disjoint-support argument makes the $D = 1$ block R a fixed function of the cutoffs at both limits. By cutoff convergence, both take identical values at the two ends, so $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$. *(ii)* $\Delta^{\min}$ is continuous on compact $[0,1]$, so by Weierstrass it attains a maximum; by (i) the endpoints share the common value and by hypothesis there is an interior point strictly above it, so the maximizer is interior. No curvature sign is used. □

Conditional Claim 1 (Strict single peak under a general-equilibrium hypothesis)**.** *Maintain the hypotheses of Proposition 16 and suppose in addition* (GE-dom)*:* $\Delta^{\min}$ *is strictly concave on* $(0,1)$*, i.e. the cutoff-shift term* $\sum_{i\in\{1,0,D\}}(\partial\Delta^{\min}/\partial k_i)(\mathrm{d}k_i/\mathrm{d}\kappa)$ *is dominated, uniformly on* $(0,1)$*, by the negative direct (integrand) term. Then* $\Delta^{\min}$ *has a unique interior maximizer* $\kappa^\dagger \in (0,1)$*.*

Proof. Under (GE-dom), $\Delta^{\min}$ is strictly concave on $(0,1)$; a strictly concave function on an interval has at most one local maximum, which is then global. With Proposition 16(ii), the maximizer is unique. □

Remark 18 (The chord is a numerical diagnostic; the peak is conditional on (GE-dom))**.** *Three points must be kept distinct. (1) The single-peak* engine *is (GE-dom),* not *the curvature condition. The cutoff-shift term is genuinely unsigned because* $\Delta^{\min}$ *is minority welfare, not the blockholder's objective, so the envelope theorem does not annihilate it; (GE-dom) is a stated, numerically maintained hypothesis, not a theorem. (2) The curvature condition is neither necessary nor sufficient for the peak: at the calibration it holds everywhere (Remark 17) yet the numerics find both humps and troughs across* $(\sigma_\xi, \bar{S})$*. (3) The chord second difference* $\mathcal{C}$ *is retained only as a numerical diagnostic; it matches the realized hump/trough orientation in* 16/20 *cells (against* 8/20 *for pointwise curvature), so it is the better heuristic—but a* 16/20 *correlation is a diagnostic, not a proof. The headline is therefore:* existence *of an interior peak with endpoint symmetry is unconditional (Lemma 25, Proposition 16); the* strict single peak *is a numerical regularity holding under (GE-dom) at the baseline, where* $\kappa^\dagger \approx 0.59$ *with amplitude* $\approx$ 14–16% *of level, and the profile flips to a shallow trough in* 4/20 *cells (all at* $\sigma_\xi = 0.60$*) through the GE cutoff-shift channel.*

E.6 Summary of what A2 buys

1. **Ordering (Proposition 15).** The four-action threshold structure survives any cost profile in the persuasion regime $C'(s) < \Lambda(s)$; A2 ($\chi > 0$) is sufficient, not necessary, with $k_0 \in (k_1, k_D)$ pinned by the boundary signs.

2. **Flat cost ($\chi = 0$).** All existence, ordering, and comparative-statics results survive; only the level of k_0 shifts.

3. **Inversion.** The ordering can invert only under entrenchment-type cost, which carries the opposite, falsifiable empirical signature (Remark 14).

4. **Interior peak (Lemma 25, Proposition 16, Conditional Claim 1).** Endpoint symmetry $\Delta^{\min}(0^+) = \Delta^{\min}(1^-)$ is derived from the noise law plus cutoff convergence; the existence of an interior maximizer is unconditional (Weierstrass); strict single-peakedness is governed by the GE cutoff-shift channel (GE-dom): Appendix F now certifies it as a theorem, up to ε-localization of the peak, on an interior region at the baseline ($\kappa^\dagger \approx 0.59$, amplitude $\approx$ 14–16%) and certifies its failure at the $\sigma_\xi = 0.60$ cells (Proposition 17).

F The general-equilibrium cutoff-shift channel: a region theorem and a counterexample

Recall the decomposition (16),

$$\frac{\mathrm{d}\Delta^{\min}}{\mathrm{d}\kappa} = \underbrace{E'(\kappa)}_{\text{(A) posterior/pricing}} + \underbrace{B(\kappa)}_{\text{(B) GE cutoff-shift}}, \qquad B(\kappa) = \sum_{i\in\{1,0,D\}} \frac{\partial\Phi}{\partial k_i}\frac{\mathrm{d}k_i}{\mathrm{d}\kappa},$$

with $\Delta^{\min}(\kappa) = \Phi(k(\kappa),\kappa)$ as in Lemma 16. Channel (A) is signed by Lemma 8 under (C*); the endpoints are pinned by Lemma 7. This appendix controls channel (B). Throughout, f denotes the signal density, $\omega = (\omega_E, \omega_H, \omega_Q, \omega_P)$ the action probabilities, and the seven feasible (X, D) cells carry posteriors $\pi(X, D)$, prices $P(X, D)$, premia $\bar{m}(X, D) = m_0 + (\widetilde{m} - m_0)\pi(X, D)$, and bid probabilities $p(X, D)$ from (1).

F.1 The pricing layer is differentiable with explicit sensitivities

Lemma 26 (Pricing-layer IFT under A5)**.** *Fix a cell* (X, D) *and view the pricing fixed point* (2) *as* $P = \delta\big[p(P,\bar{m})\,(P+\bar{m}) + (1-p(P,\bar{m}))\,\hat{V}\big]$*, an equation in* P *given the cell inputs* $u = (\pi, \hat{V})$ *with* $\bar{m} = m_0 + (\widetilde{m} - m_0)\pi$ *and* $\hat{V} = \mathbb{E}[v \mid X, D] + \widetilde{\Delta}\,\pi$*. Under Assumption* (A5) *the feedback margin*

$$\ell(P,u) \;\equiv\; \delta\Big[p + \frac{\partial p}{\partial P}\big(P + \bar{m} - \hat{V}\big)\Big]$$

satisfies $\ell < 1$ *along the equilibrium path, the solution* $P^*(u)$ *is unique and* C^1*, and*

$$\frac{\partial P^*}{\partial u_j} = \frac{\delta\,\partial_{u_j}\big[p\,(P+\bar{m}) + (1-p)\hat{V}\big]\big|_P}{1-\ell(P^*,u)}, \qquad \Big|\frac{\partial P^*}{\partial u_j}\Big| \;\le\; \frac{\delta\,\big|\partial_{u_j}[\cdot]\big|_P}{1-\bar{\ell}},$$

for any uniform margin bound $\bar{\ell} < 1$*. In particular all cell objects* $(P^*, p, \bar{m})$ *are* C^1 *functions of* (ω, κ) *through the Bayes maps of Proposition 2, with explicit derivatives by the chain rule.*

Proof. $\Phi(\cdot)$ and $\varphi(\cdot)$ are smooth, so the right side of the fixed-point equation is C^1 in (P, u). A uniform margin $\bar{\ell} < 1$ along the path is part of the maintained Assumption (A5a): the body's sufficient condition $\delta/\sigma_\xi < 1/\varphi(0)$ bounds only the feedback component of ℓ, not the direct term δp (Remark 2 in the body makes exactly this point), but the *signed* ℓ is small at the sampled equilibrium cells ($\ell \in [0.03, 0.43]$; the value gap $P+\bar{m}-\hat{V}$ enters with the sign that shrinks ℓ there); the implicit function theorem then yields uniqueness, C^1-ness, and the displayed quotient, whose denominator is $1-\ell$. The Bayes maps $\omega \mapsto \pi(X,D)$ are rational functions with positive denominators on reached cells (full-support noise), hence C^1; composition gives the final statement. $\square$

F.2 The cutoff layer: operator bound from the contraction modulus

Write the equilibrium system as $F(k,\kappa) = k - T(k,\kappa) = 0$, where T is the best-response map of Section 4 (the indifference-point update whose damped iteration the solver implements, with prices and posteriors evaluated equilibrium-consistently at (k,κ) via Lemma 26).

Lemma 27 (Cutoff-layer IFT with an inversion-free bound)**.** *Suppose Assumption* (A6) *holds in the quantified form: $T(\cdot,\kappa)$ is C^1 and $\|D_kT(k,\kappa)\|_\infty \le L < 1$ in a neighbourhood of the equilibrium path. Then $k(\kappa)$ is C^1,*

$$\frac{\mathrm{d}k}{\mathrm{d}\kappa} = \big(I - D_kT\big)^{-1}\,\partial_\kappa T, \qquad \Big\|\frac{\mathrm{d}k}{\mathrm{d}\kappa}\Big\|_\infty \;\le\; \frac{\|\partial_\kappa T(k(\kappa),\kappa)\|_\infty}{1-L},$$

and hence, with weights $W_i(\kappa) \equiv |\partial\Phi/\partial k_i|$,

$$|B(\kappa)| \;\le\; \bar{B}(\kappa) \;\equiv\; \Big(\sum_i W_i(\kappa)\Big)\,\frac{\|\partial_\kappa T(k(\kappa),\kappa)\|_\infty}{1-L}. \tag{66}$$

Every factor in $\bar{B}$ is computable from a single equilibrium solve at κ plus one finite-difference column $\partial_\kappa T$; no Jacobian inversion and no knowledge of the true path derivative $\mathrm{d}k/\mathrm{d}\kappa$ *is required.*

Proof. $D_kF = I - D_kT$ with $\|D_kT\|_\infty \le L < 1$, so $I - D_kT$ is invertible with the Neumann bound $\|(I-D_kT)^{-1}\|_\infty \le \sum_{n\ge 0} L^n = 1/(1-L)$. The IFT gives C^1-ness of $k(\kappa)$ and $\mathrm{d}k/\mathrm{d}\kappa = -(D_kF)^{-1}\partial_\kappa F = (I-D_kT)^{-1}\partial_\kappa T$ (using $\partial_\kappa F = -\partial_\kappa T$). Submultiplicativity gives the displayed bound, and (66) follows from the triangle inequality on B. The computability claim is by inspection: W_i from Lemma 28 below at the solved $(k(\kappa),\kappa)$, the modulus L from a finite-difference D_kT (or any certified upper bound), and $\partial_\kappa T$ from one κ-perturbed best-response evaluation at fixed k. $\square$

Relative to Lemma 16, whose bound $|B| \le (\sum_i |\partial\Phi/\partial k_i|)\|\mathrm{d}k/\mathrm{d}\kappa\|_\infty$ requires the true equilibrium derivative along the whole path, (66) replaces that object with the contraction modulus already maintained by Assumption (A6). This is what converts the transfer inequality from an ex-post diagnostic into an ex-ante checkable hypothesis.

F.3 The weights: explicit structure and conditional signs

Lemma 28 (Weight decomposition)**.** *Write $\Phi(k,\kappa) = \sum_{(X,D)} \mu_{X,D}(k,\kappa)\,\bar{m}(X,D)\,p(X,D)$, where $\mu_{X,D}$ is the cell mass induced by $(\omega(k),\kappa)$ and $(\bar{m},p)$ are evaluated at the cell objects of Lemma 26. Then for $i \in \{1,0,D\}$,*

$$\frac{\partial\Phi}{\partial k_i} = \underbrace{f(k_i)\sum_{(X,D)} \frac{\partial\mu_{X,D}}{\partial\omega}\cdot e_i\;\bar{m}\,p}_{\text{boundary (mass-shift) term}} \;+\; \underbrace{\sum_{(X,D)} \mu_{X,D}\,(\widetilde{m}-m_0)\Big[p + \bar{m}\,\frac{\partial p}{\partial\bar{m}}\Big]\frac{\partial\pi(X,D)}{\partial k_i}}_{\text{inference term}} \;+\; \underbrace{\sum_{(X,D)} \mu_{X,D}\,\bar{m}\,\frac{\partial p}{\partial P}\,\frac{\partial P^*}{\partial k_i}}_{\text{pricing term}}, \tag{67}$$

where e_i is the signed pair of adjacent-region mass shifts at the boundary k_i (raising k_1 moves $f(k_1)\,\mathrm{d}k$ of mass from Hold to Exit; k_0 from Quiet to Hold; k_D from Public to Quiet), $\partial\pi/\partial k_i$ and $\partial P^*/\partial k_i$ are the Bayes and pricing chain-rule derivatives through $\omega(k)$, and $\partial p/\partial\bar{m} = -\varphi(t)/\sigma_\xi$, $\partial p/\partial P = -\varphi(t)/\sigma_\xi$ with $t = (\bar{m} + K - \bar{S} + P)/\sigma_\xi$. All terms are explicit compositions of $(f, \Phi$-cdf differences, $\varphi)$ and the equilibrium cell objects. Numerically, the script evaluates the weights W_i by central finite differences of Φ in each cutoff (the decomposition above guarantees these derivatives exist and are continuous) and spot-validates the pricing-layer quotient of Lemma 26 cell by cell.

Proof. Differentiate the finite sum term by term. Cell masses depend on k only through $\omega(k)$, whose components are Gaussian probabilities of the action regions, giving the boundary terms with density factor $f(k_i)$ and adjacent-region transfer e_i (the noise mixture is k-free). The remaining dependence is through $\bar{m}(\pi)$ and $p(\bar{m}, P)$: the product rule on $\bar{m}\,p$ gives the inference bracket, and Lemma 26 gives the pricing term with $\partial P^*/\partial k_i$ well defined. Finiteness and continuity are inherited from the smooth ingredients. □

Remark 19 (Conditional signs, in the spirit of monotone comparative statics)**.** *Two cell-level sign statements are available conditionally; we use neither as an input to Theorem 3, and we record their status at the baseline equilibrium honestly (direct computation at the $\kappa = 0.5$ solve; these are* not *certified by the check script): (i)* Deterrence moderation. *The inference bracket in* (67) *is positive iff $p > \bar{m}\,\varphi(t)/\sigma_\xi$ – the cell's bid probability exceeds its deterrence slope. Note Remark 16's $a(\pi) < 0$ is derived under the surrogate $P(\pi) = \bar{m}(\pi)$, not equilibrium prices; at the actual baseline equilibrium $t > 0$ on most reached cells, and the bracket condition* fails *on the disclosed ($\pi = 1$) cells — harmlessly, since $\partial\pi/\partial k_i = 0$ there at the collapsed baseline, so those cells carry no inference weight. On the two operative inferred cells the condition holds with margins $+0.38$ at $(X, D) = (-1, 0)$ and $+0.12$ at $(0, 0)$. It is the single-crossing condition under which the cell contribution $\bar{m}\,p$ increases in the posterior π (Milgrom–Shannon, applied to the cell payoff in $(\pi; \sigma_\xi)$). (ii)* Boundary-term signs. *e_D moves mass from the disclosed branch ($\pi = 1$, κ-invariant by Appendix B.3) to the inferred $\bar{\pi}$-atom, so its sign is that of $\left[\bar{m}\,p\right]_{\bar{\pi}\text{-cells}} - \left[\widetilde{m}\,\bar{p}_1\right]$. At the baseline this difference is* positive *(per unit mass, ≈ 0.074 vs ≈ 0.0037): disclosure deters bids through the price run-up, so moving mass from Public to Quiet* raises *the premium-weighted bid mass — the opposite of the naïve "disclosure carries the premium with certainty" intuition. e_1 shifts Exit/Hold composition and operates purely through $\bar{\pi}$. These signs can sharpen the bound $\bar{B}$ cell by cell where they hold but are not needed for Theorem 3.*

F.4 The region theorem: GE-dominance as a checkable hypothesis

Theorem 3 (Strict single peak, up to ε-localization, on a certified region)**.** *Maintain Assumptions* A1–A7*, the quantified contraction of Lemma 27, and* (C*) *on $[\underline{\kappa}, \overline{\kappa}]$. Let $E(\kappa)$ denote channel (A): the antiderivative, along the equilibrium path $k(\kappa)$, of the frozen-cutoff derivative $E'(\kappa) = \partial_\kappa \Phi(k, \kappa)\big|_{k=k(\kappa)}$. Suppose additionally*

(R0) channel (A) single-peaked: *E' is positive before and negative after a unique interior crossing $\kappa_A^\dagger \in (\underline{\kappa}, \overline{\kappa})$. (Under* (C*)*, Lemma 8 delivers an interior maximum of E strictly above the endpoint values; the single-crossing form is the additional regularity recorded as* (NR1) *in the D1 record, and is verified here directly by the script's sign-change count of the channel-(A) profile.)*

and that there is $\varepsilon > 0$ such that

(R1) pointwise dominance off the peak: $\bar{B}(\kappa) < |E'(\kappa)|$ *for all* $\kappa \in [\underline{\kappa}, \overline{\kappa}] \setminus (\kappa_A^\dagger - \varepsilon, \kappa_A^\dagger + \varepsilon)$*;*

(R2) integral control through the peak: $\int_{\kappa_A^\dagger-\varepsilon}^{\kappa_A^\dagger+\varepsilon} \bar{B}(\kappa)\,\mathrm{d}\kappa \quad < \quad \min\Big\{E(\kappa_A^\dagger - \varepsilon),\, E(\kappa_A^\dagger + \varepsilon)\Big\} - \max\Big\{E(\underline{\kappa}), E(\overline{\kappa})\Big\}$.

Then $\Delta^{\min}$ is strictly increasing on $[\underline{\kappa}, \kappa_A^\dagger - \varepsilon]$, strictly decreasing on $[\kappa_A^\dagger + \varepsilon, \overline{\kappa}]$, and attains its global maximum at an interior $\kappa^\dagger \in (\kappa_A^\dagger - \varepsilon, \kappa_A^\dagger + \varepsilon)$. In particular $\mathrm{d}\Delta^{\min}/\mathrm{d}\kappa$ changes sign exactly once outside *the ε-ball, with any further sign changes (an even number on each side of the maximizer) confined to the ball. The conclusion of Hypothesis 1 is thereby discharged on $[\underline{\kappa}, \overline{\kappa}]$ in the ε-*localized *form: $\Delta^{\min}$ is single-peaked up to ε-localization of its peak; the pointwise single-crossing of the derivative* inside *the ball is not claimed (and does not follow from* (R1)–(R2)*, which control B there only in integral).*

Proof. Off the ε-ball, $|B| \le \bar{B} < |E'|$ by (R1) and Lemma 27, so $\mathrm{d}\Delta^{\min}/\mathrm{d}\kappa = E' + B$ carries the sign of E': positive on $[\underline{\kappa}, \kappa_A^\dagger - \varepsilon]$ (where $E' > 0$ by (R0)), negative on $[\kappa_A^\dagger + \varepsilon, \overline{\kappa}]$. Hence $\Delta^{\min}$ is strictly monotone on the two outer segments, any maximizer lies in the closed ε-ball, and the derivative has exactly one sign change outside the ball. The ball's edges are not maximizers: (R1) holds *at* $\kappa_A^\dagger \pm \varepsilon$ (they lie in the off-ball set as defined), so $\Delta^{\min\prime}(\kappa_A^\dagger - \varepsilon) > 0$ and $\Delta^{\min\prime}(\kappa_A^\dagger + \varepsilon) < 0$, and $\Delta^{\min}$ is C^1 by Lemma 27; hence the global maximizer is *interior* to the ball. (R2) supplies level control: integrating $\Delta^{\min\prime} = E' + B$ from either ball edge, the ball's accumulated $\bar{B}$-mass is too small to push any ball value below the outer-segment endpoints, so the maximum over the ball strictly exceeds $\Delta^{\min}(\underline{\kappa})$ and $\Delta^{\min}(\overline{\kappa})$ — the interior maximizer is global. Inside the ball (R1) is silent and (R2) bounds B only in integral, so pointwise sign changes of the derivative there cannot be excluded; the conclusions are exactly as stated. (As $\varepsilon \downarrow 0$ under continuity of $\bar{B}$ the localization tightens to $\kappa_A^\dagger$.) □

Corollary 3 (Baseline instantiation – certified interval)**.** *At the baseline calibration the script `d8_ge_dominance_check.py` evaluates L, $\|\partial_\kappa T\|_\infty$, the weights of Lemma 28, and the channel-(A) profile on $\kappa \in [0.30, 0.85]$ (23 grid points, spacing 0.025; the statements below are grid-point certificates with a continuity appeal between nodes, not interval arithmetic), and reports (JSON artifact): contraction modulus $L \le 0.836 < 1$ at every node; channel-(A) peak at $\kappa \approx 0.60$ with a single sign change of E' on the grid (*(R0)*);* (R1) *holding at* every *off-ball grid point and* (R2) *with two orders of magnitude to spare (ball integral 2.5×10^{-5} against margin 4.7×10^{-3}) under the exact-IFT bound, certifying $[0.30, 0.85]$ — a span that is* grid-limited, not failure-limited *(the exact bound also passes beyond both ends of the tested grid); and – the operative result – the* inversion-free *bound* (66) *alone certifying $[0.35, 0.825]$, whose endpoints* are *genuinely bracketed by failures (R1 fails at 0.325 and 0.85 under the loose bound). On these intervals the equilibrium hump is a theorem in the ε-localized sense of Theorem 3, conditional on the maintained Assumptions* A1–A7, (C*)*, and the grid-verified* (R0)*; Hypothesis 1 is discharged there in that sense and remains a (falsifiable) regularity in the residual endpoint neighbourhoods. Two further notes for honesty: the entire baseline path sits on the collapsed-Hold manifold ($k_0 = k_1$ at all 23 nodes), so the IFT machinery is applied to the effective two-cutoff system, with persistence of the collapse observed at* *every node; and the loose-bound margins at the certified boundary are thin (*2.9% *at* 0.35*,* 6.1% *at* 0.825*), so sub-grid violations there, while not found in finer probes, are not formally excluded.*

F.5 The counterexample: GE-dominance is false as a global claim

Proposition 17 (A certified trough with channel (A) single-peaked)**.** *Consider the robustness cells of the calibration family with $\sigma_\xi = 0.60$ (Appendix E, Remark 18). There: (i) the curvature condition holds everywhere – indeed it cannot fail on the family (Remark 17 gives $a(\pi) < 0$, hence $p' < 0$ and $p'' < 0$, so both the cell payoff $g(\pi) = \bar{m}(\pi)p(\pi)$ and Lemma 8's object $h(\pi) = \pi\, p(\pi)$, with $h'' = 2p' + \pi p'' < 0$, are concave:* (C*) *holds) – so channel (A) has an interior maximum above equal endpoints by Lemma 8, and the script verifies the single-crossing form* (R0) *cell by cell; (ii) the full-equilibrium profile $\Delta^{\min}(\kappa)$ exhibits an interior* trough*: the script certifies troughs at $(\sigma_\xi, \bar{S}) = (0.60, 1.44), (0.60, 1.54), (0.60, 1.64)$ with interior minimizers $\kappa \in \{0.275, 0.35, 0.475\}$ and depths up to 2.8×10^{-3}, channel (A) single-peaked in every cell, and $\int |B|$ exceeding the channel-(A) amplitude in every cell (e.g.* 0.021 *vs* 0.0069 *at $\bar{S} = 1.64$); hence (iii) $B(\kappa)$ overturns channel (A) on these cells, and the conclusion of Hypothesis 1* fails*. Consequently no proof of the unconditional hump can dispense with a restriction on the GE channel: the region hypothesis of Theorem 3 is not removable, and at the $\sigma_\xi = 0.60$ cells the theorem's certification* must *fail — given (i), (ii), and the trough, no $(\varepsilon, \mathrm{R1}, \mathrm{R2})$ configuration can hold there whenever the contraction and* (R0) *hypotheses do (modus tollens on Theorem 3).*

Proof. (i) is Remark 17 plus the concavity bridge displayed in the statement and Lemma 8. (ii) is the certified computation: the script re-solves the cells at the standard residual gate (5×10^{-3}) on the solver's reference path, evaluates $\Delta^{\min}$ on a fine κ grid, computes channel (A) by κ-perturbation at frozen cutoffs and B as the residual of the total derivative, and reports an interior minimum with $\int |B|$ exceeding the channel-(A) amplitude (JSON artifact). (iii) follows from (i)+(ii) by the decomposition (16): with equal endpoints and (A) single-peaked in the single-crossing sense of (R0), a trough requires $E' + B$ to change sign in the opposite pattern, impossible unless $|B| > |E'|$ on a set of positive measure. The final claims are immediate. □

Remark 20 (Economic boundary: where and why GE feedback wins)**.** *Channel (A)'s amplitude is a deterrence object: it is the chord gap of the concave cell payoff $g(\pi) = \bar{m}(\pi)p(\pi)$ (Lemma 8's $h(\pi) = \pi\, p(\pi)$ is concave under the same condition), whose scale is governed by the bid-incidence sensitivity $|\partial p/\partial \bar{m}| = \varphi(t)/\sigma_\xi$. As σ_ξ grows, entry becomes insensitive to the activism premium, the chord gap flattens, and the fixed-cutoff hump shrinks – while the cutoff system's response to κ (the B weights, driven by trading rents and disclosure composition) does not shrink in step. GE feedback therefore dominates precisely in the high-σ_ξ corner, which is where the troughs live. The testable content: the liquidity hump in minority gains should be strongest where takeover entry is most sensitive to expected resistance (low effective σ_ξ), and can invert where entry is premia-insensitive.*

F.6 What this buys downstream

The machinery applies verbatim to the welfare objects: W_B and W_{bid} have their own weight sums $W_i^{(B)}$, $W_i^{(\text{bid})}$ (computed by the same decomposition as Lemma 28), so the bundle $\mathcal{B}(\kappa)$ of Proposition 7 obeys $|\mathcal{B}(\kappa) - \partial_\kappa W_B|_k - \mathrm{d}W_{\text{bid}}/\mathrm{d}\kappa| \leq (\sum_i W_i^{(B)})\, \|\partial_\kappa T\|_\infty/(1-L)$, putting a computable band around the liquidity-planner wedge of Proposition 7 and the disclosure-planner terms of Proposition 8. We do not re-derive the welfare propositions here; the point is that their conditional clauses ((W*) and the unsigned GE terms) inherit the same certify-or-falsify structure, replacing ”unsigned” with ”signed up to a computable band.”

F.7 Ledger

Proved: Lemmas 26–28, Theorem 3, and Proposition 17(iii) given (ii). *Computable-verified at the calibration (JSON artifacts):* the contraction modulus L; the certified interval of Corollary 3 with margins; the $\sigma_\xi = 0.60$ trough and its channel decomposition. *Open:* a calibration-free analytic modulus for (A6); a global analytic sign for B (now known to be impossible in the strong form: Proposition 17); endpoint neighbourhoods (both E' and B degenerate at the exact grid edges, as flagged in Appendix B.14).

G Tables

D	Order flow X	$P(X,D)$	$\pi(X,D)$	$p(X,D)$	$m(X,D)$
0	-2	0.84	0.00	0.81	0.10
0	-1	1.06	0.41	0.55	0.17
0	0	1.23	0.74	0.33	0.23
0	1	1.39	1.00	0.17	0.28
1	0	1.90	1.00	0.01	0.28
1	1	1.90	1.00	0.01	0.28
1	2	1.90	1.00	0.01	0.28

Table G.1: Baseline numerical equilibrium (values rounded to two decimals). Parameters: $\mu = 1$, $\sigma_v = 0.5$, $\sigma_\varepsilon = 0.5$, $\delta = 0.95$, $\Delta = 0.25$, $C_0 = 0.12$, $\chi = 0.5$, $\kappa = 0.5$, $m_0 = 0.10$, $m_1 = 0.30$, $\bar{S} = 1.44$, $K = 0.15$, $\sigma_\xi = 0.40$, $\rho = 0.9$. Cutoffs: $k_1 = k_0 \approx 0.82$ and $k_D \approx 2.26$.

X	$\pi(X,0)$	$\pi_{\text{ND}}(X)$	$\pi_{\text{NR}}(X,0,R=0)$	$\pi_{\text{NR}}(X,0,R=1)$
-1	0.41	0.41	0.19	0.68
0	0.74	0.74	0.48	0.89
1	1.00	1.00	1.00	1.00

(q,a)	π_{FI}
$(-1,0)$	0.00
$(0,0)$	0.00
$(0,1)$	1.00
$(1,1)$	1.00

Table G.2: Illustration of posterior engagement probabilities under alternative disclosure signals (Section 8), evaluated at the baseline calibrated equilibrium cutoffs and $\kappa = 0.5$. For NR, the rumor parameters are set to $(\eta_1, \eta_0) = (0.75, 0.25)$.

Parameter	Symbol	Value	Interpretation
Fundamentals			
Prior mean	μ	1.00	Normalized baseline value
Fundamental volatility	σ_v	0.50	Moderate uncertainty
Signal noise	σ_ε	0.50	Informative but noisy signal
Engagement			
Base engagement cost	C_0	0.12	12% of baseline value
Cost sensitivity	χ	0.50	Moderate signal-dependence
Success probability	ρ	0.90	High success rate
Value improvement	Δ	0.25	25% improvement if successful
Takeover			
Base premium	m_0	0.10	10% premium without activism
Activism premium	m_1	0.30	30% premium with activism
Baseline synergy	$\bar{S}$	1.44	
Bidding cost	K	0.15	
Synergy volatility	σ_ξ	0.40	
Other			
Discount factor	δ	0.95	$\delta = \exp(-r\tau)$ between pricing and settlement
Noise intensity	κ	0.50	Baseline (varies in analysis)

Table G.3: Baseline parameterization for numerical analysis.

H Figures

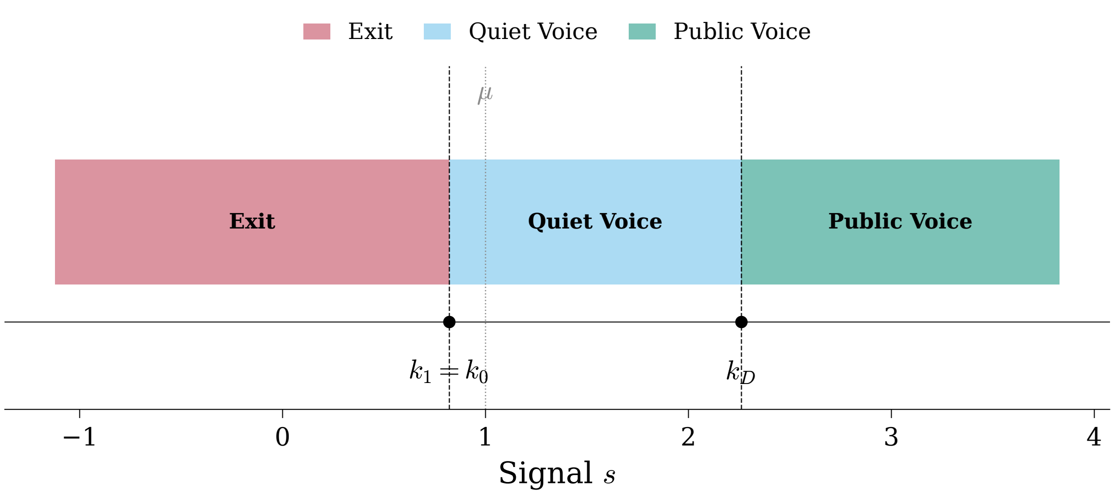

Figure H.1: Equilibrium cutoff structure. The blockholder follows a threshold strategy with weakly ordered cutoffs. In general, the signal space can be partitioned into up to four regions (Exit, Hold, Quiet Voice, Public Voice), with disclosure occurring only under Public Voice. In the baseline numerical example, $k_0 = k_1$, so the strategy reduces to Exit, Quiet Voice, and Public Voice.

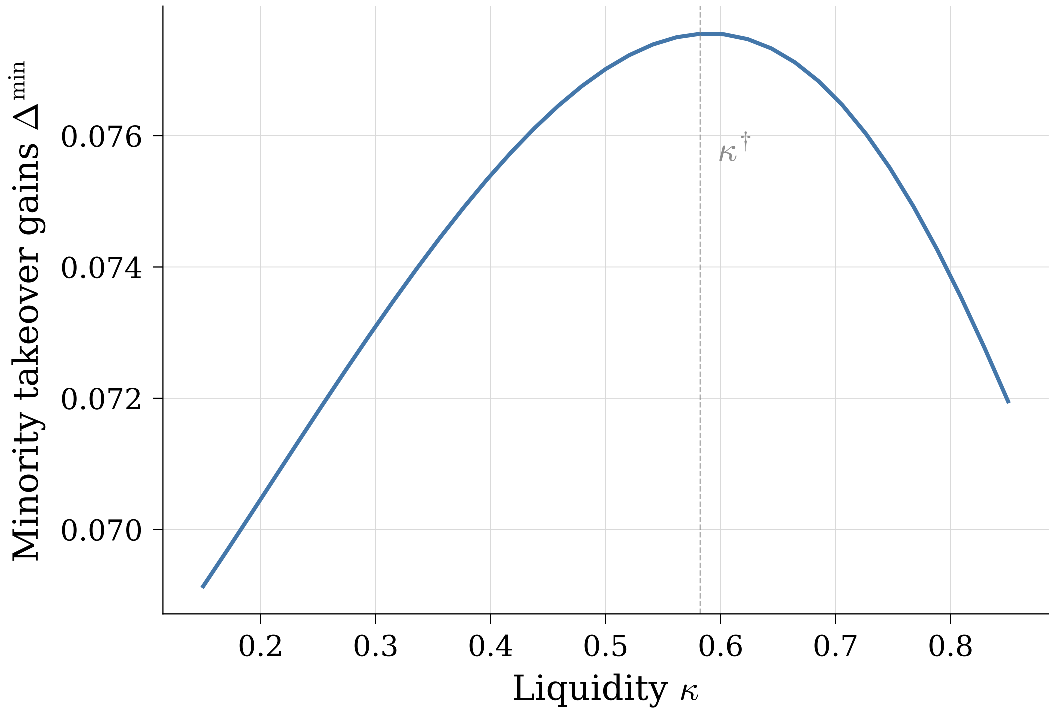

Figure H.2: Nonmonotonic effect of liquidity on expected minority takeover gains. The vertical dashed line marks the turning point $\kappa^\dagger$ (the maximizer of $\Delta^{\min}$ on the plotted grid) in the baseline calibration.

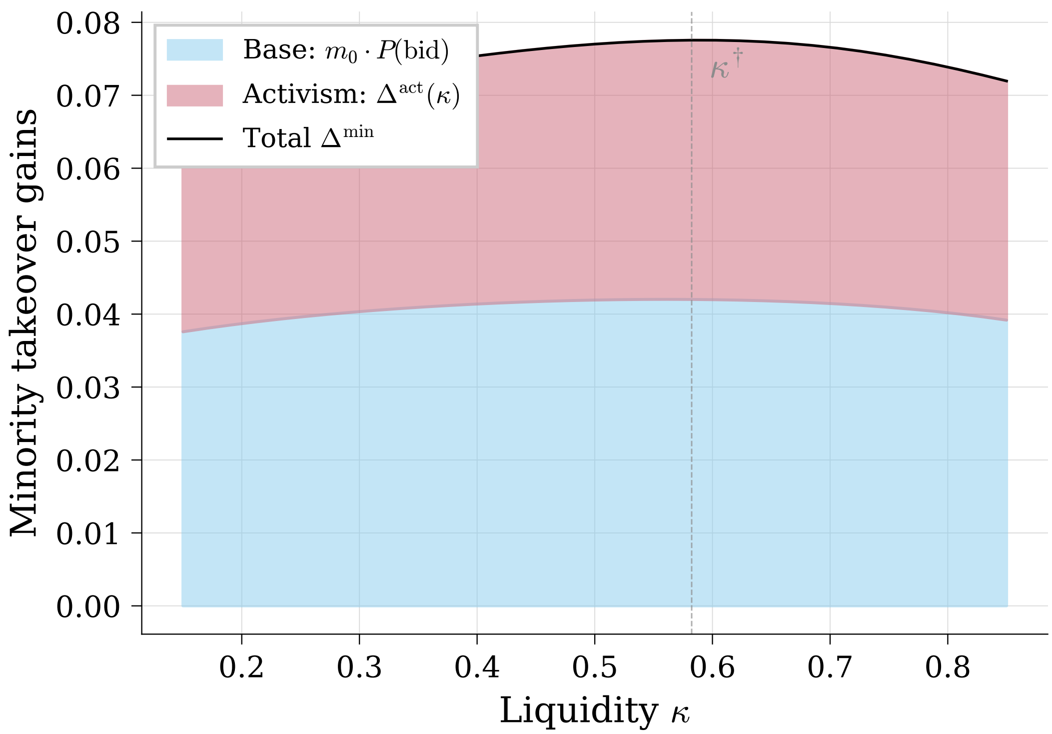

Figure H.3: Decomposition of minority takeover gains. The base component $m_0 \cdot \mathbb{P}(\text{bid})$ and the activism component $\Delta^{\text{act}}(\kappa)$ vary smoothly with liquidity in the baseline calibration; the dashed line marks $\kappa^\dagger$.

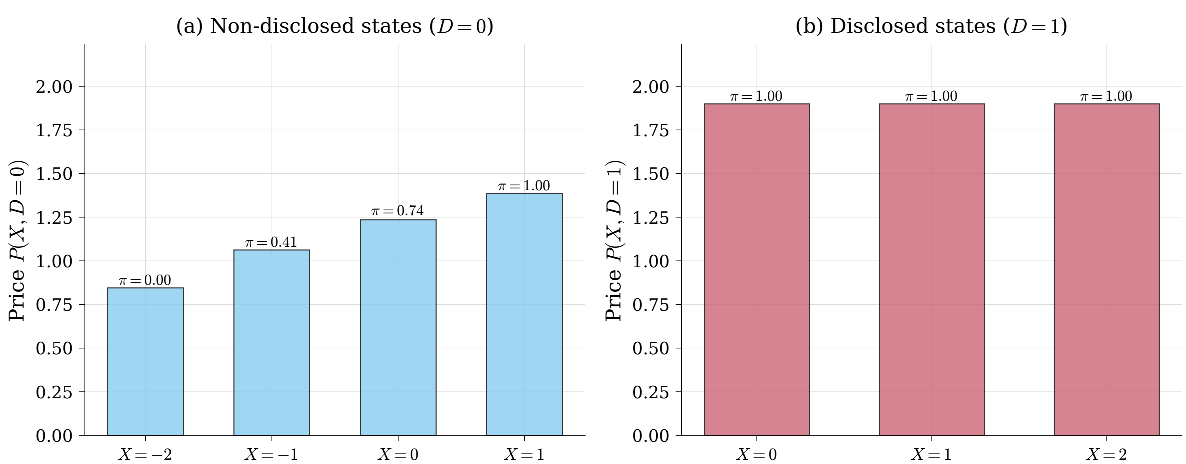

Figure H.4: Equilibrium prices by state (X, D) in the baseline numerical example. Left panel: nondisclosed states ($D = 0$). Right panel: disclosed states ($D = 1$). Bars are shown for on-path states under the calibrated strategy. Numbers above bars indicate the posterior engagement probability $\pi(X, D)$.

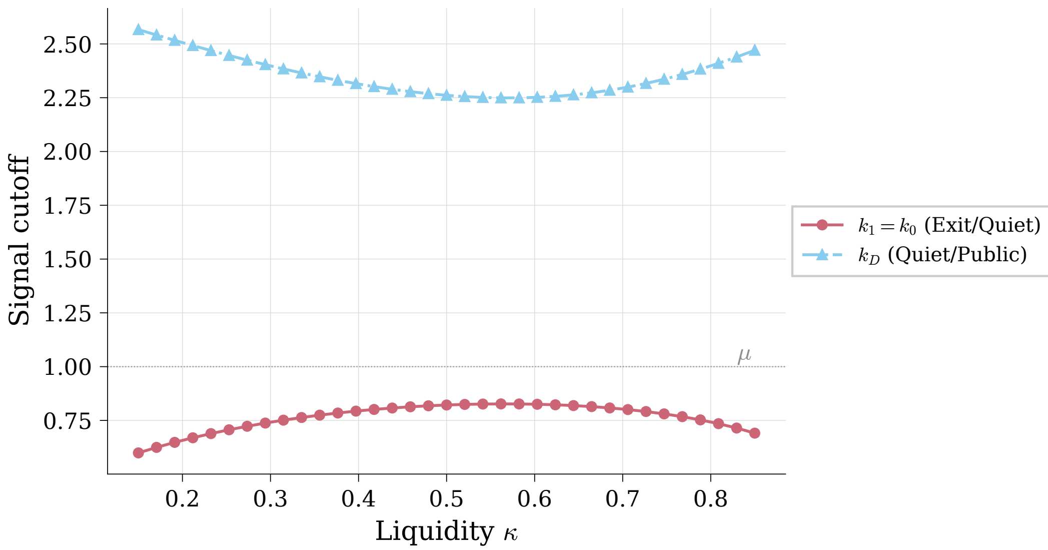

Figure H.5: Equilibrium cutoffs as functions of liquidity. The ordering $k_1 \leq k_0 \leq k_D$ is preserved; regions may collapse. The horizontal dashed line marks the prior mean μ.

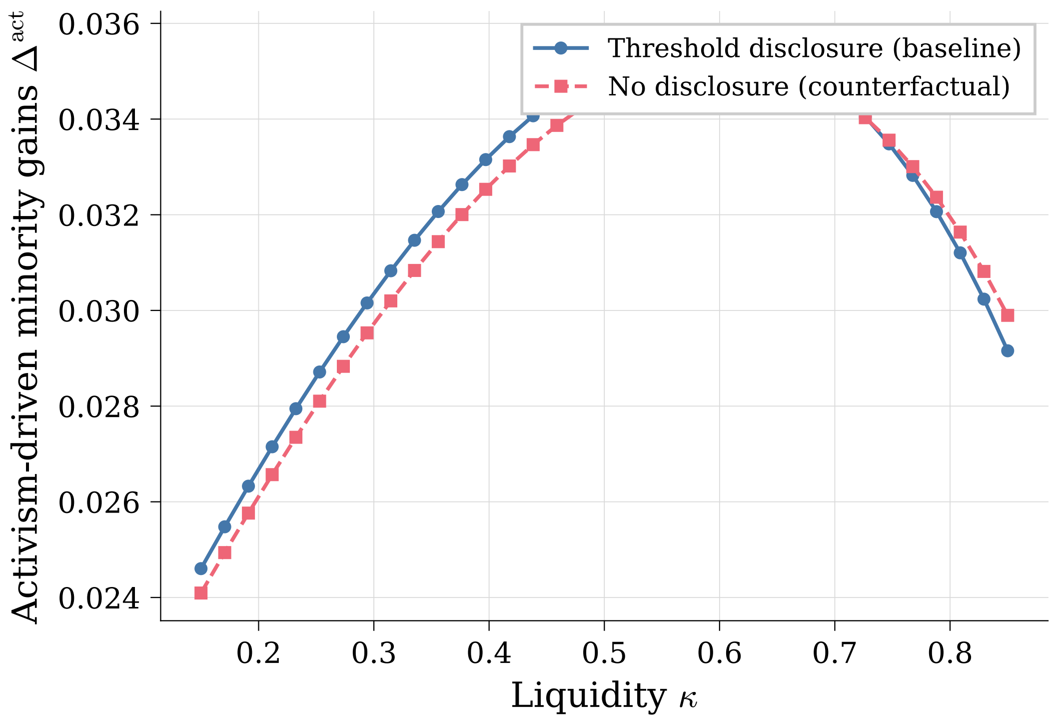

Figure H.6: Disclosure attenuation through the inference channel (partial equilibrium in κ). We fix the blockholder's cutoff strategy at the baseline calibrated equilibrium ($\kappa = 0.5$) and vary liquidity κ only through the order-flow noise distribution. The blue curve plots $\Delta^{\mathrm{act}}(\kappa)$ under threshold disclosure with fixed cutoffs. The rose curve plots the counterfactual "no-disclosure" benchmark with the same fixed cutoffs, in which the market conditions only on order flow X.

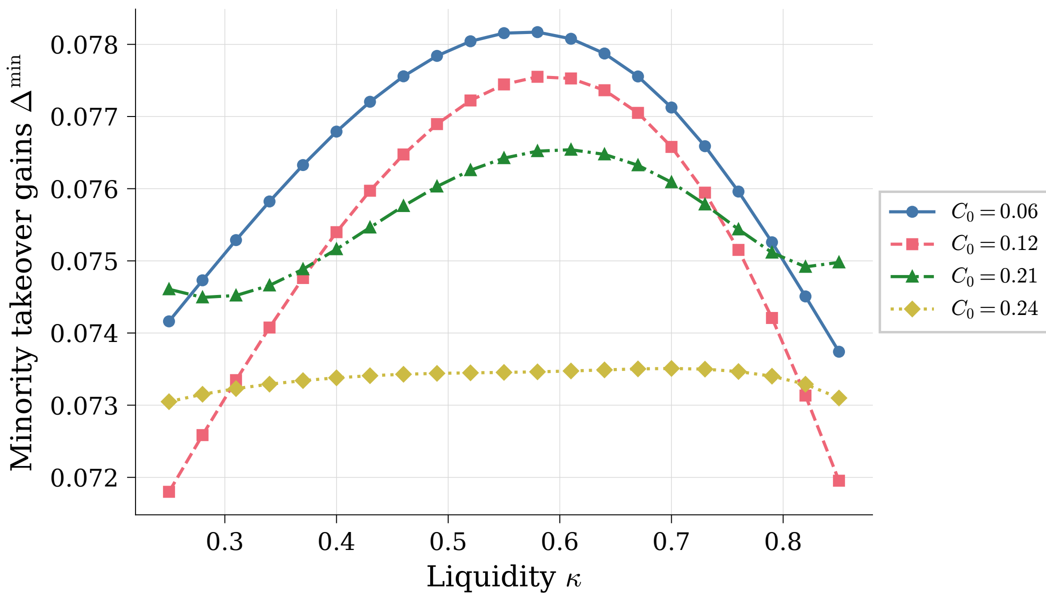

Figure H.7: Sensitivity to engagement cost C_0. Expected minority takeover gains remain broadly nonmonotone in κ across cost levels (computed over $\kappa \in [0.25, 0.85]$), with the hump flattening at high C_0 as voice becomes rare. Higher costs shrink the voice regions and reduce the activism component, but can increase bid incidence; the net effect on $\Delta^{\min}$ is therefore ambiguous in general.

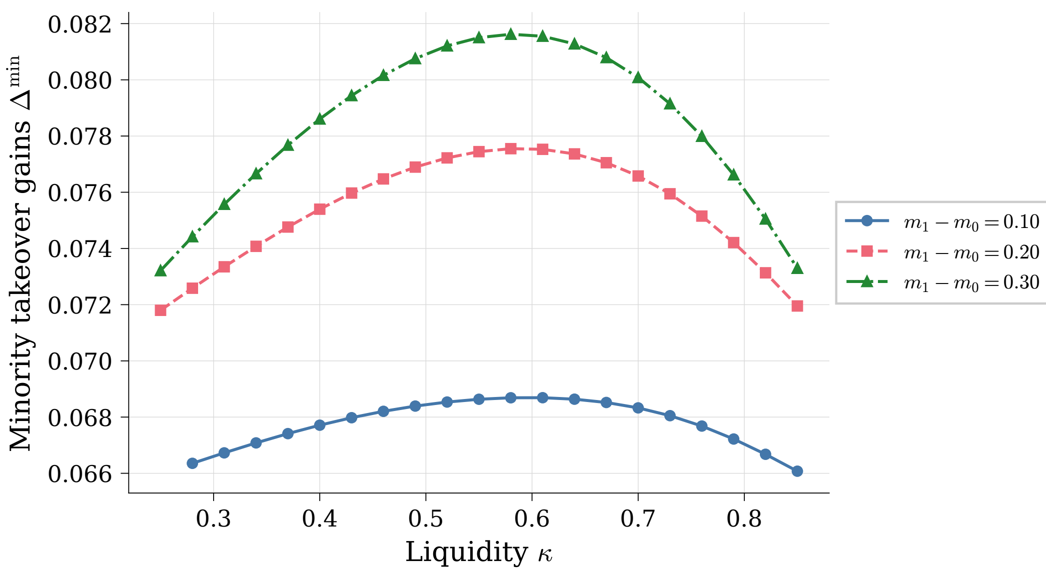

Figure H.8: Sensitivity to premium wedge $(m_1 - m_0)$. A larger wedge raises the premium conditional on bidding but can deter bids; expected minority takeover gains can therefore be nonmonotone in the wedge (computed over $\kappa \in [0.25, 0.85]$).

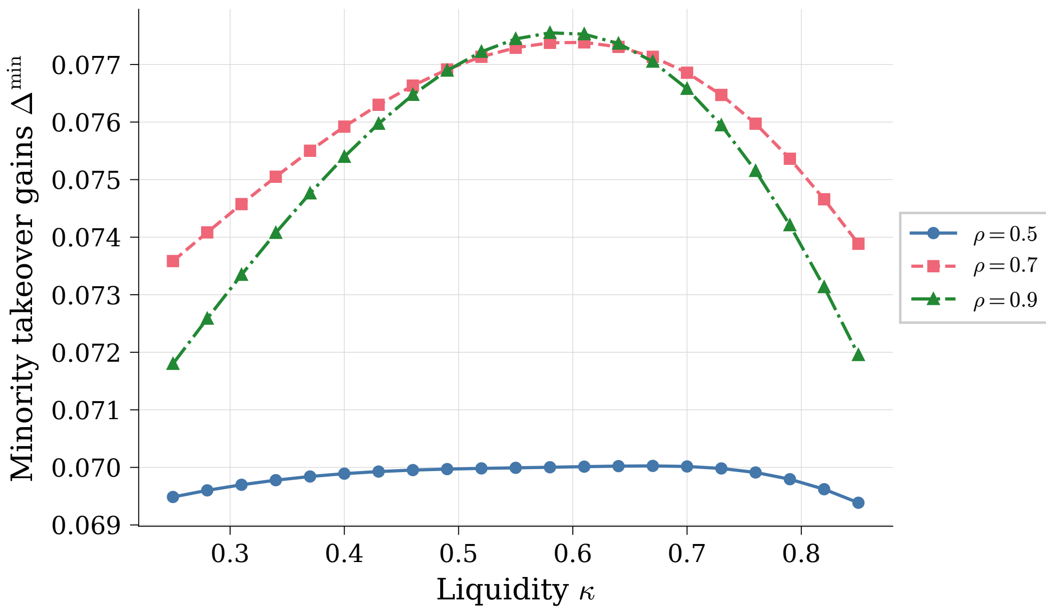

Figure H.9: Sensitivity to engagement success probability ρ. Lowering the expected success rate naturally shrinks the magnitude of minority takeover gains, flattening the curve. However, the qualitative nonmonotonicity remains robust, driven by the structural trade-off between bid incidence and order-flow inference.

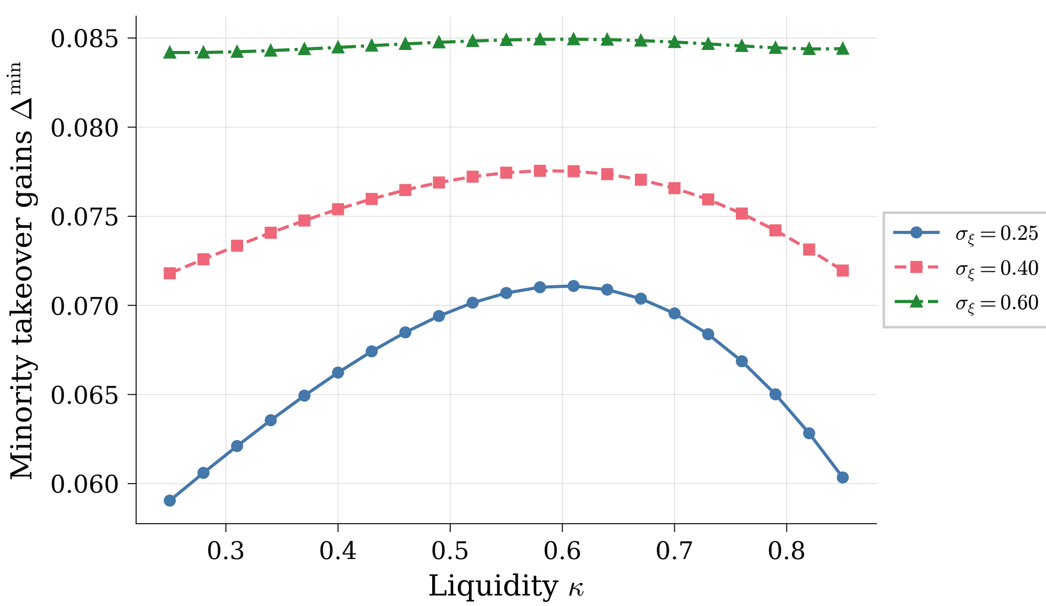

Figure H.10: Sensitivity to bidder synergy volatility σ_ξ. Increasing bidder heterogeneity raises overall bid incidence, which uniformly shifts expected minority gains upward while preserving the inference-driven hump shape. Larger values of σ_ξ also strictly relax the condition governing the price-to-entry feedback loop.

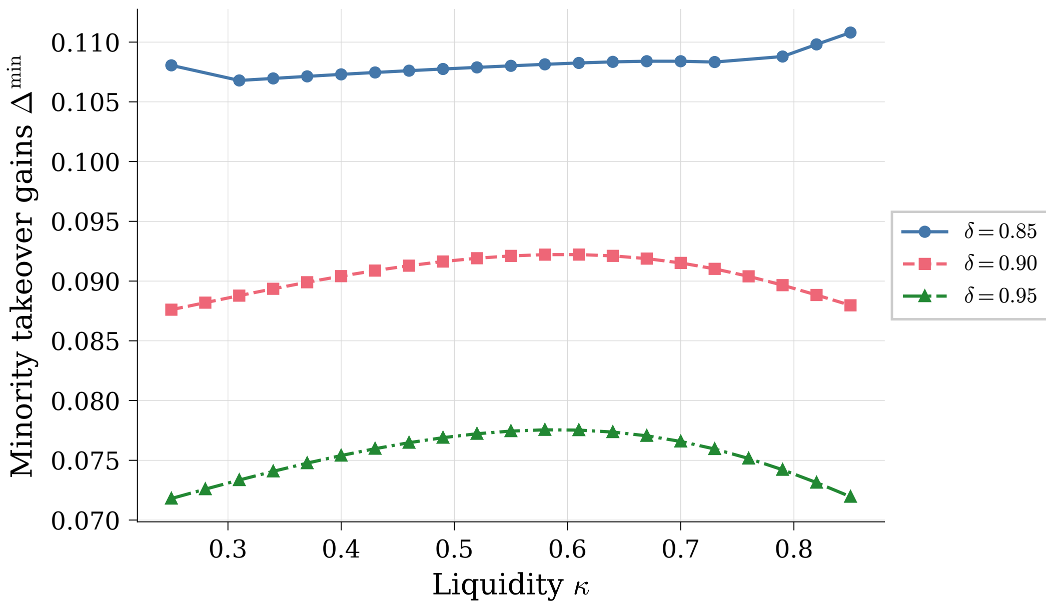

Figure H.11: Sensitivity to the discount factor δ. Lower discount factors mechanically reduce equilibrium prices, increasing bid incidence as the bidder faces lower acquisition costs. Although lower δ weakens engagement incentives, the bid-probability channel dominates, producing a uniform upward shift in minority gains and a progressive flattening of the hump shape.

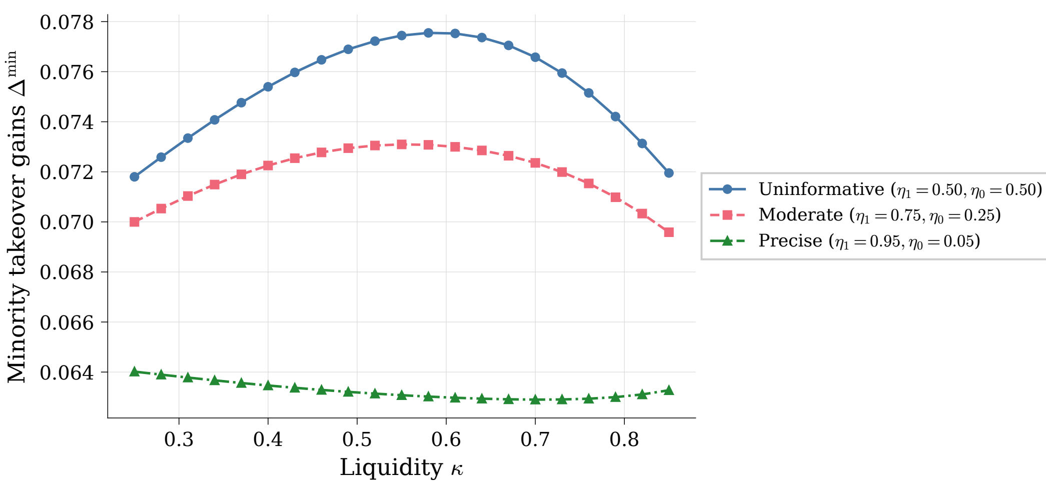

Figure H.12: Disclosure attenuation via noisy rumors. Under the intermediate information regime, the market receives an imperfect signal of hidden engagement. As the public rumor becomes more precise ($\eta_1 - \eta_0$ increases), engagement becomes more observable in nondisclosed states, progressively flattening the sensitivity of the activism premium to market liquidity.

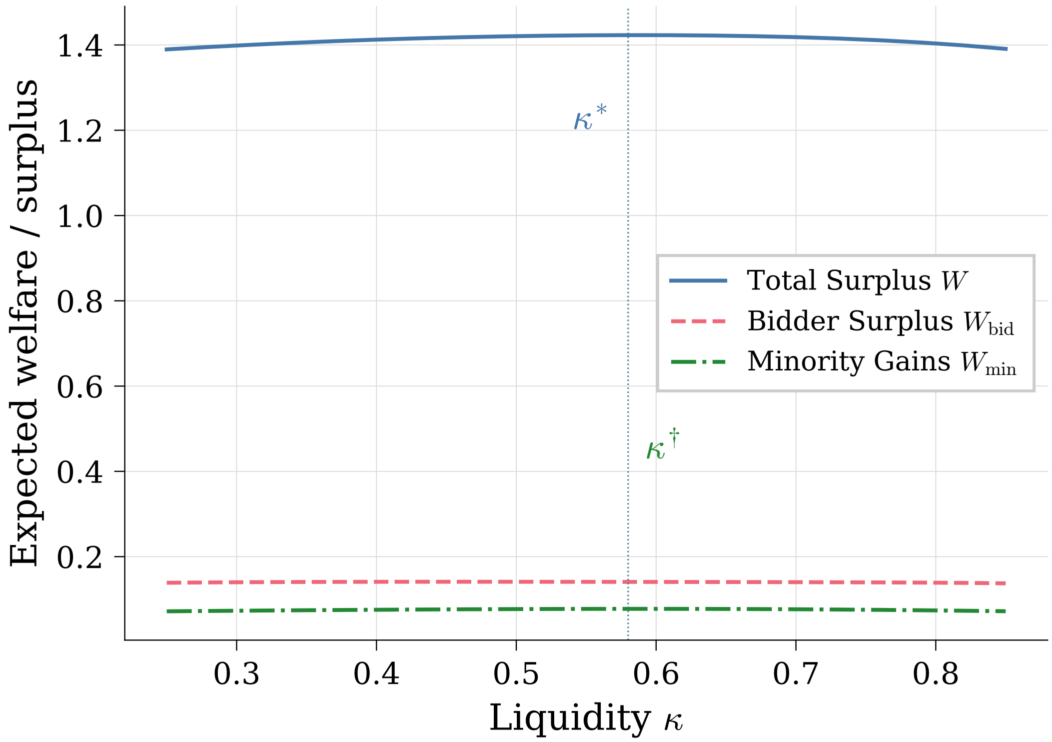

Figure H.13: **Welfare and the liquidity hump.** Expected total surplus W and its components—blockholder payoff W_B, minority gains Δ^{min}, and bidder surplus W_{bid}—against noise-trading intensity κ. By Lemma 7 minority value creation is endpoint-symmetric, $\Delta^{\text{min}}(0) = \Delta^{\text{min}}(1) > 0$: it does *not* vanish at the extremes. The interior hump is *conditional*—it obtains under condition (W*) and inverts to a trough when the condition fails—and the total W peaks at κ^*, which may differ from the minority optimum $\kappa^\dagger$ (the planner's wedge of Proposition 7). At the baseline calibration the two optima nearly coincide.

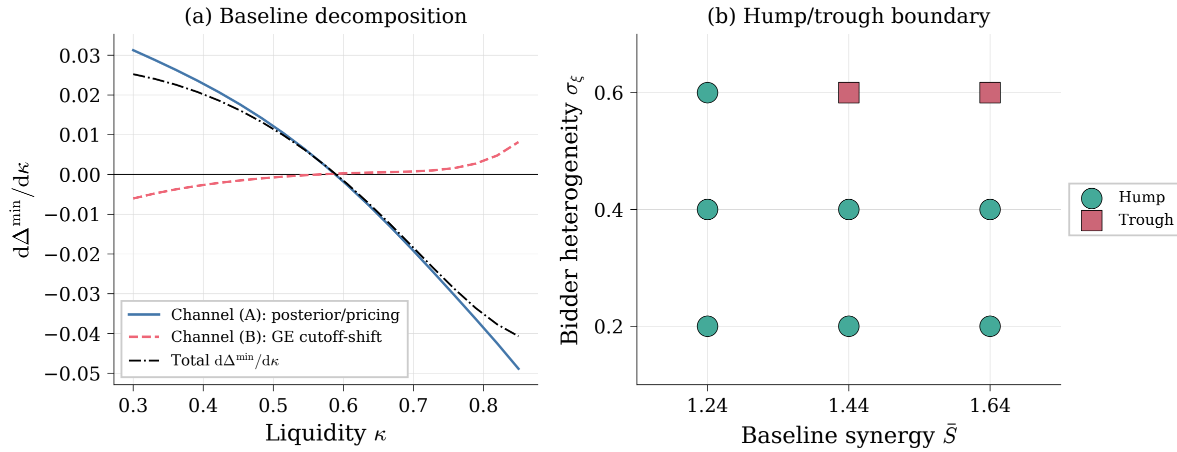

Figure H.14: **The GE cutoff-shift channel (Appendix F).** Left: decomposition of $\mathrm{d}\Delta^{\min}/\mathrm{d}\kappa$ into the posterior/pricing channel (A) and the GE cutoff-shift channel (B) along the baseline equilibrium path; channel (A) dominates throughout the certified region of Corollary 3, so the hump is a theorem there (up to ε-localization of the peak, Theorem 3). Right: hump/trough classification over the $(\bar{S}, \sigma_\xi)$ robustness family. The troughs at $\sigma_\xi = 0.60$ are the certified counterexamples of Proposition 17: channel (A) is single-peaked in every cell, so the trough orientation is produced entirely by the GE channel.

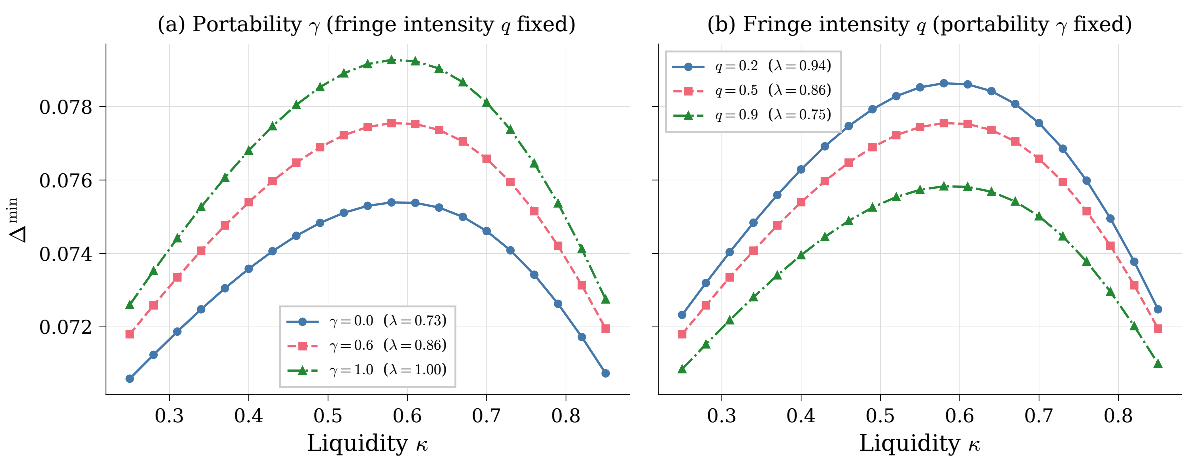

Figure H.15: **Minority gains under the microfounded wedge (Appendix D).** Equilibrium $\Delta^{\min}(\kappa)$ when the premium wedge is generated by the disagreement-node tender game, $m_1 - m_0 = (1-\theta)\lambda\rho\Delta_{\text{eng}}$ with $\lambda = 1 - q(1-\gamma)\psi$. Left: sweeping the portability γ of the activist's improvement under fringe acquirers ($\gamma = 0$: superseding acquirers erode appropriability). Right: sweeping the fringe-raid intensity q. The legend reports the implied appropriability λ; the sensitivity analysis of Figure H.8 is thereby re-expressed in observable primitives.